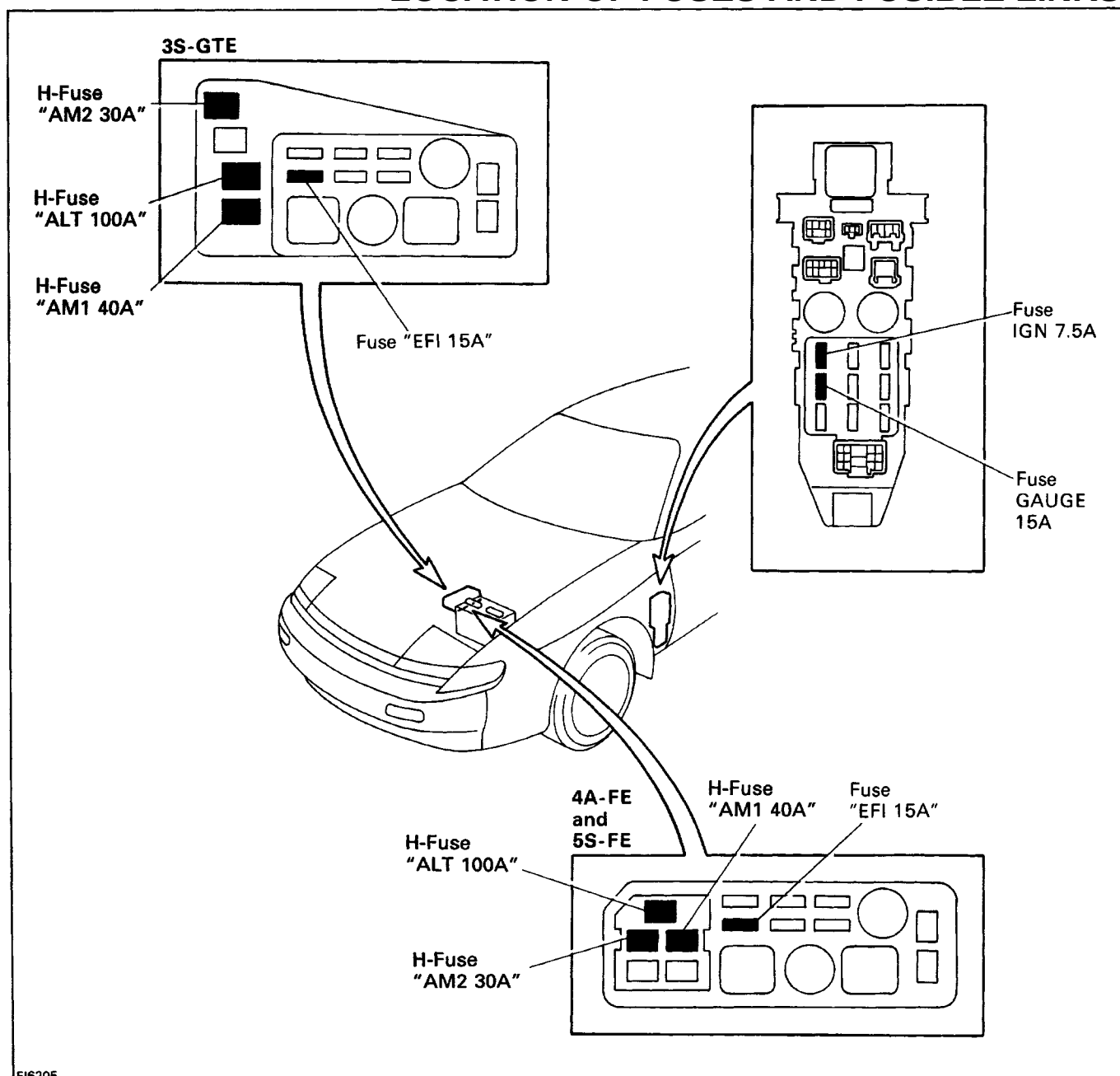


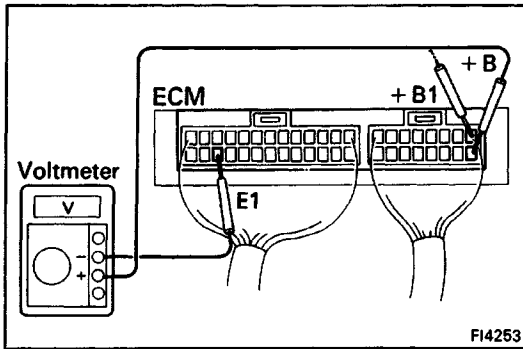
TROUBLESHOOTING WITH VOLT/OHMMETER

HINT:

- The following troubleshooting procedures are designed for inspection of each separate system, and therefore the actual procedure may vary somewhat. However, troubleshooting should be performed while referring to the inspection methods described in this manual.
- Before beginning inspection, it is best to first make a simple check of the fuses, fusible links and the condition of the connectors.
- The following troubleshooting procedures are based on the supposition that the trouble lies in either a short or open circuit within the computer.
- If engine trouble occurs even though proper operating voltage is detected in the computer connector, then it can be assumed that the ECM is faulty and should be replaced.

LOCATION OF FUSES AND FUSIBLE LINKS





MFI SYSTEM CHECK PROCEDURE (4A-FE)

HINT:

- Perform all voltage measurements with the connectors connected.
- Verify that the battery voltage is 11 V or more when the ignition switch is in "ON" position.

Using a voltmeter with high impedance (10 k Ω /V minimum), measure the voltage at each terminal of the wiring connectors.

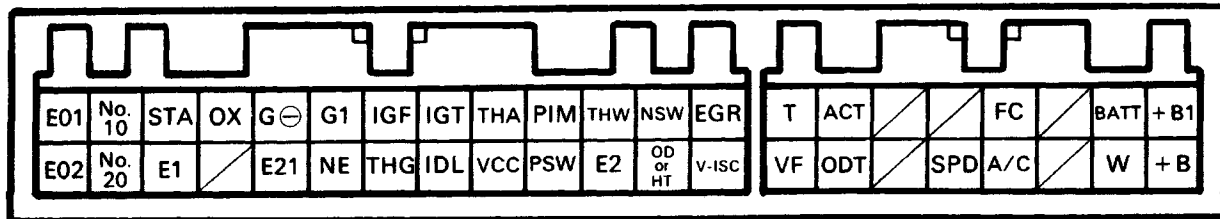
Terminals of Engine ECM (4A-FE)

Symbol	Terminal Name	Symbol	Terminal Name	Symbol	Terminal Name
E01	POWER GROUND	IGT	IGNITER	ACT	A/C AMPLIFIER
E02	POWER GROUND	IDL	THROTTLE POSITION SENSOR	*2ODT	O/D SOLENOID
No.10	INJECTOR	THA	INTAKE AIR TEMP. SENSOR		—
No.20	INJECTOR	VCC	VACUUM SENSOR		
STA	STARTER SWITCH	PIM	VACUUM SENSOR		
E1	ENGINE GROUND	PSW	THROTTLE POSITION SENSOR	SPD	SPEED SENSOR
OX	OXYGEN SENSOR	THW	ENGINE COOLANT TEMP. SENSOR	FC	CIRCUIT OPENING RELAY
	—	E2	SENSOR GROUND	A/C	A/C COMPRESSOR
G $\ominus$	DISTRIBUTOR GROUND	N SW	PARK/NEUTRAL POSITION SWITCH		—
E21	SENSOR GROUND	*1 OD *2 H T	O/D SOLENOID OXYGEN SENSOR HEATER		
G1	DISTRIBUTOR	EGR	EGR VSV	BATT	BATTERY
NE	DISTRIBUTOR	V-ISC	ACV VSV	W	MALFUNCTION INDICATOR LAMP
IGF	IGNITER	T	DATA LINK CONNECTOR 1	+B1	MFI MAIN RELAY
" THG	EGR GAS TEMP. SENSOR	VF	DATA LINK CONNECTOR 1	B	MFI MAIN RELAY

ECM Terminals

*1 CALIF. only

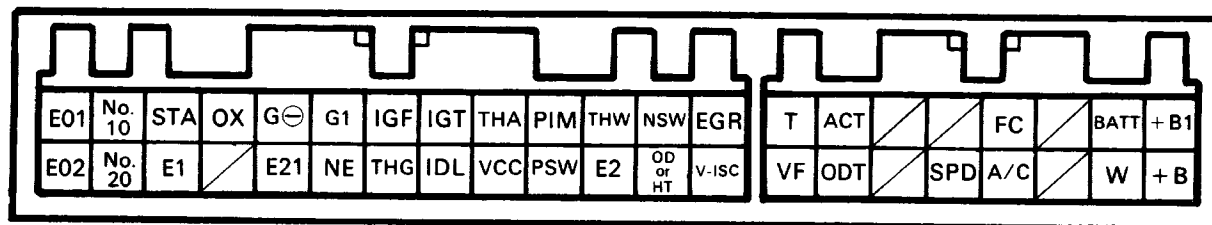
*2 Ex. CALIF.

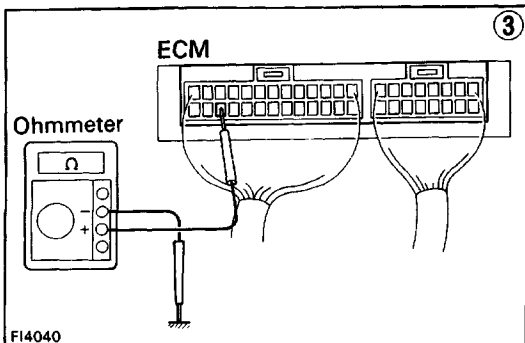
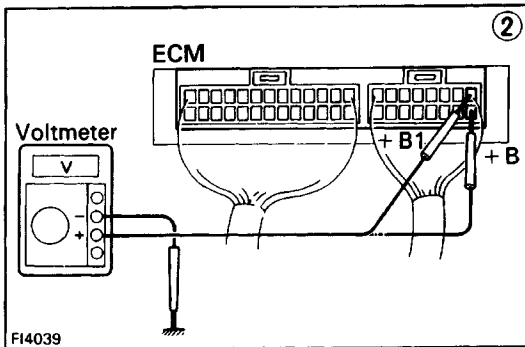
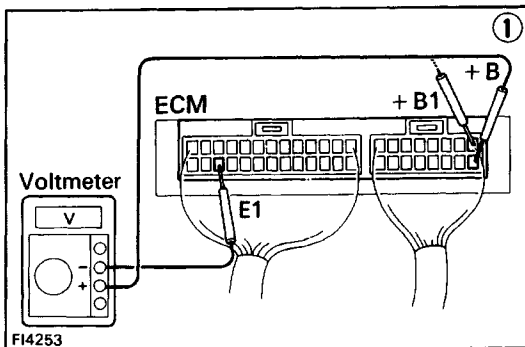
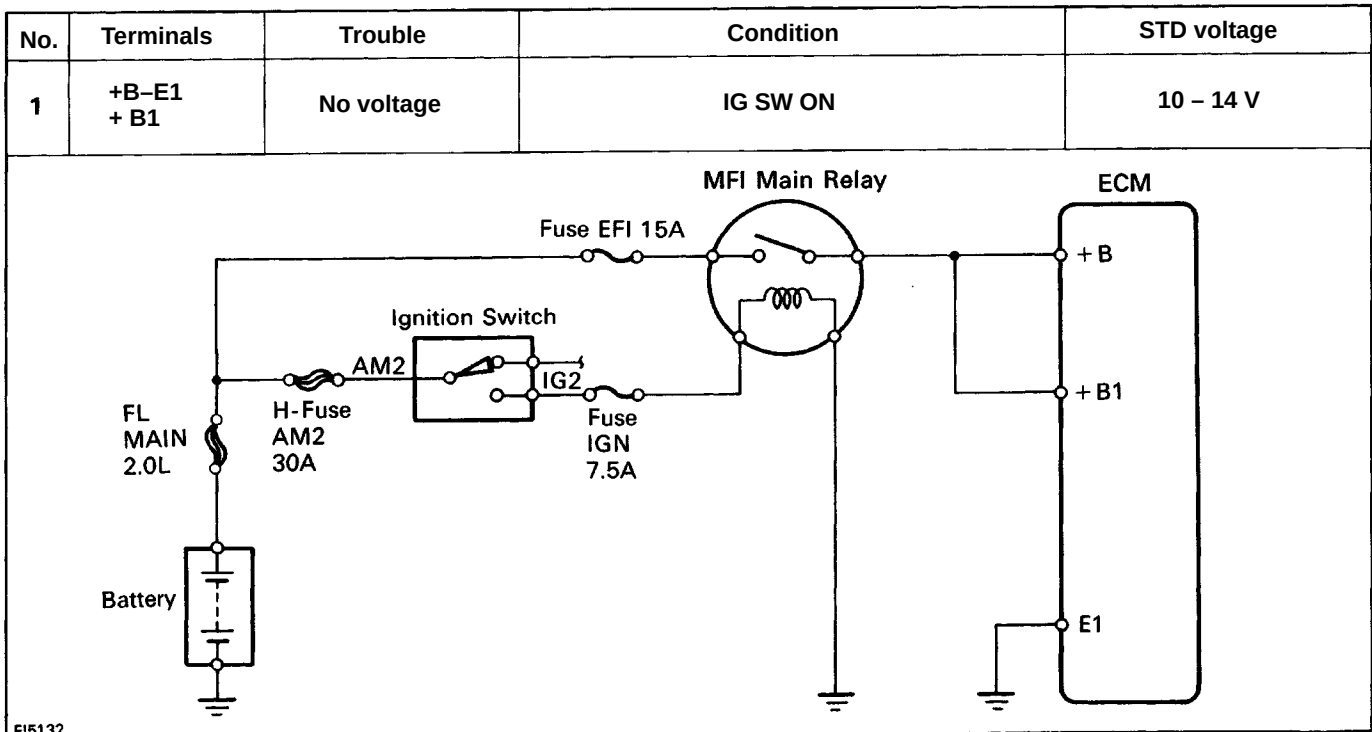


Voltage at ECM Wiring Connectors (4A-FE) (4A-FE)

No.	Terminals	Condition		STD voltage (V)	See page
1	+B-E1 + B1	IG SW ON		10-14	FI-61
2	BATT – E 1	–		10-14	FI-62
3	IDL – E2	IG SW ON	Throttle valve open	10-14	FI-63
	PSW – E2		Throttle valve fully closed	10-14	
4	PIM – E2	IG SW ON		3.3-3.9	FI-64
	VCC – E2			4.5-5.5	
5	No. 10 E01 No. 20 E02			10-14	FI-65
6	THA – E2	IG SW ON	Intake air temp. 20°C (68°F)	1 –3	FI-66
7	THW – E2		Engine coolant temp. 80°C (176°F)	2.0-2.8	FI-67
8	STA – E1	Cranking		6-14	FI-68
9	IGT – E1	Cranking or idling		0.7-1.0	FI-69
10	W – E1	No trouble (malfunction indicator lamp off) and engine running		10-14	FI-70
11	A/C – E1	IG SW ON	Air conditioning ON	8-14	FI-71

ECM Terminals





(1) There is no voltage between ECM terminals + B or + B1 and E1. (IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1 and body round. (IG SW ON)

NO

OK

(3) Check wiring between ECM terminal E1 and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Check fuses, H-fuse, fusible link and ignition switch.

BAD

Repair or replace.

OK

Check MFI main relay.
(See page FI-216)

BAD

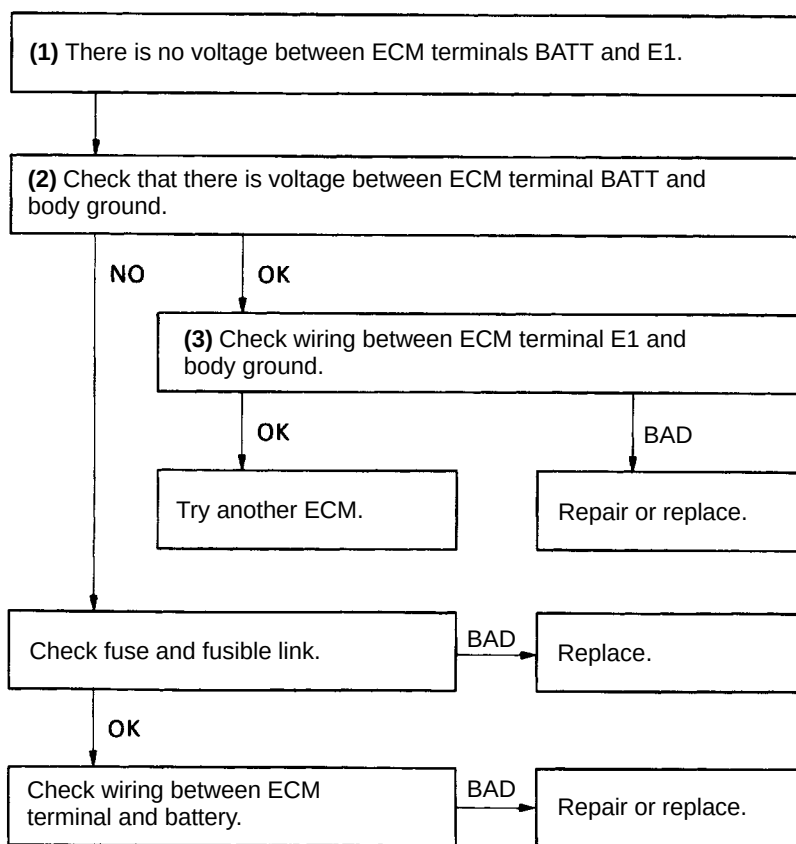
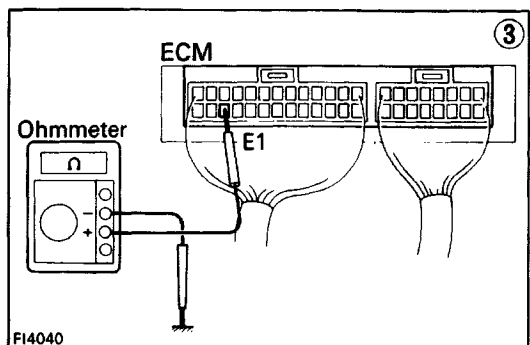
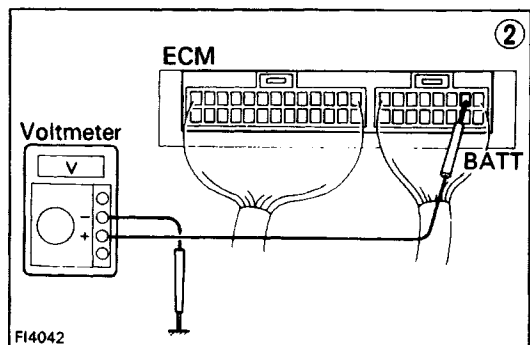
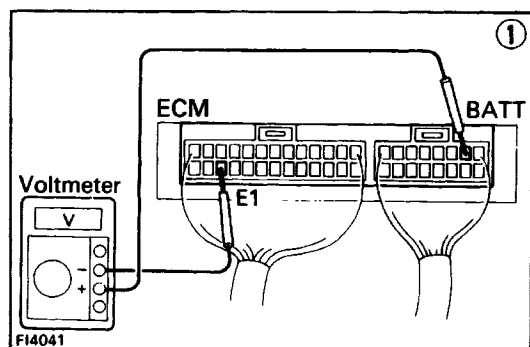
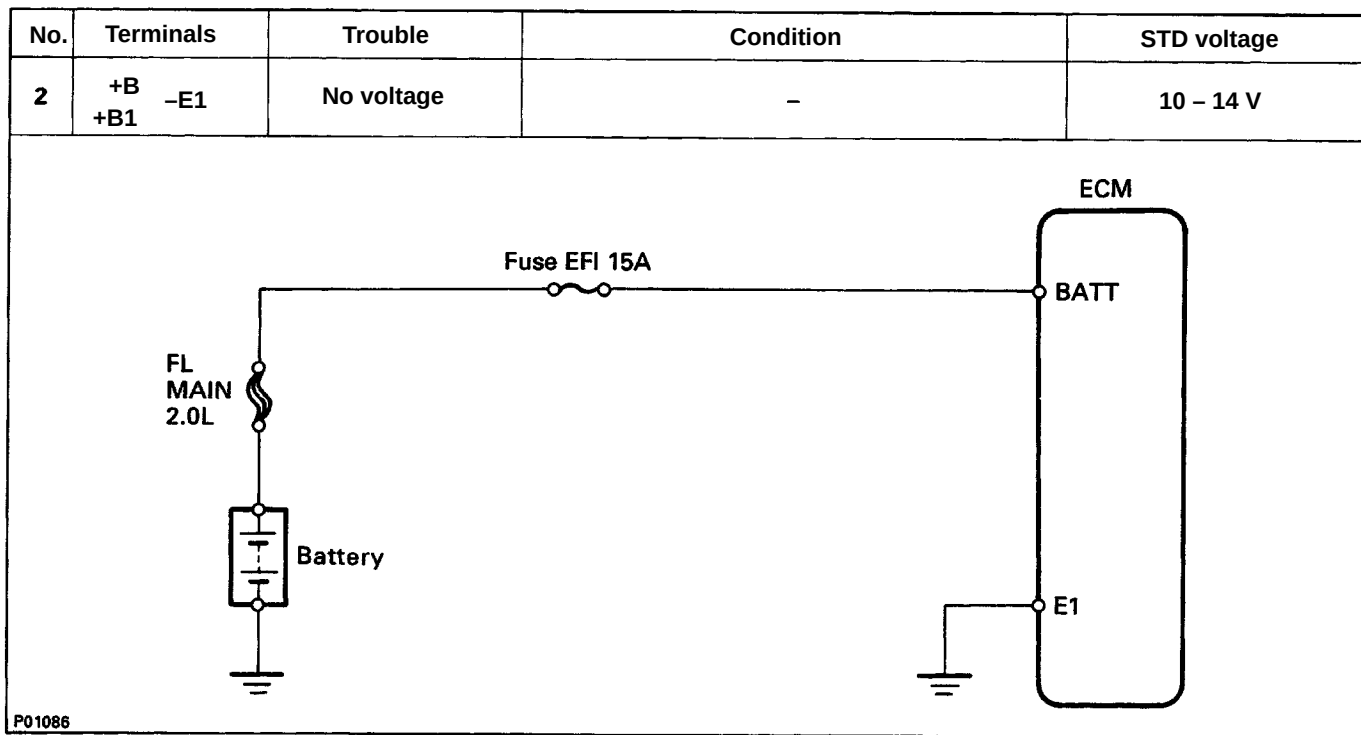
Replace.

OK

Check wiring between MFI main relay and battery.

BAD

Repair or replace.



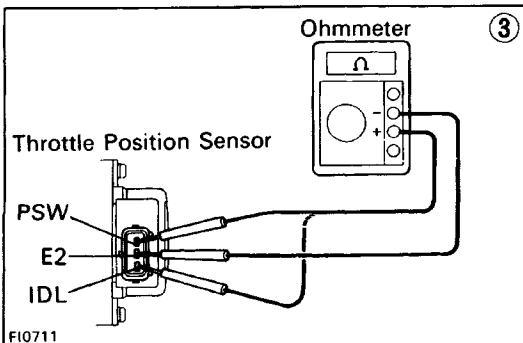
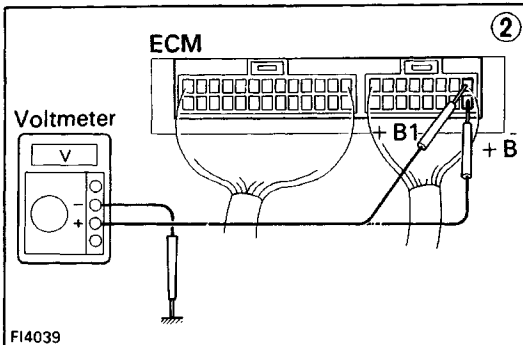
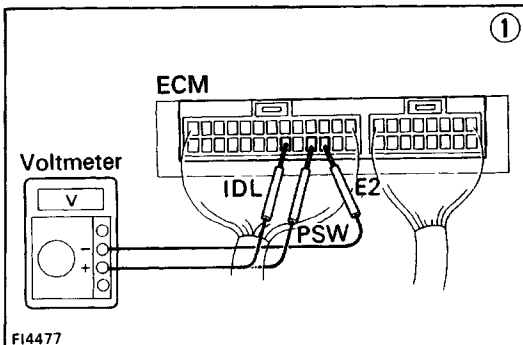
No.	Terminals	Trouble	Condition		STD voltage
3	IDL – E2	No voltage	IG SW ON	Throttle valve open	10 – 14 V
	PSW – E2			Throttle valve fully closed	10 – 14 V

Throttle Position Sensor

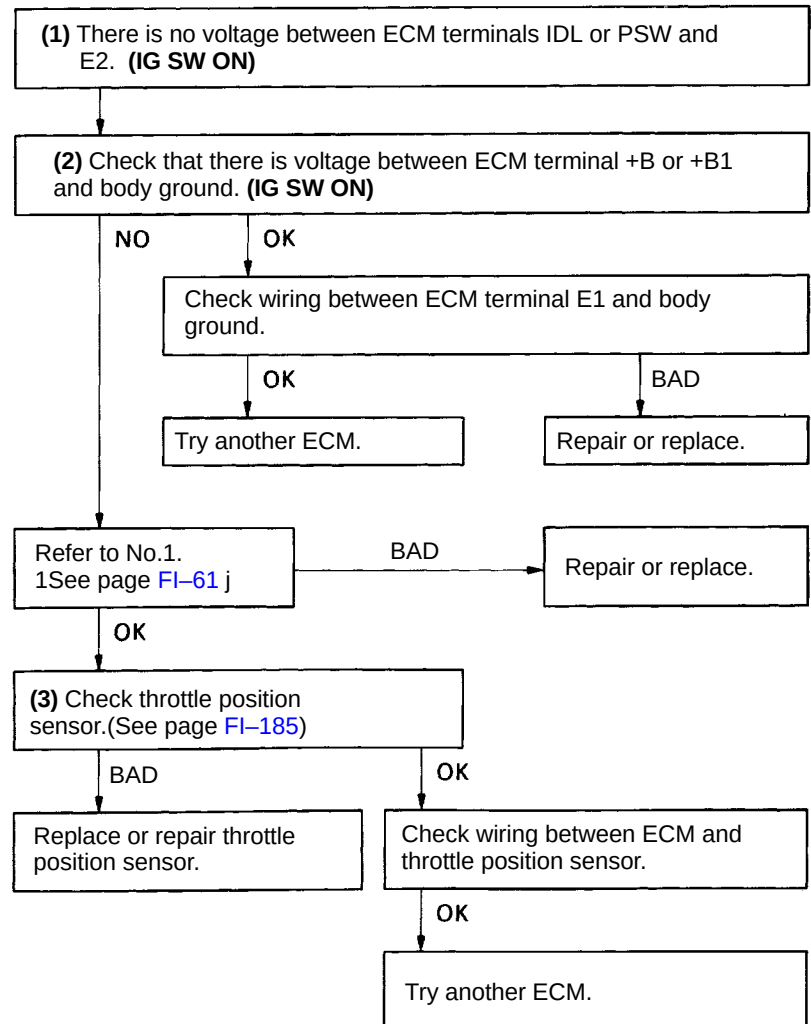
ECM

IDL, E2, PSW, +B, IDL, E2 (E21), PSW, E1

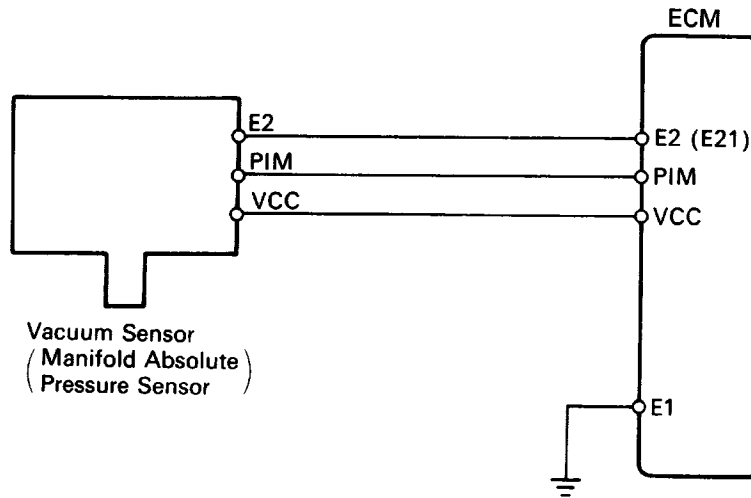
P02927



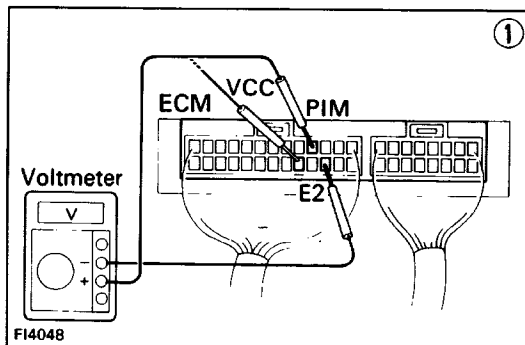
• IDL-E2, PSW-E2



No.	Terminals	Trouble	Condition	STD voltage
4	PIM – E2	No voltage	IG SW ON	3.3 – 3.9 V
	VCC–E2			4.5 – 5.5 V



FI1226



• PIM – E2, VCC – E2

(1) There is no voltage between ECM terminals PIM or VCC and E2. (IG SW ON)

(2) Check that there is voltage between ECM terminal + B (+B1) and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-61)

(3) Check wiring between ECM terminal E1 and body ground.

OK

BAD

Check vacuum sensor.
(See page FI-233)

Repair or replace.

BAD

OK

Replace vacuum sensor.

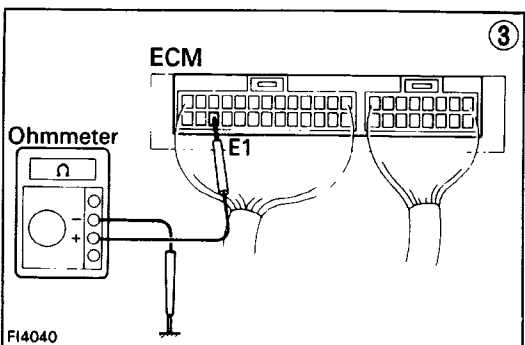
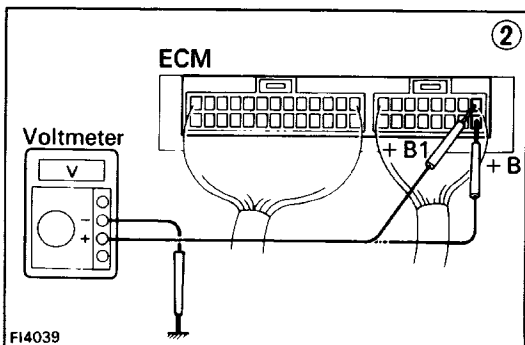
Check wiring between ECM and vacuum sensor.

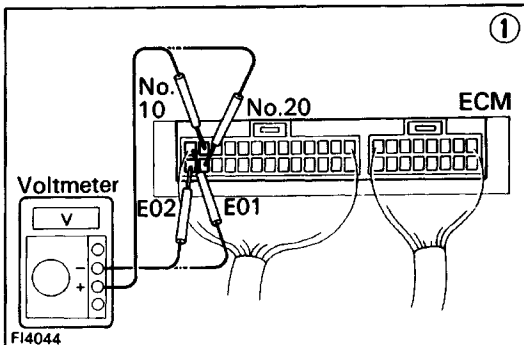
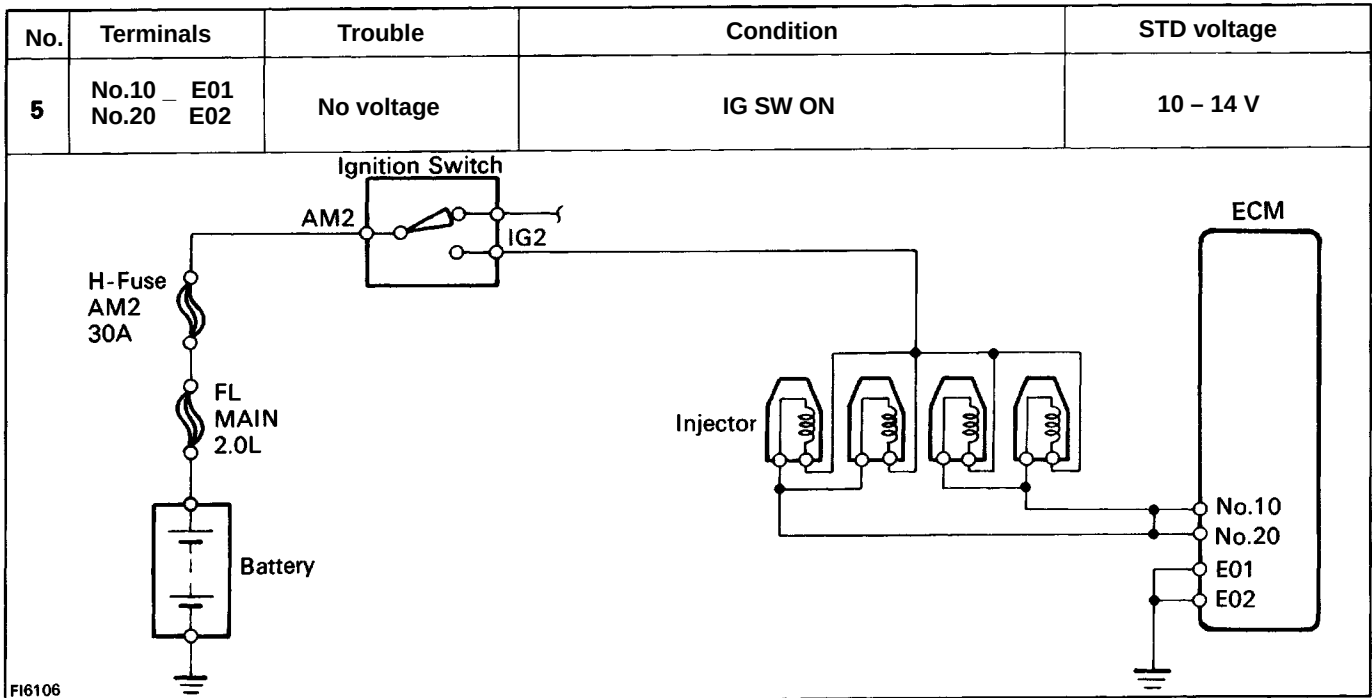
OK

BAD

Try another ECM.

Repair or replace.





(1) There is no voltage between ECM terminals No.10 and / or No.20 and E01 and / or E02. **(IG SW ON)**

(2) Check that there is voltage between ECM terminal No.10 and / or No. 20 and body ground.

NO

OK

Check wiring between ECM terminal E01 and / or E02 and body ground.

OK

Try another ECM.

Repair or replace.

Check H-fuse, fusible link and ignition switch.

BAD

Repair or replace.

OK

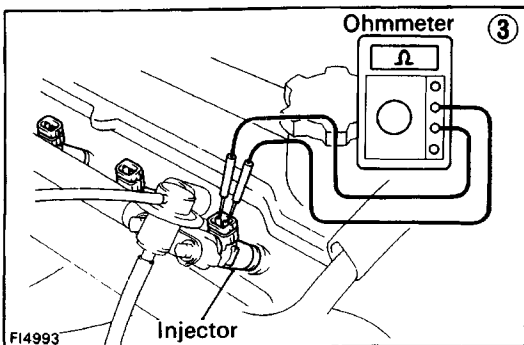
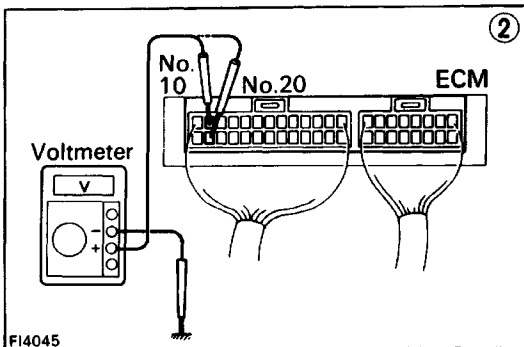
(3) Check resistance of each injector.
STD resistance. Approx. 13.8Ω

OK

Check wiring between ECM terminal No-10 and / or No.20 and battery.

BAD

Repair or replace.



No.	Terminals	Trouble	Condition		STD voltage
6	THA- E2	No voltage	IG SW ON	Intake air temperature 20°C (68°F)	1 – 3 V

Intake Air Temp. Sensor

ECM

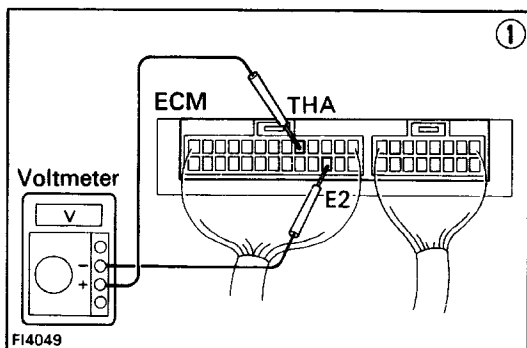
+ B (+ B1)

THA

E2 (E21)

E1

FI3572



(1) There is no voltage between ECM terminals THW and E2.
(IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1
and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-61)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

(3) Check intake air temp. sensor.
(See page FI-232)

Repair or replace.

BAD

OK

Replace intake air
temp. meter.

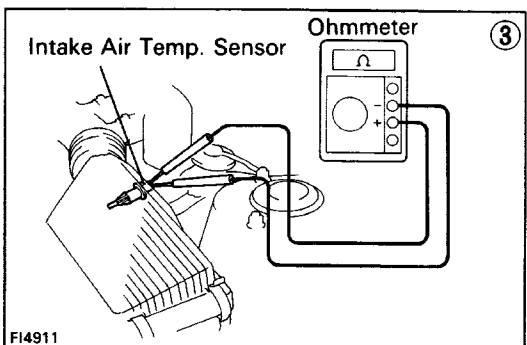
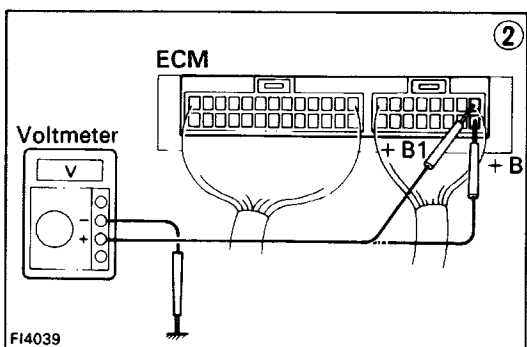
Check wiring between ECM and
intake air temp. sensor.

OK

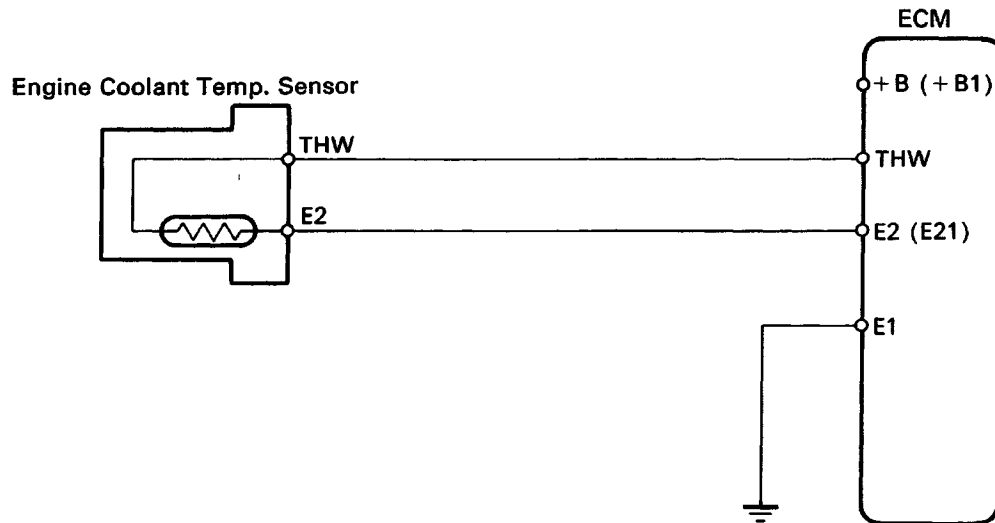
BAD

Try another ECM.

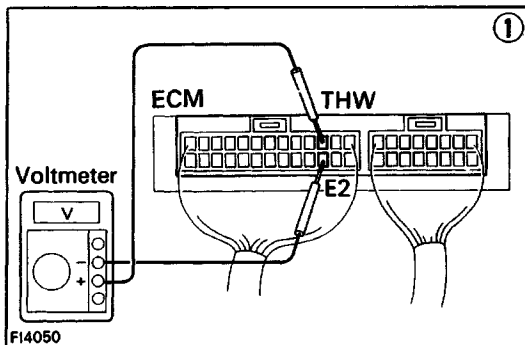
Repair or replace.



No.	Terminals	Trouble	Condition		STD voltage
7	THW- E2	No voltage	IG SW O N	Engine coolant temperature 80°C (176°F)	0.1 – 1.0 V



FI3572



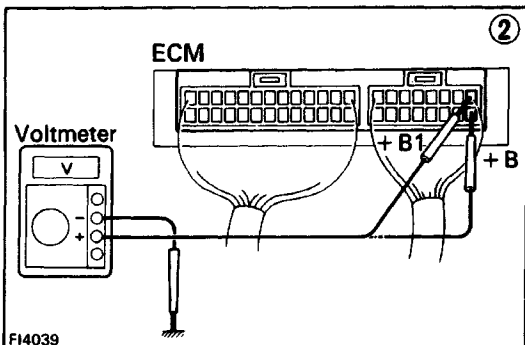
(1) There is no voltage between ECM terminals THW and E2. (IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1 and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-61)



Check wiring between ECM terminal E1 and body ground.

OK

BAD

(3) Check engine coolant temp. sensor. (See page FI-231)

Repair or replace.

BAD

OK

Replace engine coolant temp. sensor

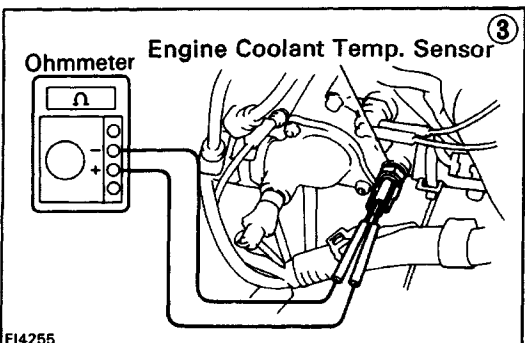
Check wiring between ECM and engine coolant temp. sensor.

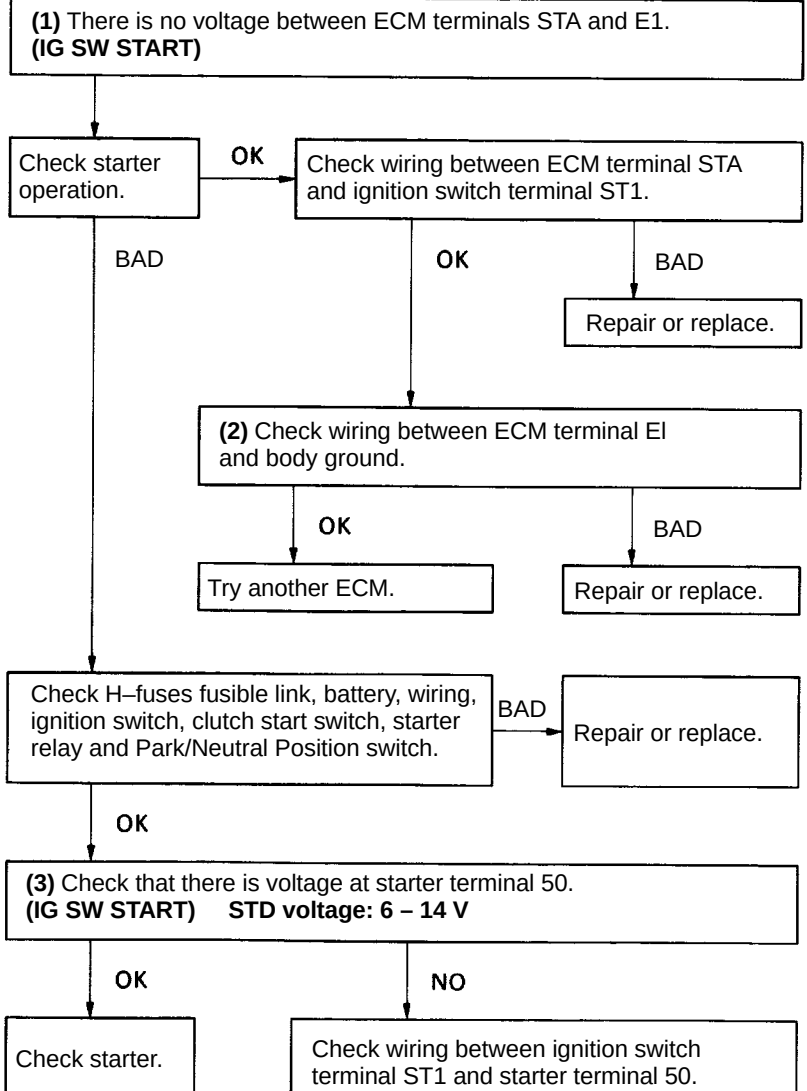
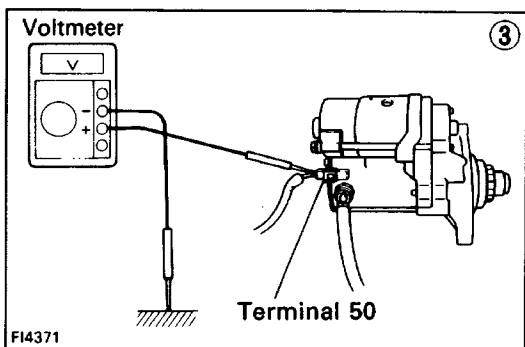
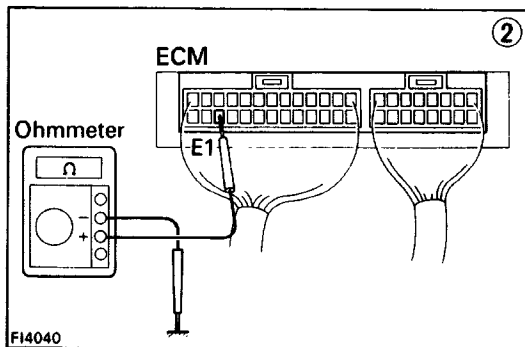
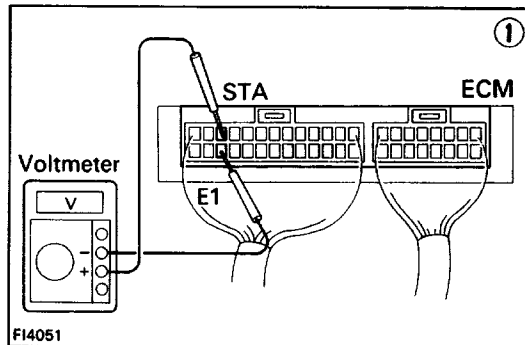
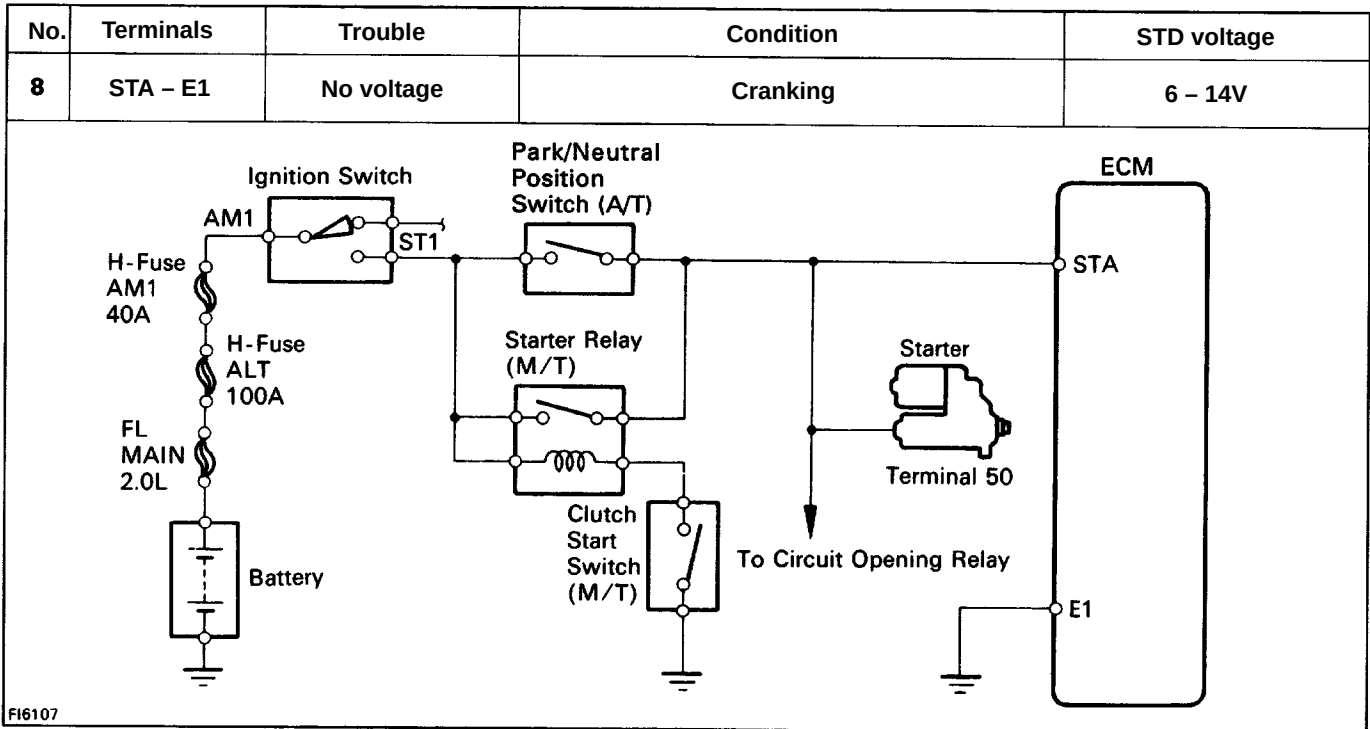
Try another ECM.

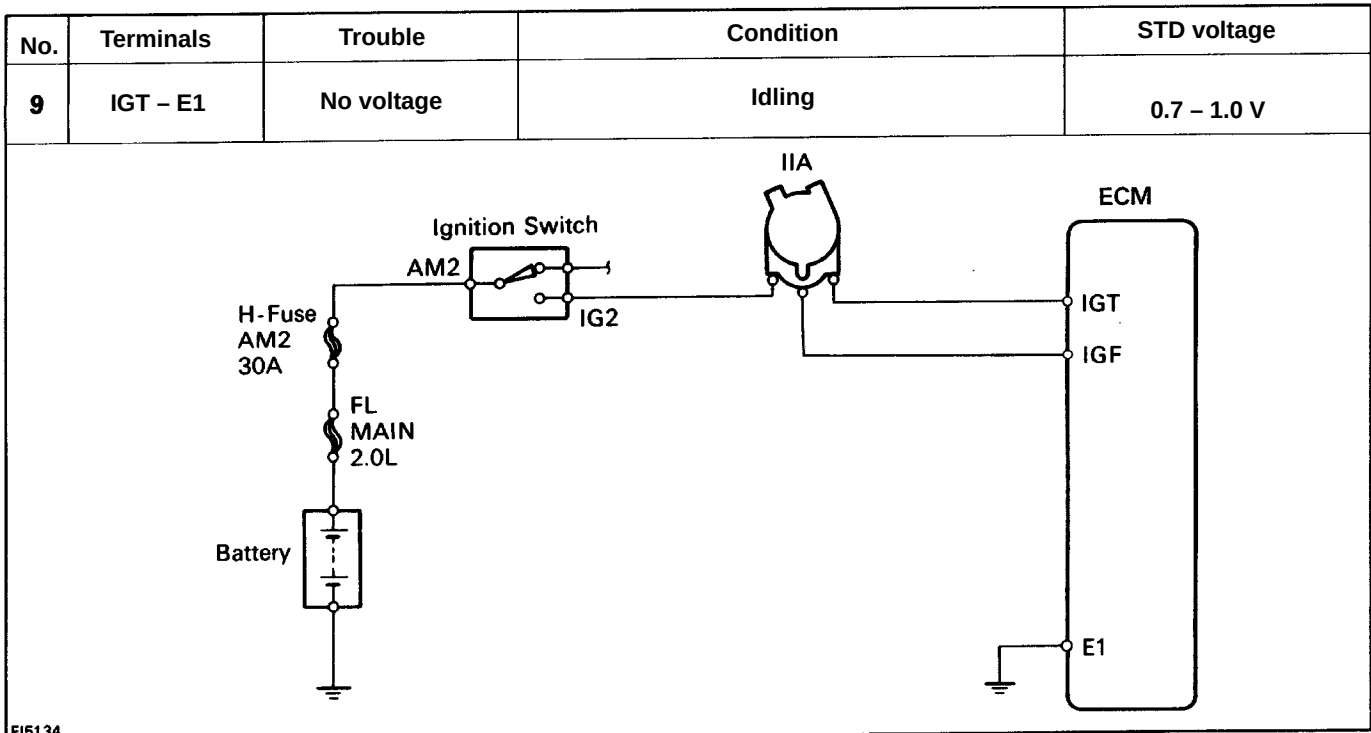
OK

BAD

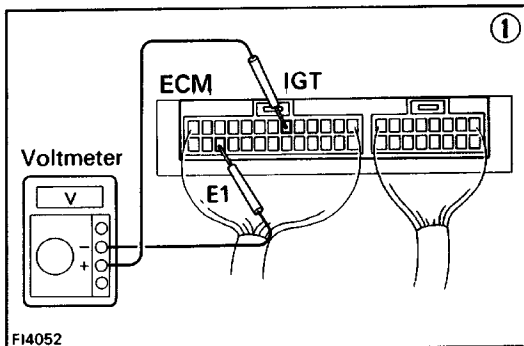
Repair or replace.



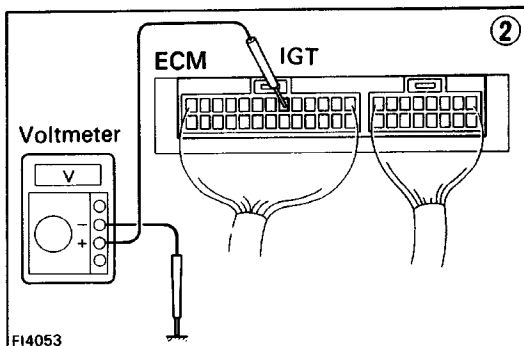




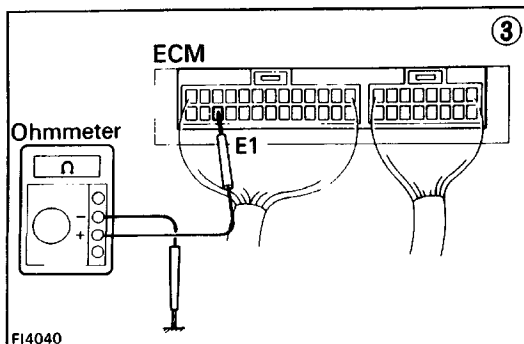
FI5134



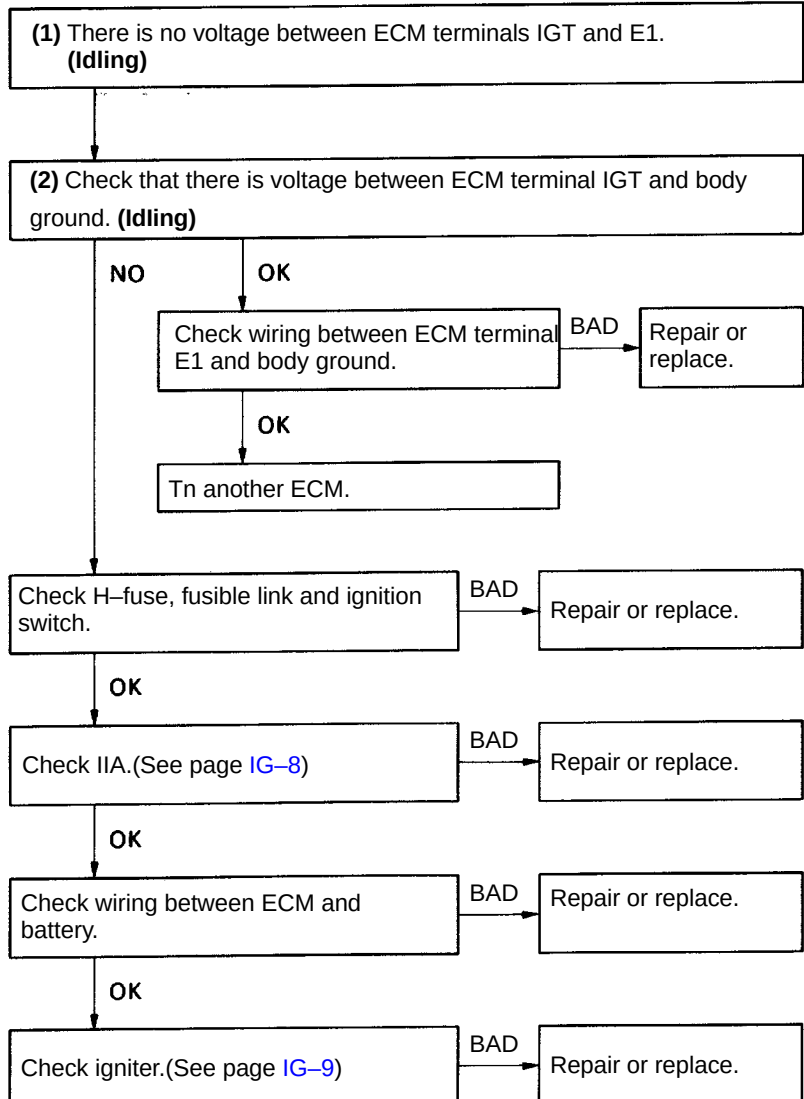
FI4052

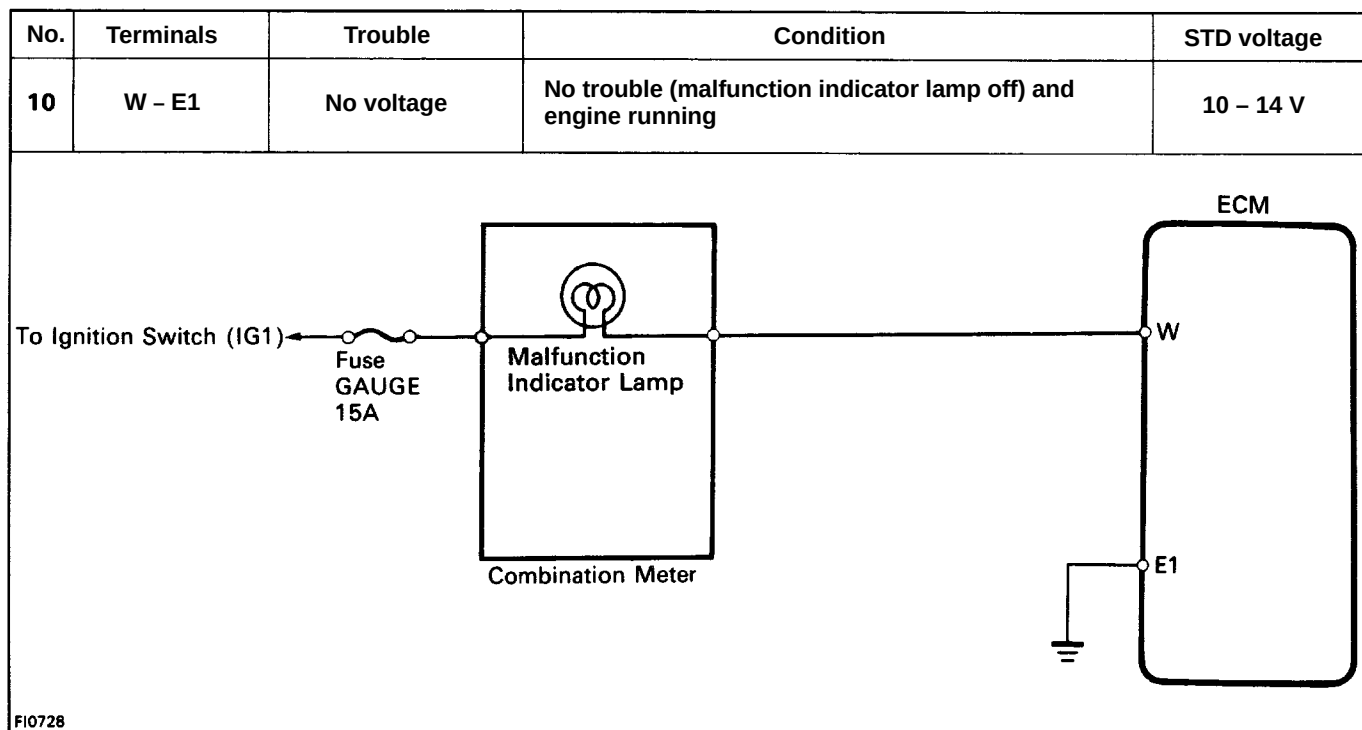


FI4053

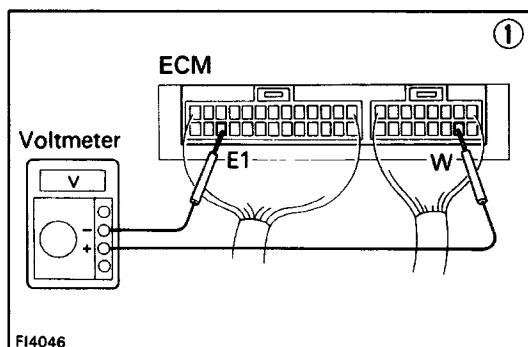


FI4040

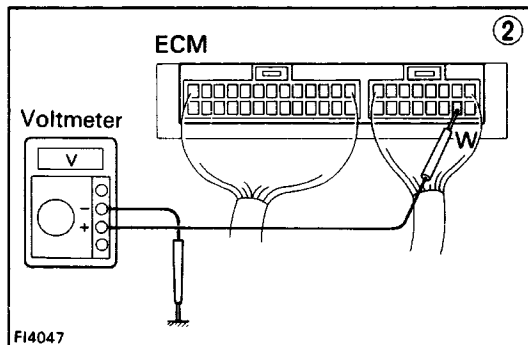




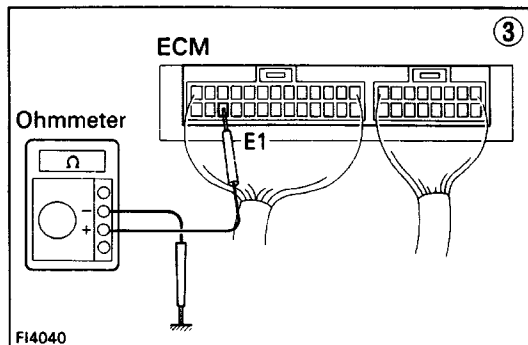
FI0728



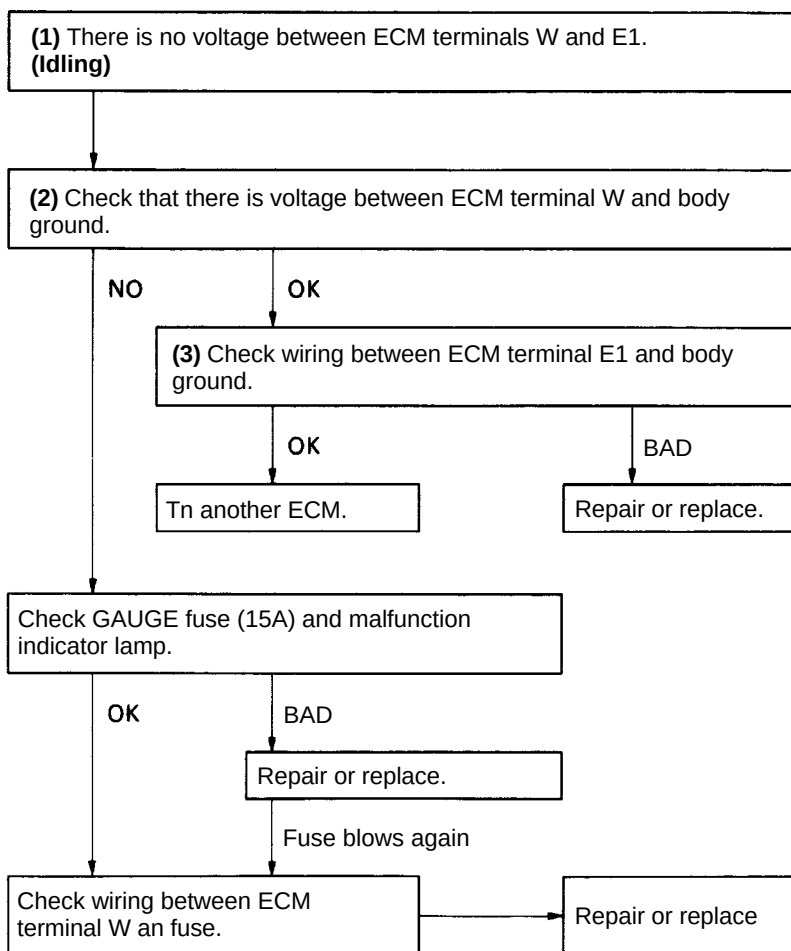
FI4046



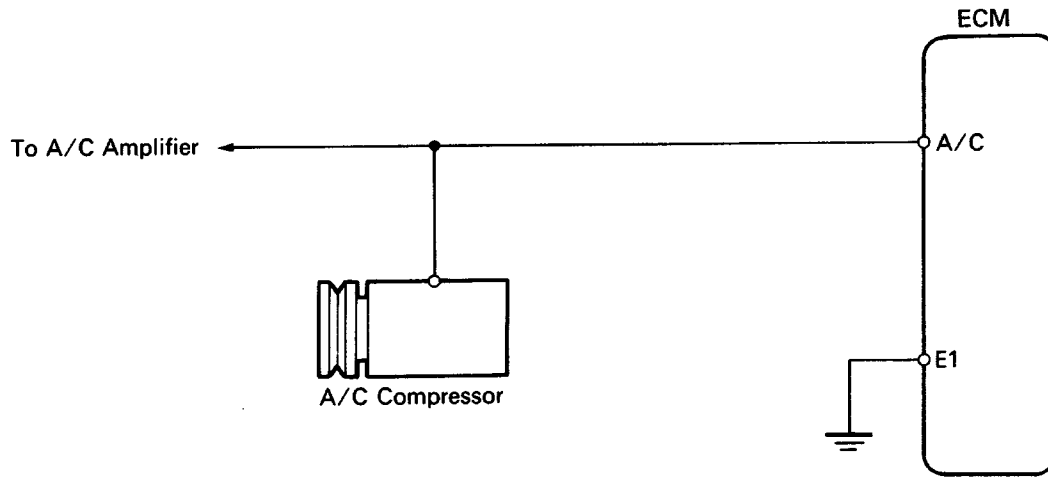
FI4047



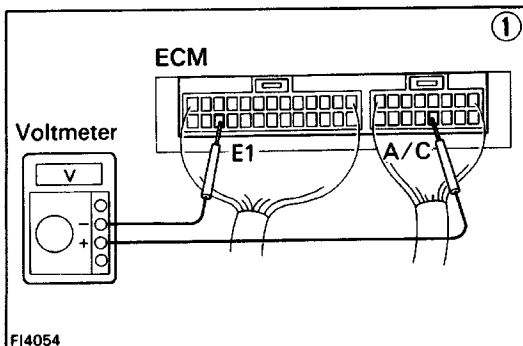
FI4040



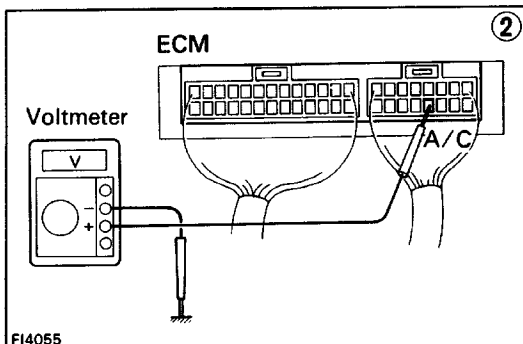
No.	Terminals	Trouble	Condition	STD voltage
11	A/C – E 1	No voltage	Air conditioning ON	8–14 V



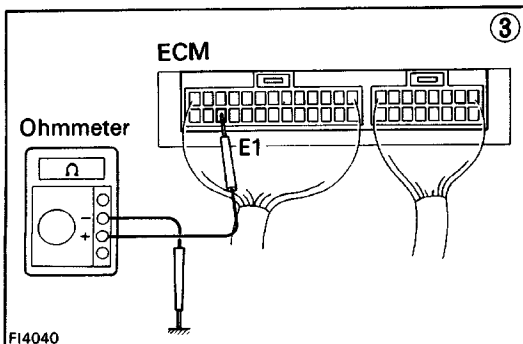
FI0922



FI4054



FI4055



FI4040

(1) There is no voltage between ECM terminals A/C and E1.
(Air conditioning on)

(2) Check that there is voltage between ECM terminal A/C and body ground.

NO

OK

(3) Check wiring between ECM terminal E1 and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Check compressor running.

OK

Check wiring between ECM terminal A/C and amplifier.

BAD

Repair or replace.

Check that there is voltage between amplifier terminal and body ground.

BAD

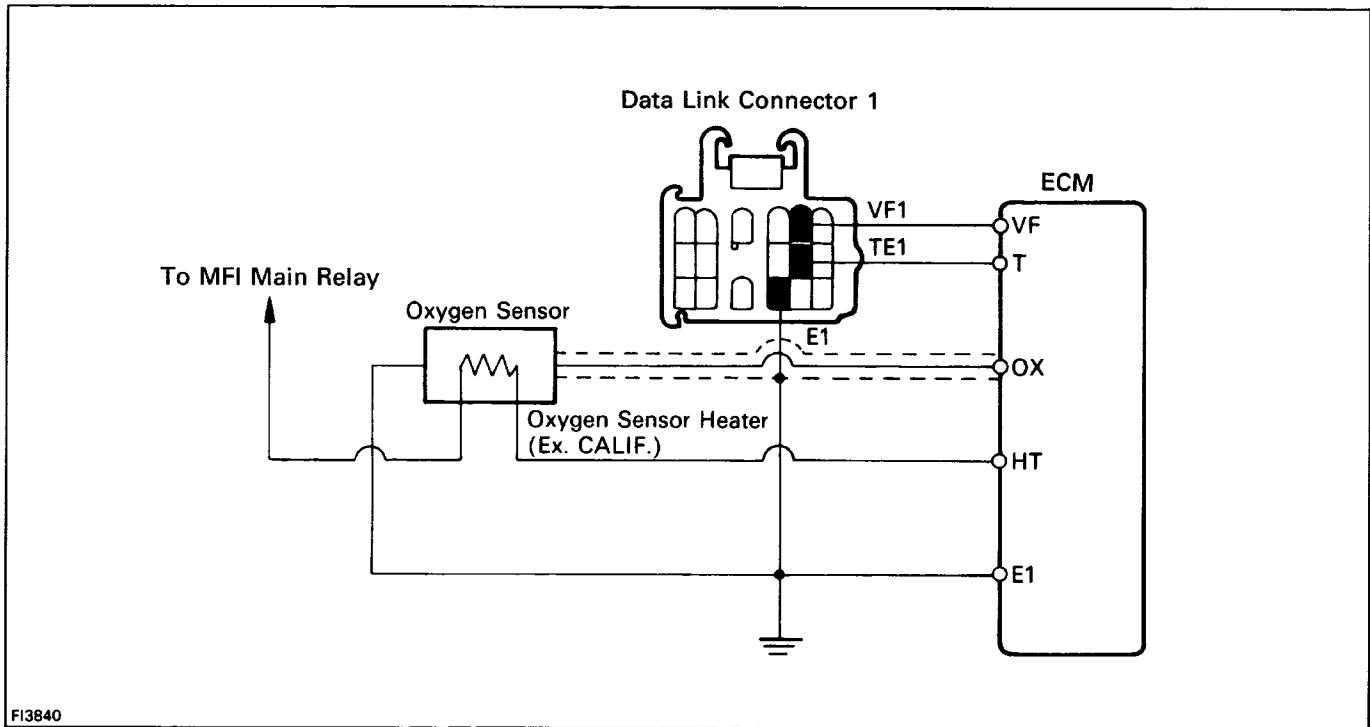
Repair or replace.

OK

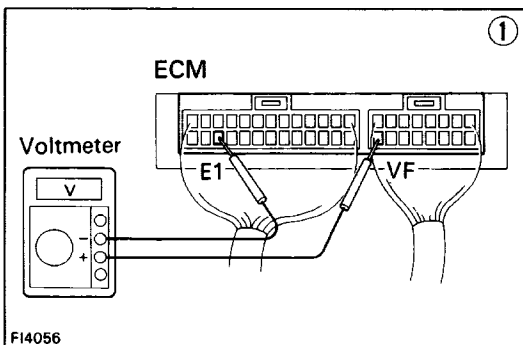
Check wiring between amplifier and ECM or compressor.

BAD

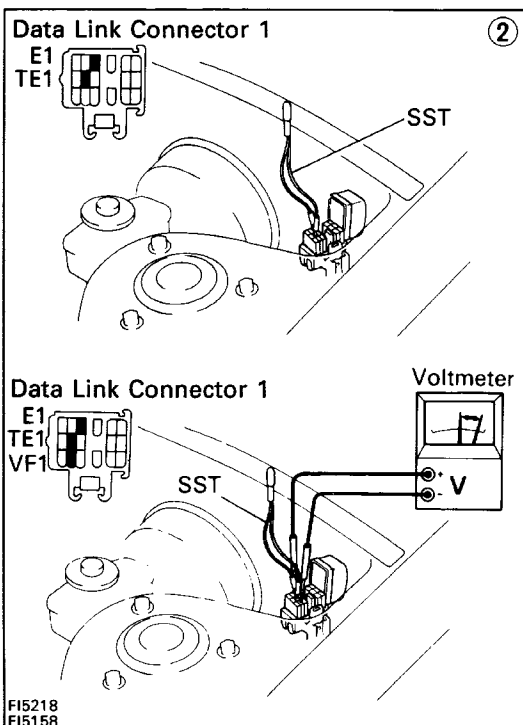
Repair or replace.



FI3840



FI4056

FI5218
FI5158

(1) There is no voltage between ECM terminals VF and E1.

Check that there is voltage between ECM terminal VF and body ground.

NO

OK

Check wiring between ECM terminal E1 and body ground.

OK

BAD

Try another— ECM.

Repair or replace.

Is air leaking into air induction system?

YES

Repair air leak.

NO

Check spark plugs. (See page IG-7)

BAD

Repair or replace.

OK

Check IIA and ignition system. (See page IG-4)

BAD

Repair or replace.

OK

Check fuel pressure. (See page FI-125)

BAD

Repair or replace.

OK

Check injectors. (See page FI-155)

BAD

Repair or replace.

OK

Check vacuum sensor. (See page FI-233)

BAD

Repair or replace.

OK

(2) Check operation of oxygen sensor. (See page FI-237)

OK

System Normal

BAD

Check wiring between oxygen sensor and ECM.

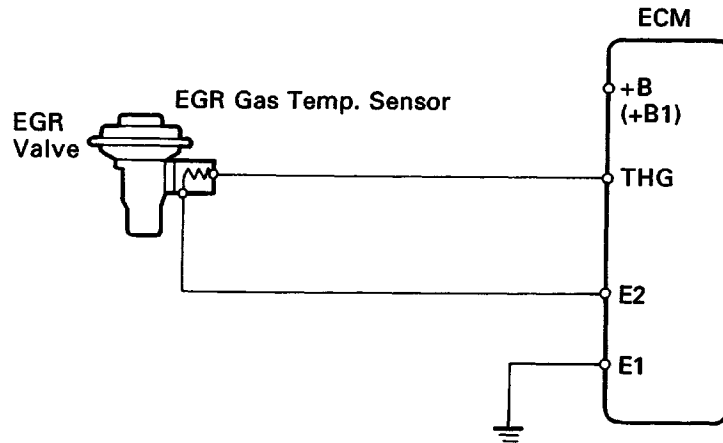
BAD

Repair wiring.

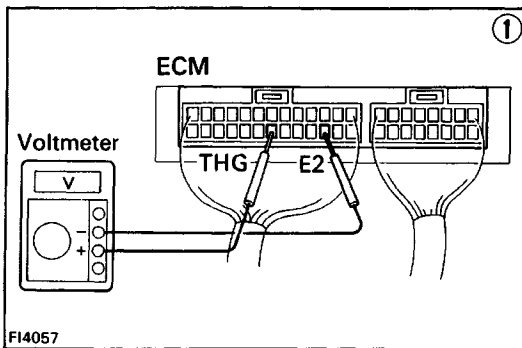
OK

Replace oxygen sensor.

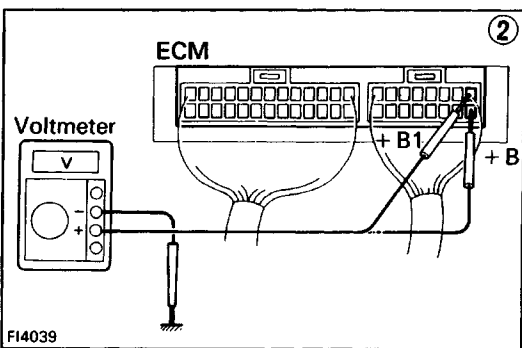
CALIF. only



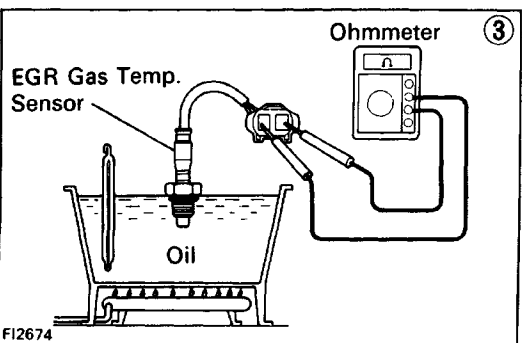
FI2680



FI4057



FI4039



FI2674

(1) There is no voltage between ECM terminals THG and E2.
(IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1
and body ground.

OK

NO

Refer to No.1.
(See page FI-61)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

Repair or replace.

Check EGR system.
(See page EC-8)

BAD

Repair or replace.

OK

(3) Check EGR gas temp.
sensor. (See page FI-236)

BAD

Replace EGR gas
temp. sensor.

OK

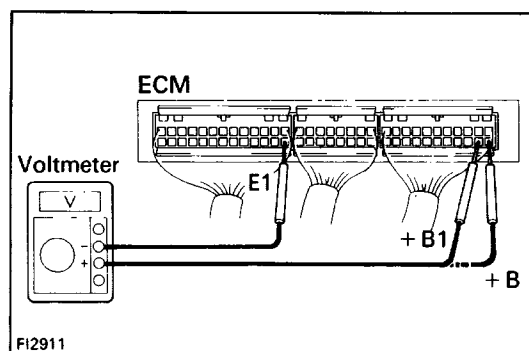
Check wiring between ECM and EGR
gas temp. sensor.

OK

Try another ECM.

BAD

Repair or replace.

**SFI SYSTEM CHECK PROCEDURE (3S-GTE)****HINT:**

- Perform all voltage measurements with the connectors disconnected.
 - Verify that the battery voltage is 11 V or more when the ignition switch is in "ON" position.
- Using a voltmeter with high impedance (10 kΩ/V minimum), measure the voltage at each terminal of the wiring connectors.

Terminals of ECM (3S-GTE)

Symbol	Terminal Name	Symbol	Terminal Name	Symbol	Terminal Name
E01	POWER GROUND	TPC	TURBOCHARGING PRESSURE VSV	AC	A/C MAGNET SWITCH
E ₂	POWER GROUND		—	ACT	A/C AMPLIFIER
No.1	INJECTOR (No.1)	T-VIS	T-VIS VSV	SPD	SPEED SENSOR
No.3	INJECTOR (No.3)	E1	ENGINE GROUND		—
No.2	INJECTOR (No.2)	VF	DATA LINK CONNECTOR 1		—
No.4	INJECTOR (No.4)				—
RSO	IAC VALVE			ATS	A/C AMPLIFIER
	—	TE1	DATA LINK CONNECTOR 1		—
RSC	IAC VALVE	OX	OXYGEN SENSOR	FPR	FUEL PUMP RELAY
	—	TE2	DATA LINK CONNECTOR 1		—
HT	OXYGEN SENSOR HEATER	PI M	TURBOCHARGING PRES-SURE SENSOR	W	MALFUNCTION INDICATOR LAMP
	—	KNK	KNOCK SENSOR		—
STJ	COLD START INJECTOR	THW	ENGINE COOLANT TEMP. SENSOR	STP	STOP LIGHT SWITCH
IGT	IGNITER	IDL	THROTTLE POSITION SENSOR		—
EGR	EGR VSV	THA	AIR TEMP. SENSOR		—
	—	VTA	THROTTLE POSITION SENSOR	FC	CIRCUIT OPENING RELAY
G2	DISTRIBUTOR	VS	VOLUME AIR FLOW METER	E LS	HEADLIGHT DEFOGGER
G1	DISTRIBUTOR	*THG	EGR GAS TEMP. SENSOR	+B I	SFI MAIN RELAY
NE	DISTRIBUTOR	VC	SENSOR POWER SOURCE	BATT	BATTERY
G ⊖	DISTRIBUTOR	E2	SENSOR GROUND	+B	SFI MAIN RELAY
IGF	IGNITER	STA	STARTER SWITCH	*CALIF. only	
	—		—		

ECM Terminals

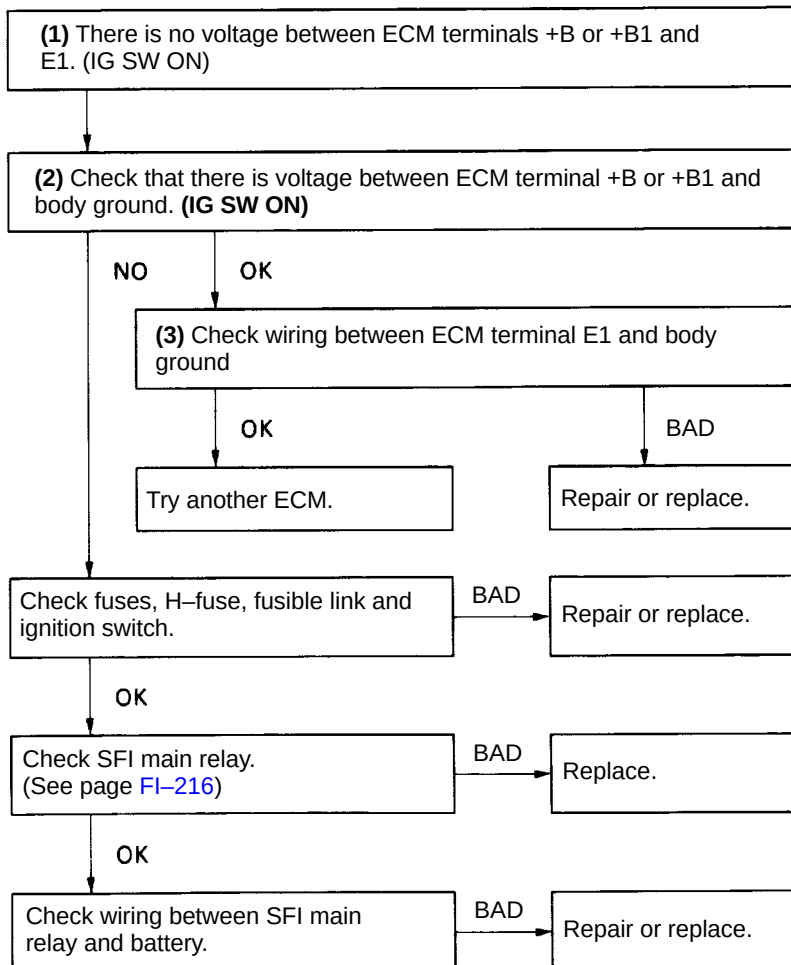
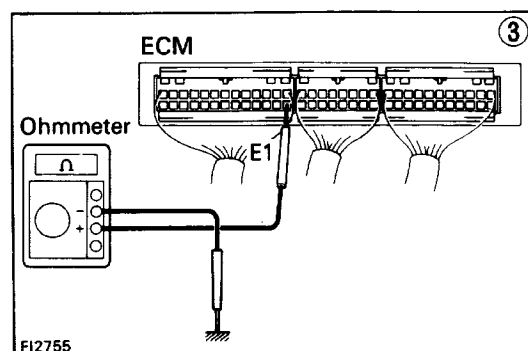
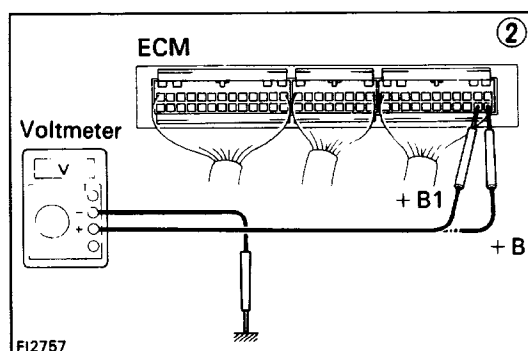
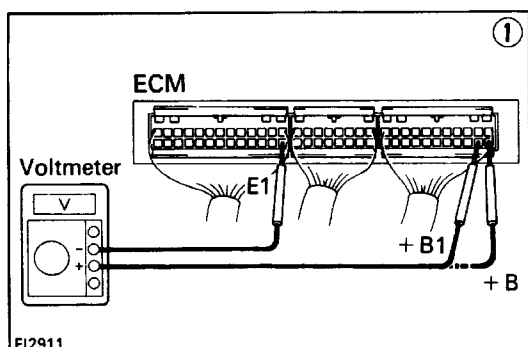
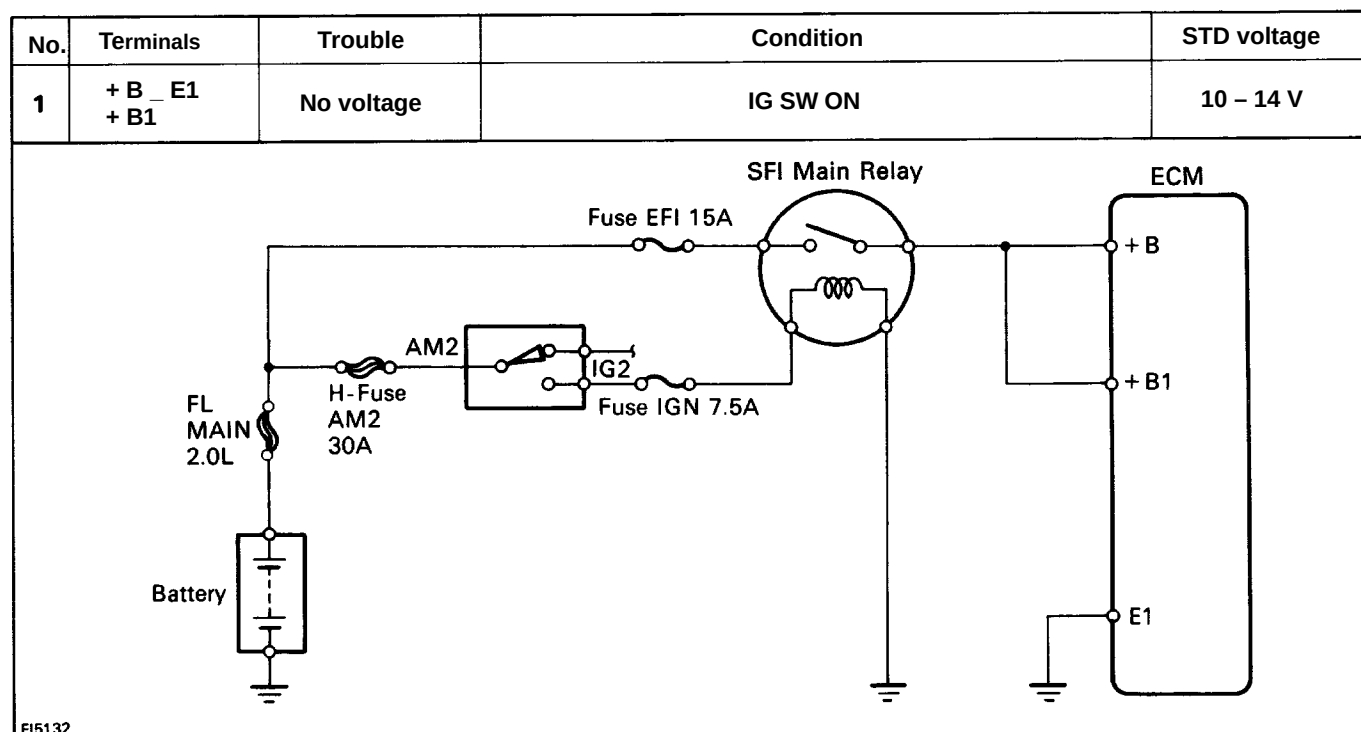
E01	No.1	No.2	RSO	RSC	HT	STJ	EGR	G2	NE	IGF	TPC	TVIS	VF	OX	PI M	THW	THA	VS	VC	STA	AC	SPD	ATS	FPR	W	STP	ELS	BATT
E02	No.3	No.4				IGT		G1	G ⊖		E1		TE1	TE2	KNK	IDL	VTA	THG	E2		ACT					FC	+B1	+B

Voltage at ECM Wiring Connectors (3S-GTE)

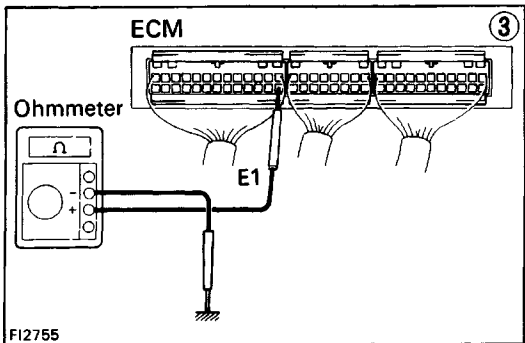
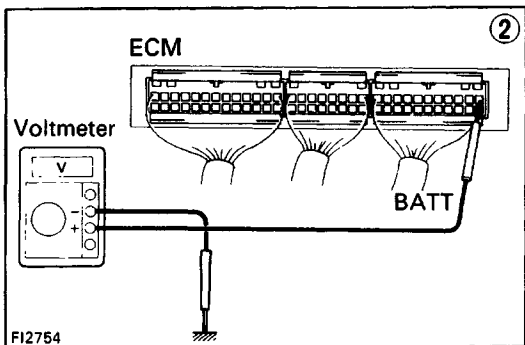
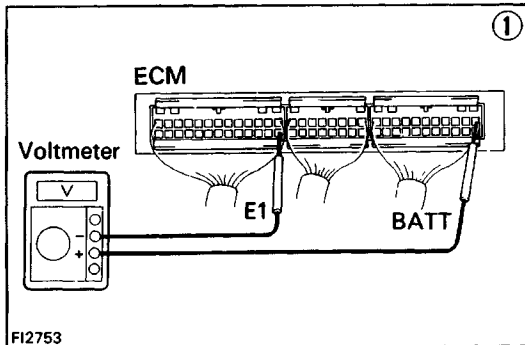
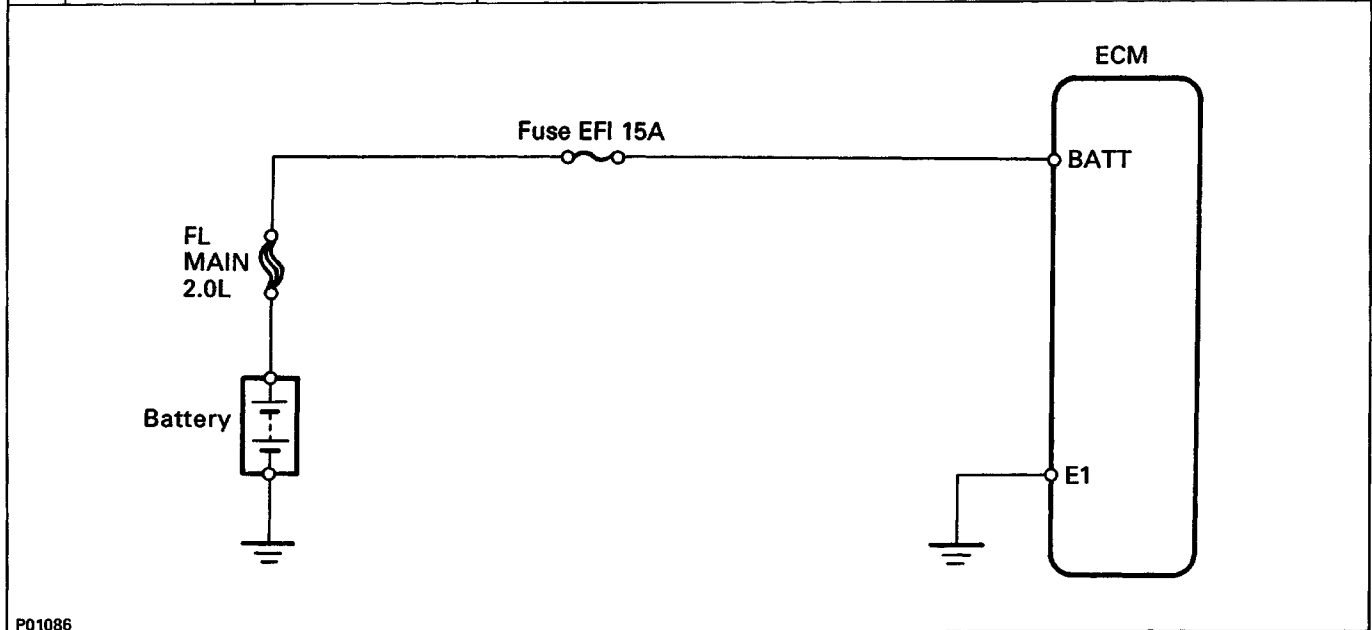
No.	Terminals	Condition		STD voltage (V)	See page
1	+ B – E1 + B1	IG SW ON		10–14	FI-7s
2	BATT – E1	–		10–14	FI-77
3	I D L – E2	IG SW ON	Throttle valve open	4.5–5.5	FI-7 s
	VC – E2		–	4.5–5.5	
	VTA – E 2		Throttle valve fully closed (Throttle opener must be cancelled first)	0.1 –1.0	
			Throttle valve fully open	3.2–4.2	
4	VC – E2	IG SW ON	–	4.5–5.5	FI-80
	VS – E2		Measuring plate fully closed	3.7–4.3	
			Measuring plate fully open	0.2–0.5	
		Idling	1.6–4.1		
	3,000 rpm	1.0–2.0			
	5	No:1 No.2 E01 No.3 E02 No.4	IG SW ON		10–14
6	THA – E2	IG SW ON	Intake air temp. 20°C (68°F)	1 –3	FI-82
7	TH W – E2		Engine coolant temp. 80°C (176°F)	0.1 –1.1	FI-83
8	STA – E 1	Cranking		6–14	FI-84
9	IGT – E1	Cranking or idling		0.8–1.2	FI-85
10	RSC – E1 RSO	IG SW ON	ECM connectors disconnected	8–14	FI-85
11	W –EI	No trouble (malfunction indicator lamp off) and engine running		10–14	F1-87
12	PIM – E2	IG SW ON		2.5–4.5	FI-88
	VC – E2			4.5–5.5	
13	AC – E 1	IG SW ON	Air conditioning ON	8–14	FI-89

ECM Terminals

E01	No.1	No.2	RSO	RSC	HT	STJ	EGR	G2	NE	IGF	TPC	TVIS	VF	OX	PIM	THW	THA	VS	VC	STA	AC	SPD	ATS	FPR	W	STP	ELS	BATT	
E02	No.3	No.4				IGT		G1	G⊖			E1	TE1	TE2	KNK	IDL	VTA	THG	E2		ACT						FC	+B1	+B



No.	Terminals	Trouble	Condition	STD voltage
2	BATT – E1	No voltage	–	10 – 14 V



(1) There is no voltage between ECM terminals BATT and E1.

(2) Check that there is voltage between ECM terminal BATT and body ground.

NO OK

(3) Check wiring between ECM terminal E1 and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Check fuse and fusible link.

BAD → Replace.

OK

Check wiring between ECM terminal and battery.

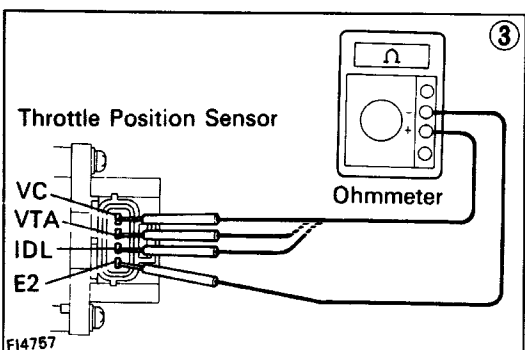
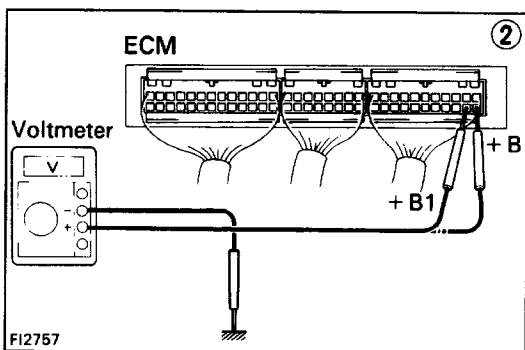
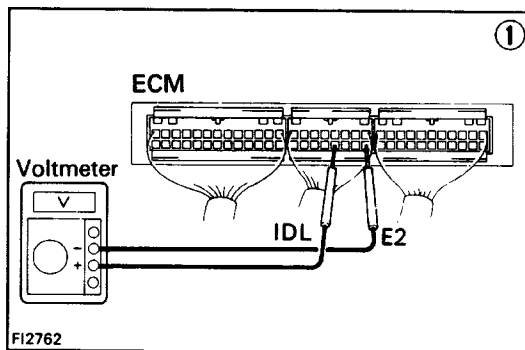
BAD → Repair or replace.

No.	Terminals	Trouble	Condition		STD voltage
3	IDL – E2	No voltage	IG SW ON	Throttle valve open	4.5 – 5.5 V
	VC – E2				4.5 – 5.5 V
	VTA – E2			Throttle valve fully closed (Throttle opener must be cancelled first)	0.1 – 1.0 V
				Throttle valve fully open	3.2 – 4.2 V

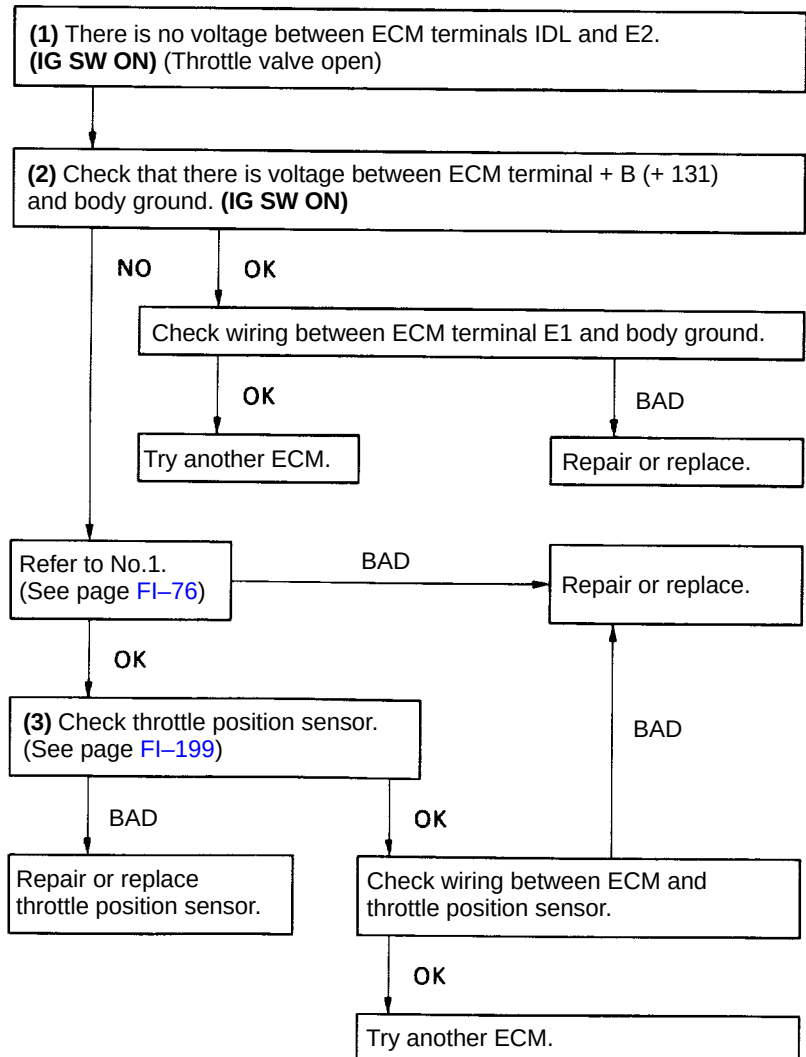
Throttle Position Sensor

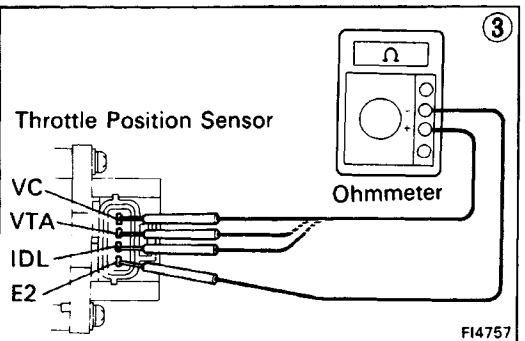
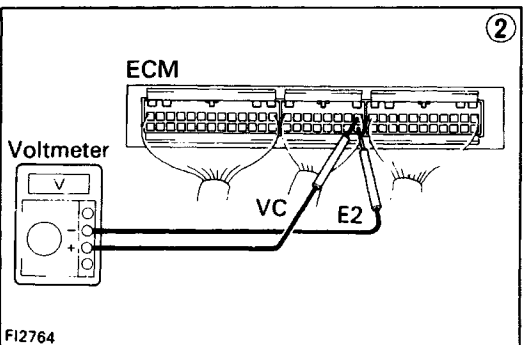
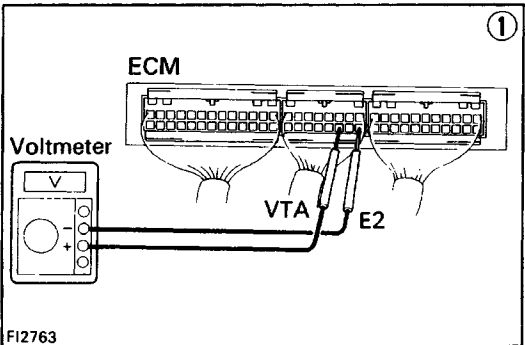
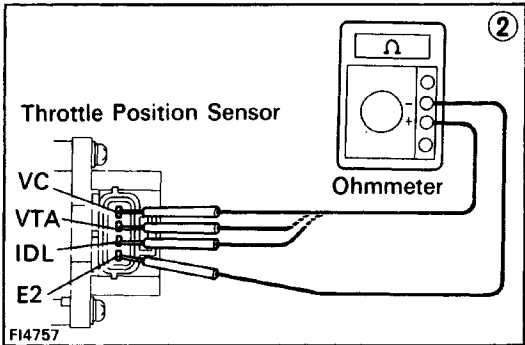
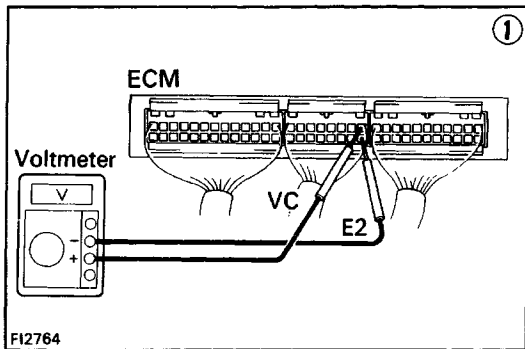
ECM

P01419

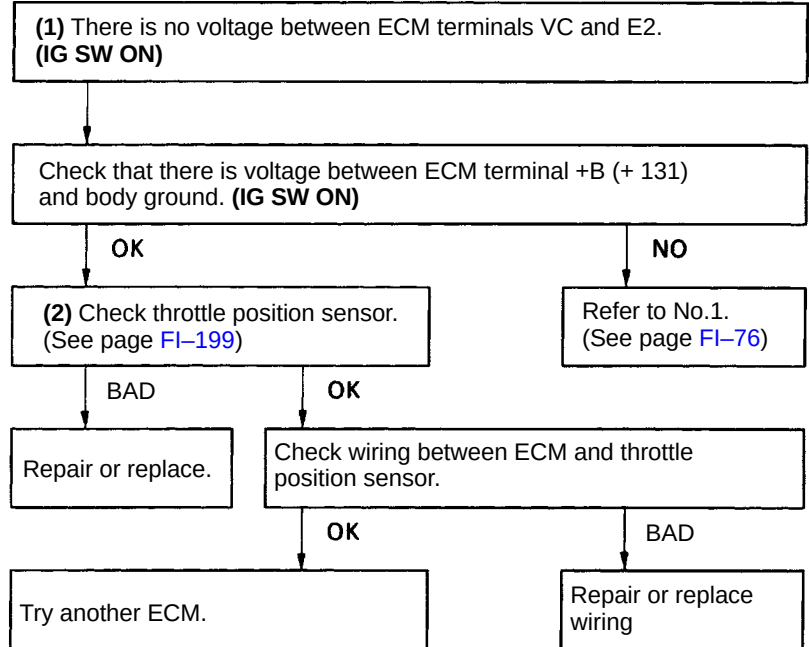


• IDL – E2

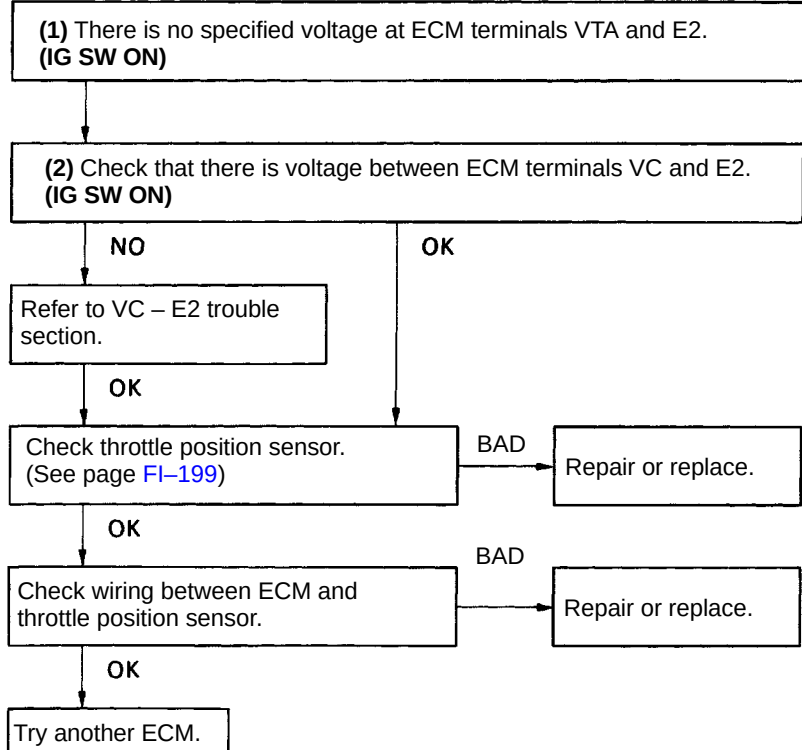




• VC – E2



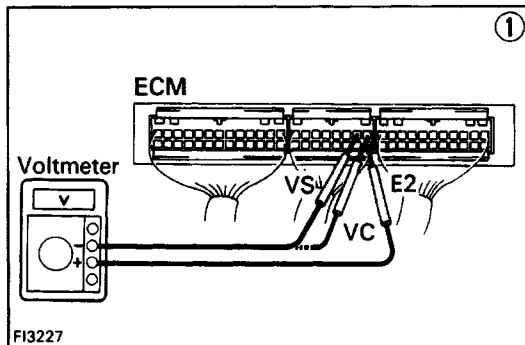
• VTA – E2



No.	Terminals	Trouble	Condition		STD voltage
4	VC – E2	No voltage	IG SW ON	–	4.5 – 5.5 V
	VS – E2			Measuring plate fully closed	3.7 – 4.3 V
				Measuring plate fully open	0.2 – 0.5 V
	Idle		1.6 – 4.1 V		
	3,000 rpm		1.0 – 2.0 V		

The diagram shows a Volume Air Flow Meter connected to an ECM. The meter has four terminals: E2, VS, VC, and a common ground. The ECM has five terminals: +B (+B1), E2, VS, VC, and E1. Wires connect E2 to +B (+B1), VS to VS, VC to VC, and E1 to ground.

FI6032



(1) There is no specified voltage at ECM terminals VC or VS and E2. (IG SW ON)

(2) Check that there is voltage between ECM terminal +B (+B1) and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-76)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

(3) Check volume air flow meter.
(See page FI-183)

BAD

OK

Repair or replace
volume air flow
meter

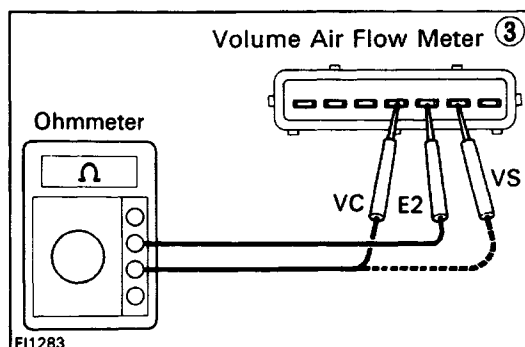
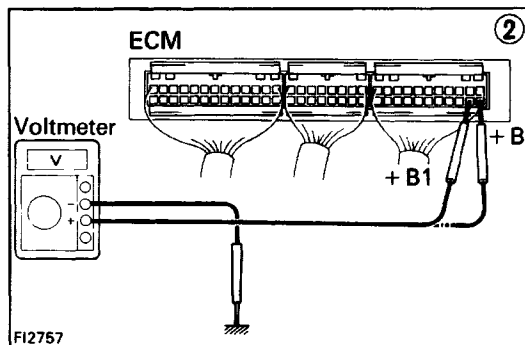
Check wiring between ECM and
volume air flow meter.

OK

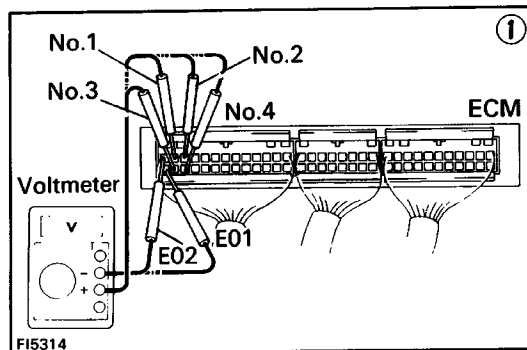
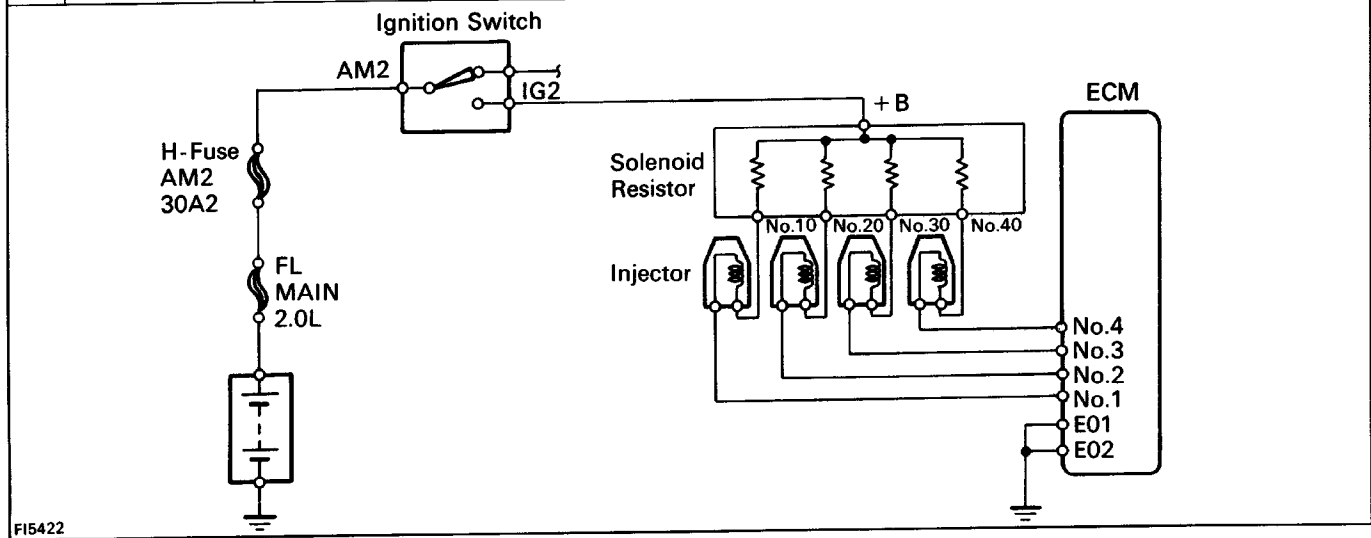
BAD

Try another ECM.

Repair or replace.



No.	Terminals	Trouble	Condition	STD voltage
5	No. 1 No. 2 – E01 No. 3 – E02 No. 4	No voltage	IG SW ON	10–14V



(1) There is no voltage between ECM terminals No.1, No-2, No.3 and/ or No.4 and E01 and/ or E02. (IG SW ON)

Check that there is specified voltage between solenoid resistor terminal +B and body ground. **STD voltage: 10 – 14V**

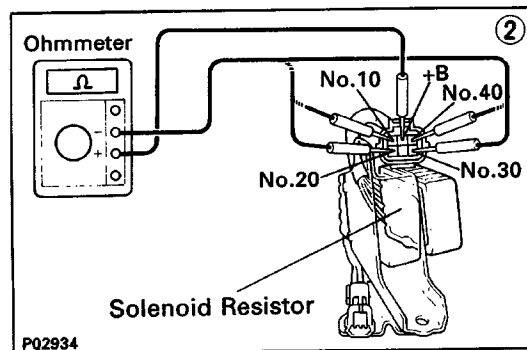
OK

NO

Check H-fuse, fusible link, wiring and ignition switch.

BAD

Repair or replace.



(2) Check resistance between +B of solenoid resistor and each other terminal. **STD resistance: 4 – 6 Ω**

OK

NO

Replace resistor.

(3) Check resistance of each injector. **STD resistance: 2 – 4 Ω**

OK

BAD

Replace injector.

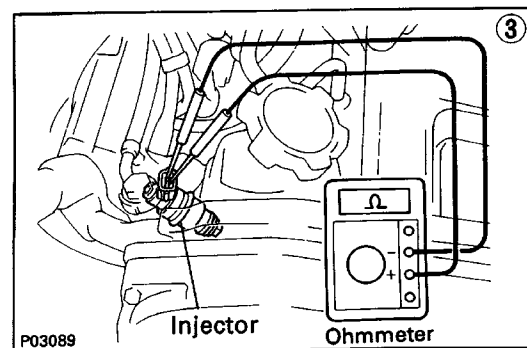
Check wiring between ECM and resistor

BAD

Repair or replace wiring

OK

Try another ECM.

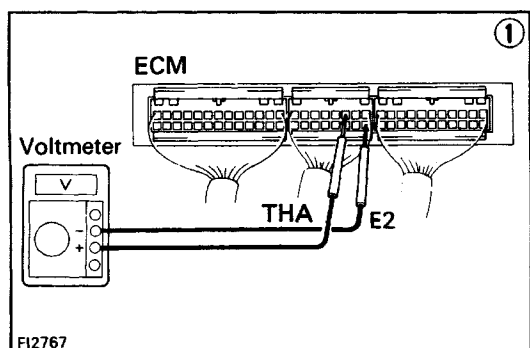


No.	Terminals	Trouble	Condition		STD voltage
6	THA – E2	No voltage	IG SW ON	Intake air temperature 20°C (68°F)	1 – 3 V

Intake Air Temp. Sensor
(Volume Air Flow Meter)

ECM
+B (+B1)
E2
THA
E1

FI6030



(1) There is no voltage between ECM terminals THA and E2.
(IG SW ON)

(2) Check that there is voltage between ECM terminal +B (+B1)
and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-76)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

(3) Check intake air temp. sensor
(See page FI-183)

Repair or replace.

6AD

OK

Replace volume
air flow meter—.

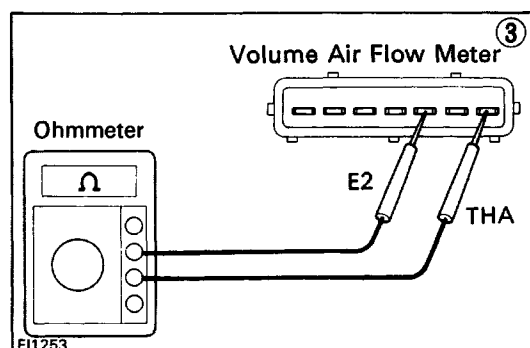
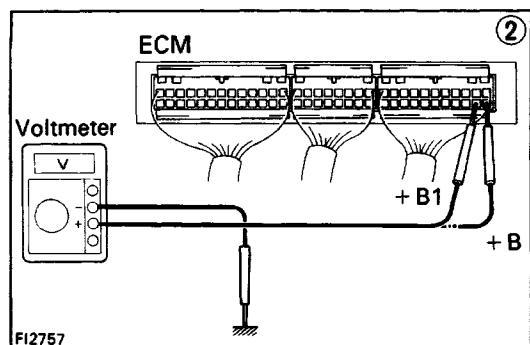
Check wiring between ECM and
intake air temp. sensor.

OK

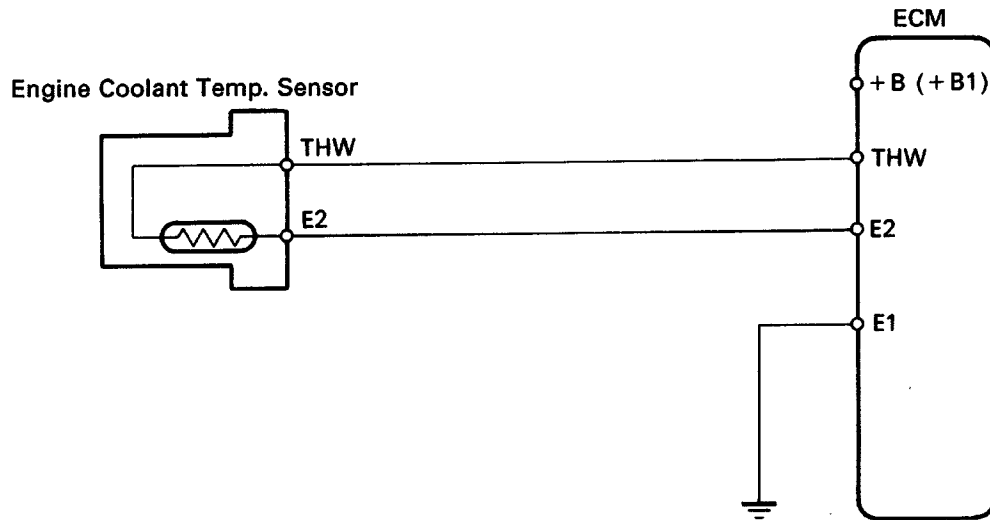
BAD

Try another ECM.

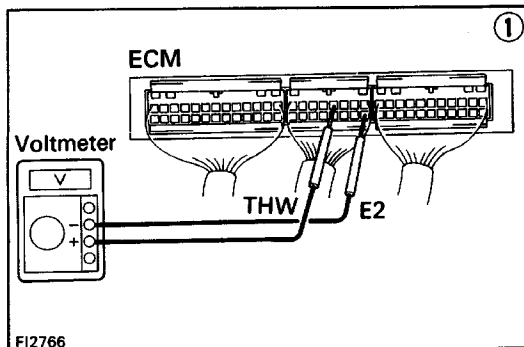
Repair or replace.



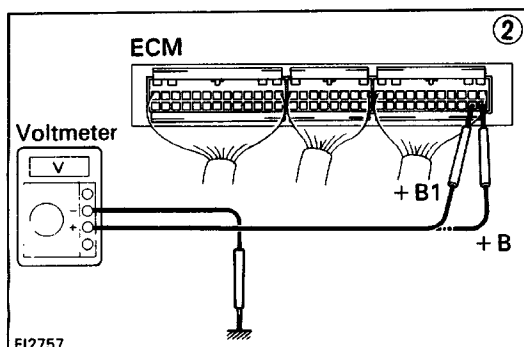
No.	Terminals	Trouble	Condition		STD voltage
7	THW – E2	No voltage	IG SW ON	Engine coolant temperature 80°C (176°F)	0.1 – 1.1 V



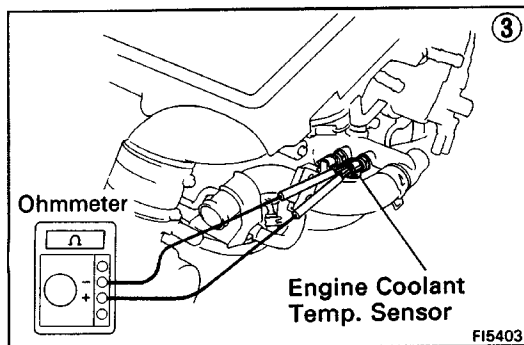
FI3572



FI2766



FI2757



FI5403

(1) There is no voltage between ECM terminals THW and E2 (IG SW ON)

(2) Check that there is voltage between ECM terminal + B 1+ B1 and body ground.(IG SW ON)

OK

NO

Refer to No1.
(See page FI-76)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

Check engine coolant temp. sensor. (See page FI-231)

Repair or replace.

6AD

OK

Replace engine coolant temp. sensor.

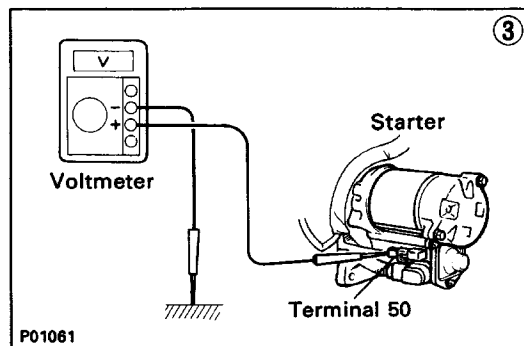
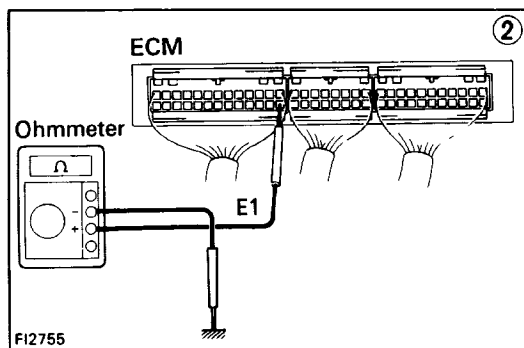
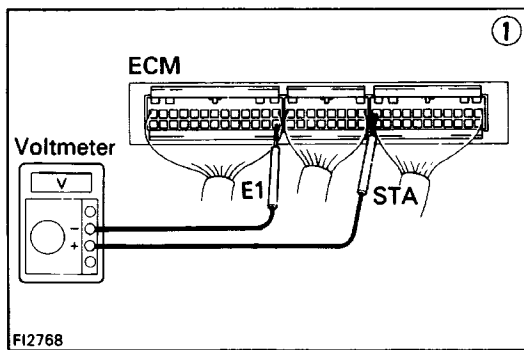
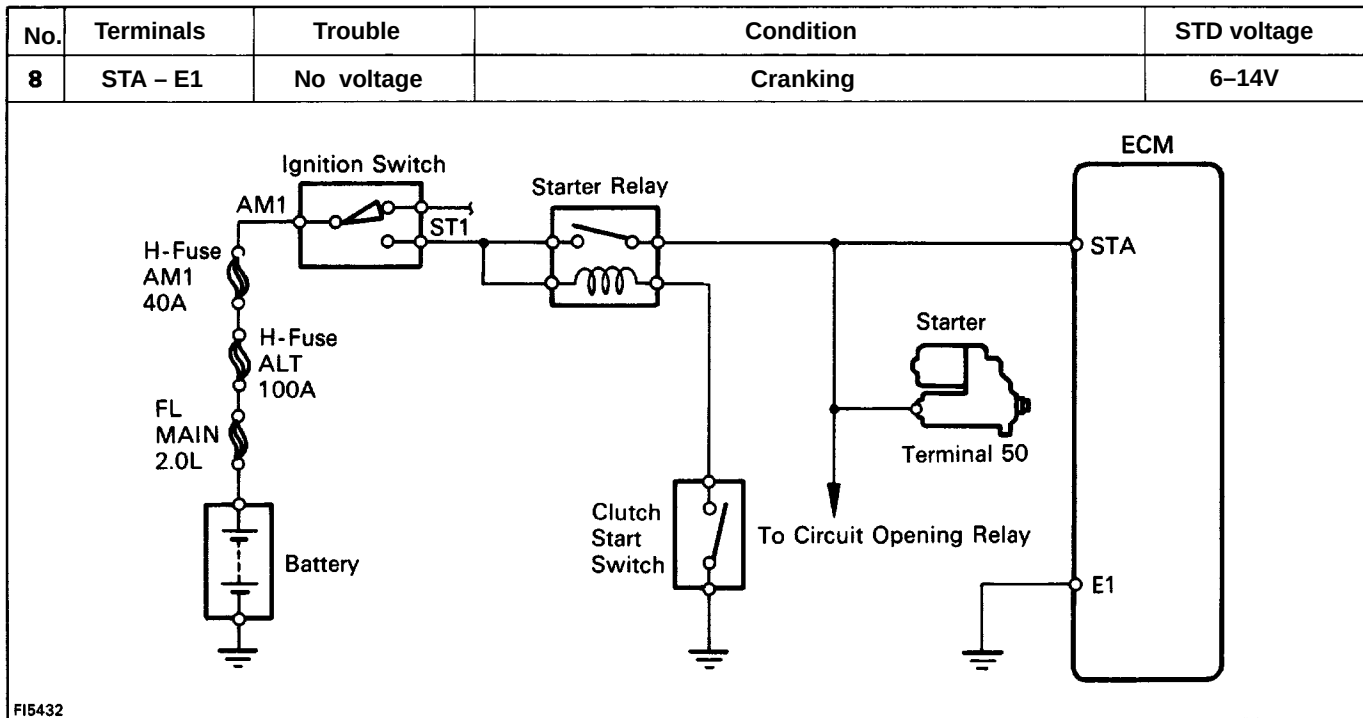
Check wiring between ECM and engine coolant temp. sensor.

OK

BAD

Try another ECM.

Repair or replace.



(1) There is no voltage between ECM terminals STA and E1.
(IG SW ON)

Check starter
operation.

OK

Check wiring between ECM terminal
STA and ignition switch terminal ST1.

BAD

OK

BAD

Repair or replace.

(2) Check wiring between ECM terminal E1 and body
ground.

OK

BAD

Try another ECM.

Repair or replace.

Check H-fuses, fusible link, battery,
wiring, ignition switch, clutch start
switch and starter relay.

BAD

Repair or replace.

OK

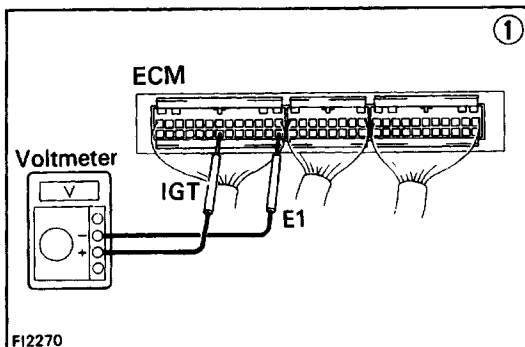
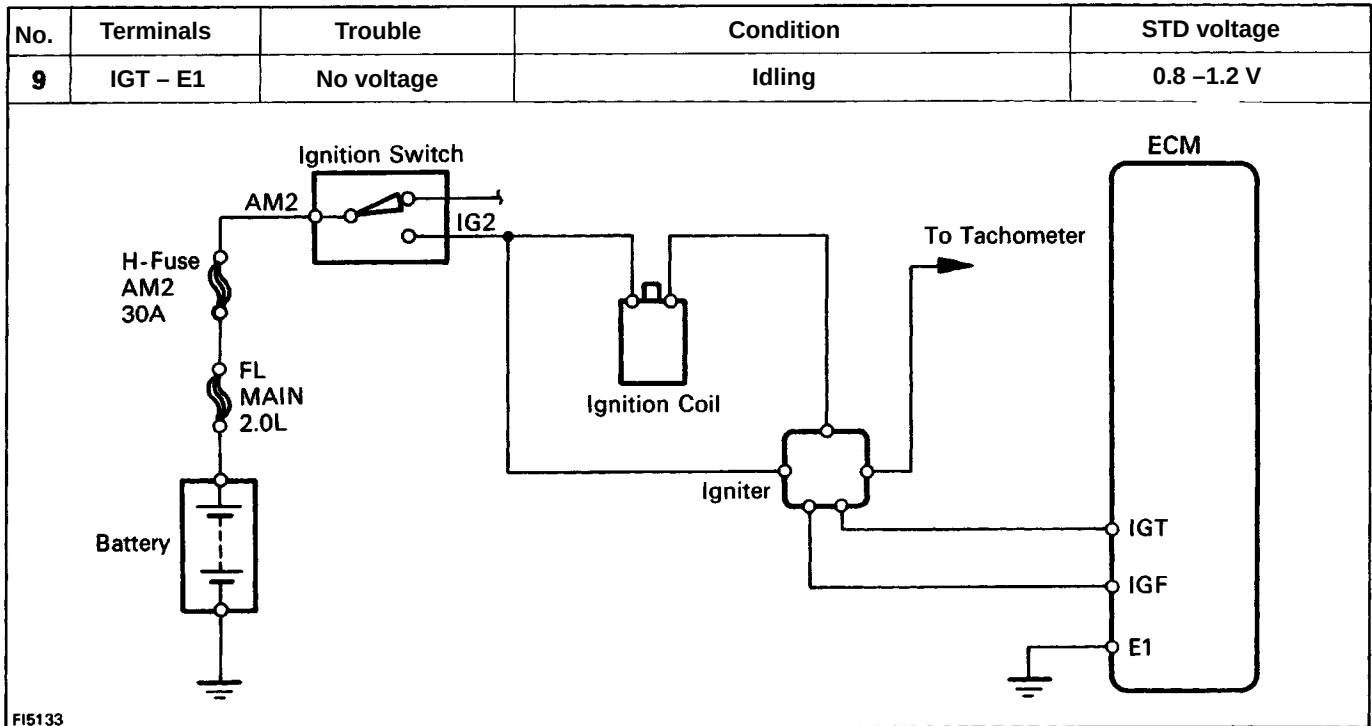
(3) Check that there is voltage at starter terminal 50.
(IG SW START) STD voltage: 6 – 14 V

OK

Check starter.

NO

Check wiring between ignition switch
terminal ST1 and starter terminal 50.



(1) There is no voltage between ECM terminals IGT and E1. (idling)

(2) Check that there is voltage between ECM terminal IGT and body ground. (idling)

NO OK

(3) Check wiring between ECM terminal E1 and body ground.

BAD → Repair or Replace.

OK

Try another ECM

Check H-fuse, fusible link and ignition switch.

BAD → Repair or replace.

OK

Check distributor. (See page IG-14)

BAD → Repair or replace.

OK

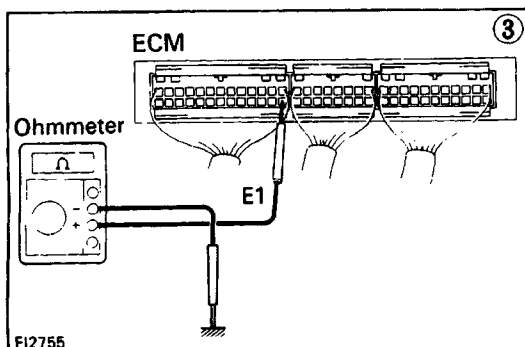
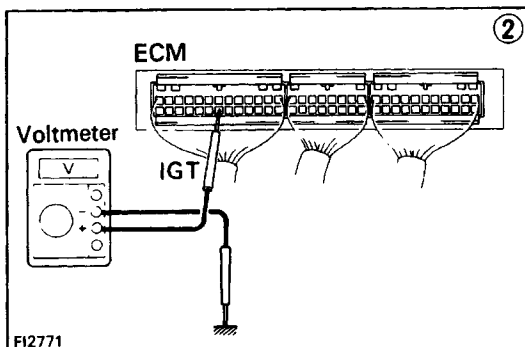
Check wiring between ECM and battery.

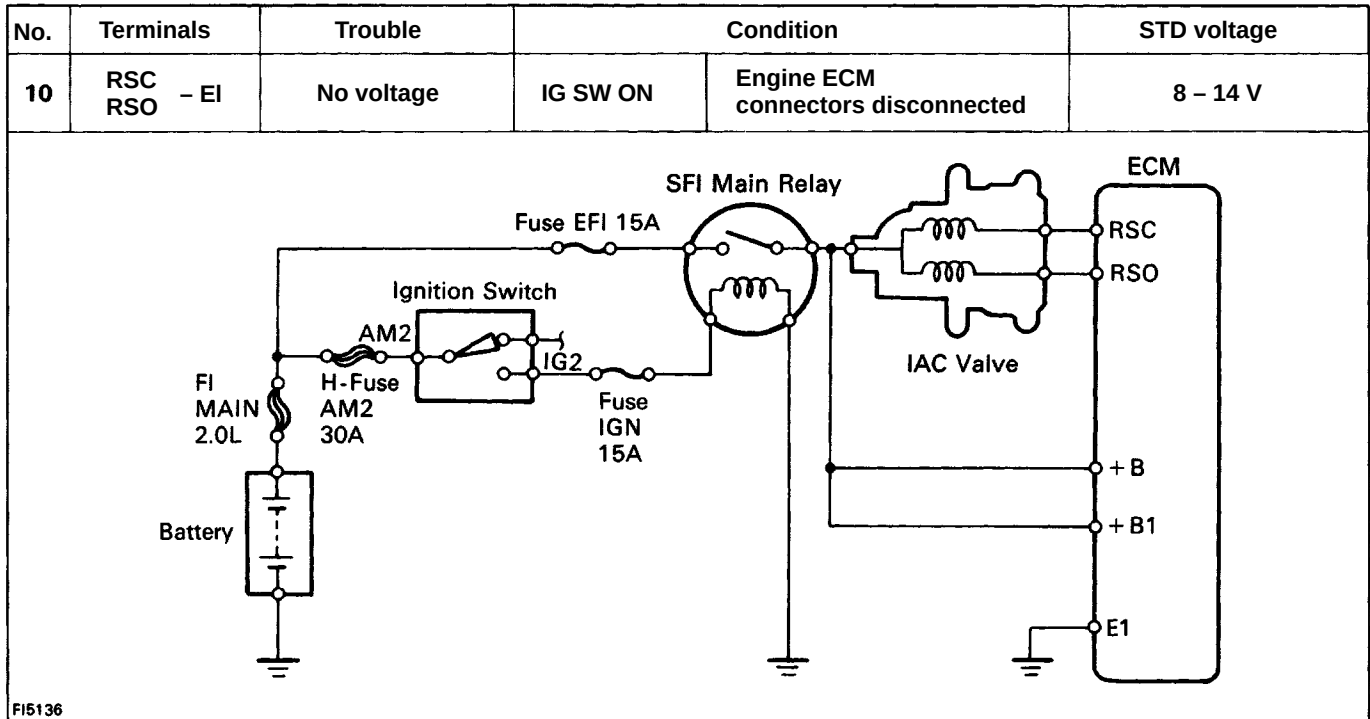
BAD → Repair or replace.

OK

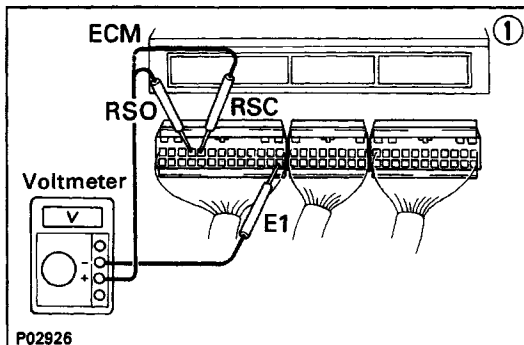
Check igniter. (See page IG-14)

BAD → Repair or replace.

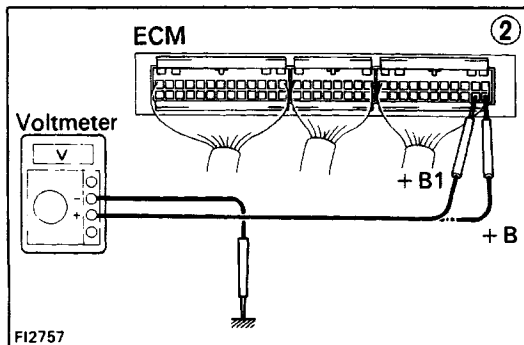




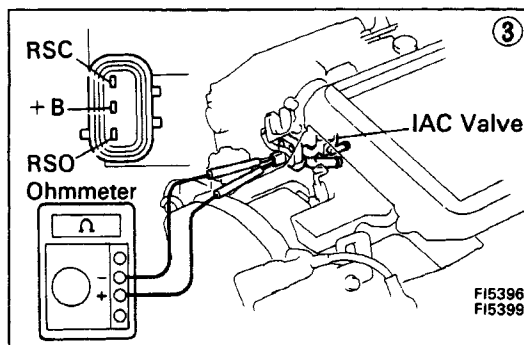
FI5136



P02926



FI2757

FI5396
FI5399

(1) There is no voltage between ECM terminals RSC or RSO and E1. (IG SW ON)

(2) Check that there is voltage between ECM terminal + B (+B1) and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-76)

(3) Check resistance between IAC valve terminals + B and RSC or RSO.
STD resistance: 19.3 - 22.3Ω

BAD

Replace IAC valve.

OK

Check wiring between ECM and IAC valve.

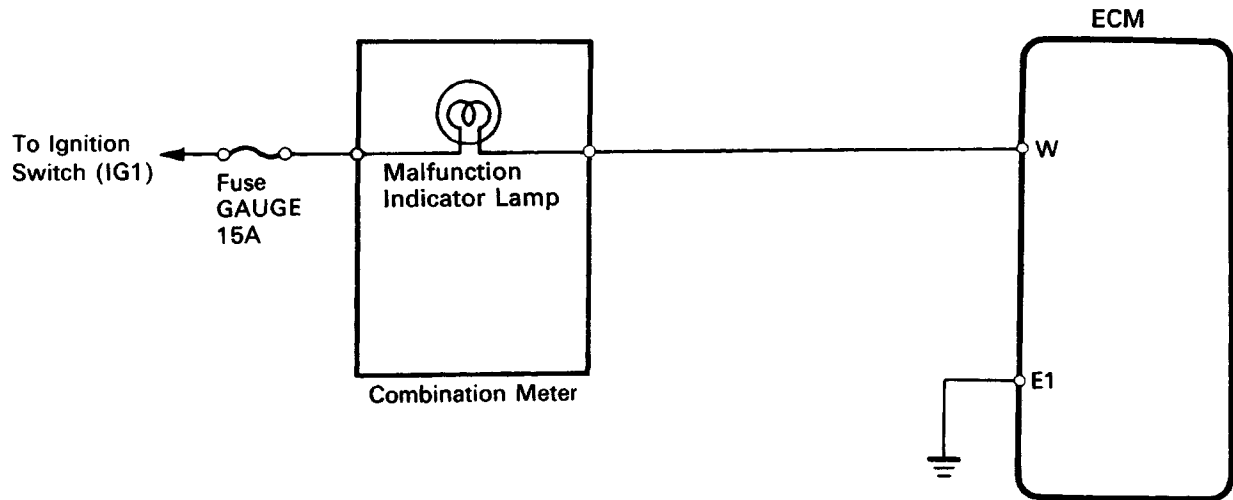
BAD

Repair or replace wiring.

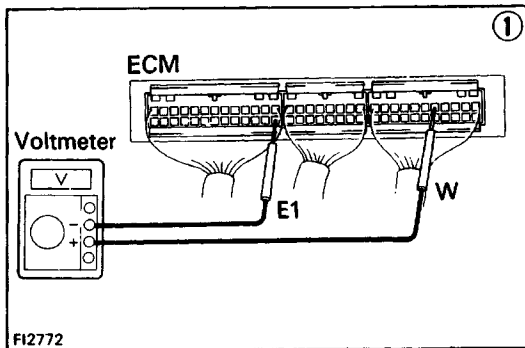
OK

Try another ECM.

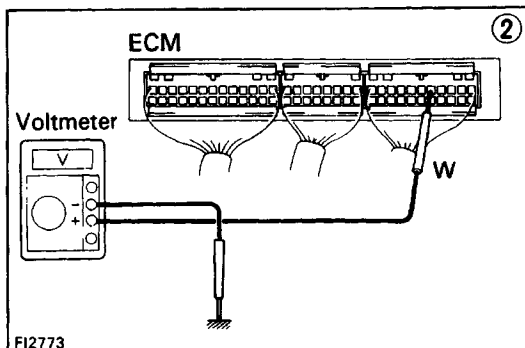
No.	Terminals	Trouble	Condition	STD voltage
11	W – E1	No voltage	No trouble (malfunction indicator lamp off) and engine running.	10–14V



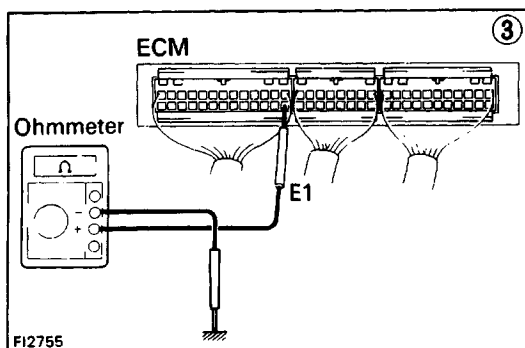
FI0728



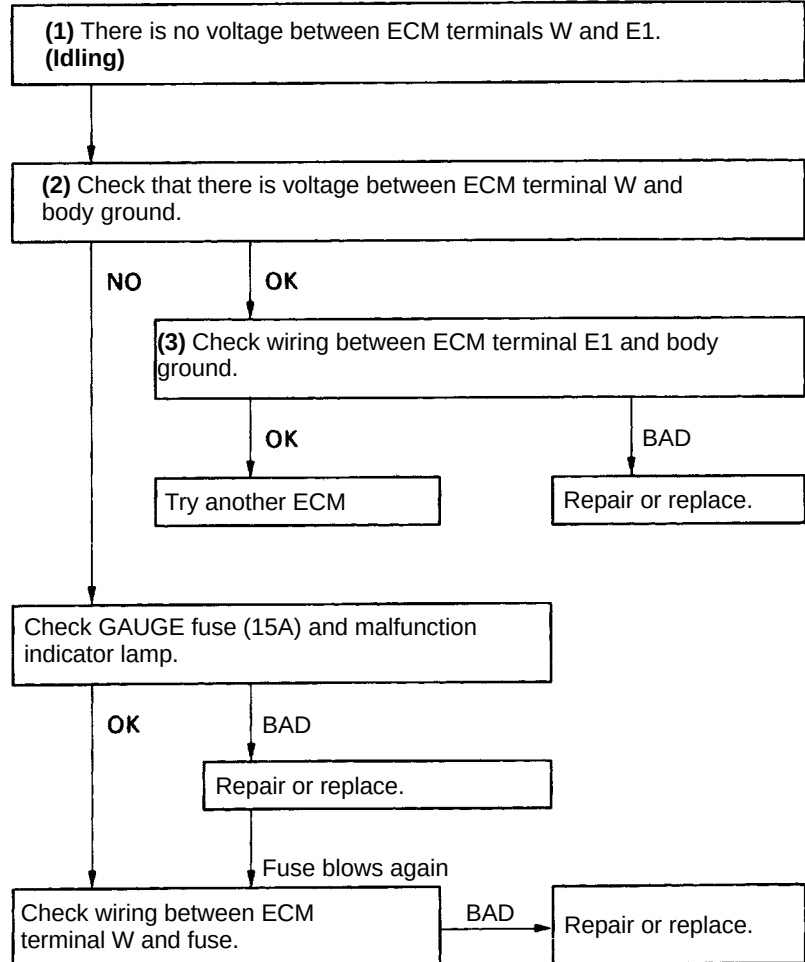
FI2772



FI2773



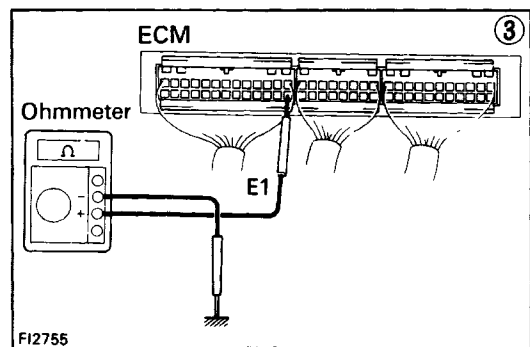
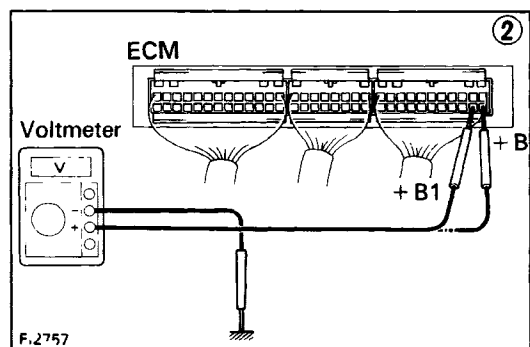
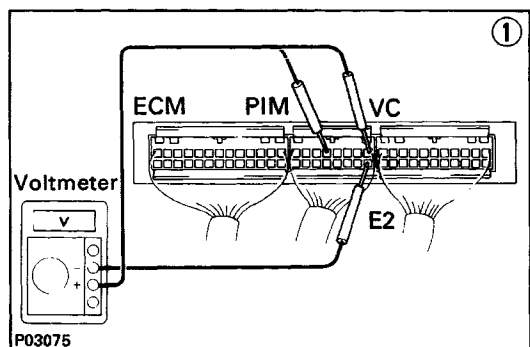
FI2755



No.	Terminals	Trouble	Condition	STD voltage
12	PIM-E2	No voltage	IG SW ON	2.5 – 4.5 V
	VC – E2			4.5 – 5.5 V

The diagram shows a Turbocharging Pressure Sensor connected to an ECM. The sensor has three terminals: E2, PIM, and VC. The ECM has four terminals: E2, PIM, VC, and E1. E1 is connected to ground. The sensor is connected to the ECM as follows: E2 to E2, PIM to PIM, and VC to VC.

FI1226



• PIM-E2, VC-E2

(1) There is no voltage between ECM terminals PIM or VC and E2. (IG SW ON)

(2) Check that there is voltage between ECM terminal + B or + 131 and body ground. (IG SW ON)

OK

NO

Refer to No-1 .
(See page FI-76)

(3) Check wiring between ECM terminal E1 and body ground.

OK

BAD

Check turbocharging pressure sensor. (See page FI-235)

Repair or replace.

BAD

OK

Replace turbocharging pressure sensor.

Check wiring between ECM and turbocharging pressure sensor.

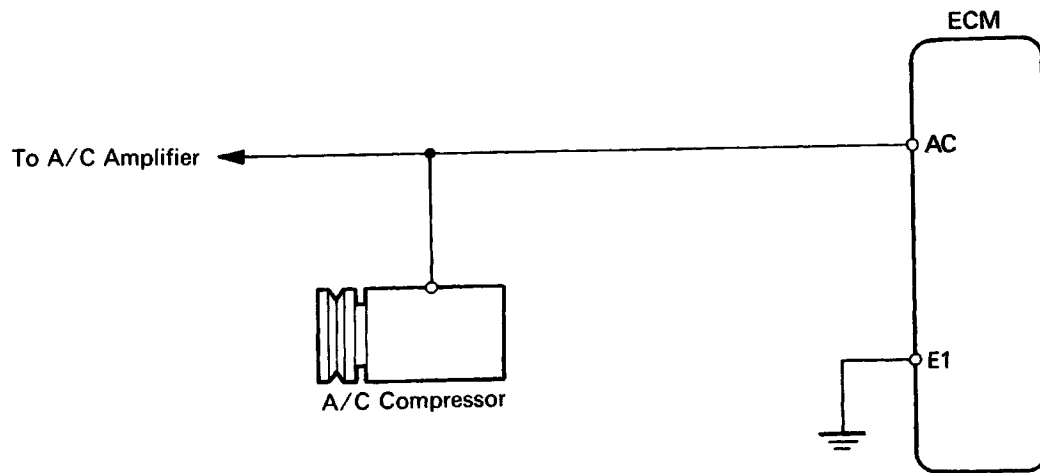
OK

BAD

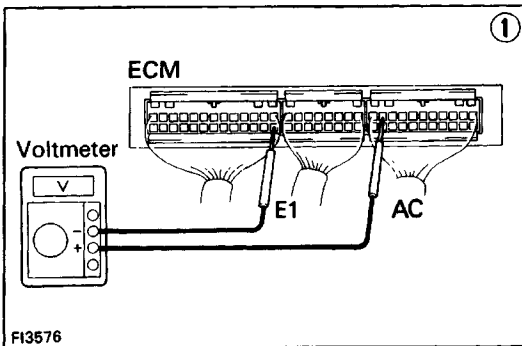
Try another ECM.

Repair or replace.

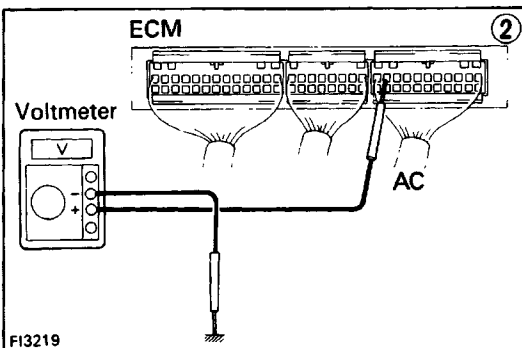
No.	Terminal	Trouble	Condition	STD voltage
13	AC – E1	No voltage	Air conditioning ON	8 – 14V



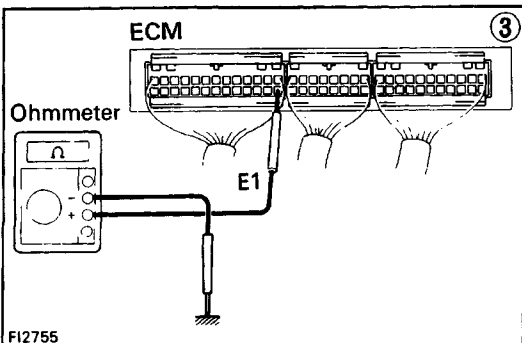
FI0922



FI3576



FI3219



FI2755

(1) There is no voltage between ECM terminals AC and E1.
(Air conditioning ON)

(2) Check that there is voltage between ECM terminal AC and body ground.

NO

OK

(3) Check wiring between ECM terminal E1 and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Check compressor running.

OK

BAD

Check wiring between ECM terminal AC and amplifier.

BAD

Repair or replace.

Check that there is voltage between amplifier terminal and body ground.

BAD

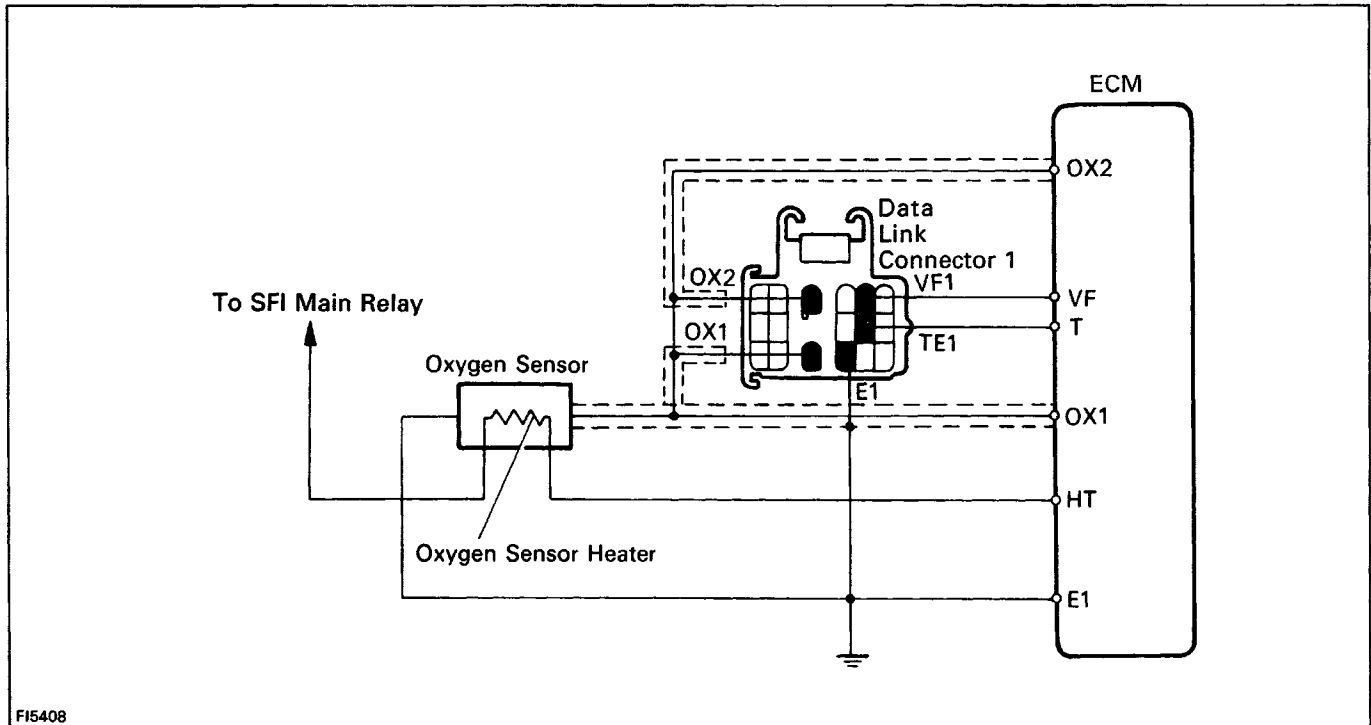
Repair or replace.

OK

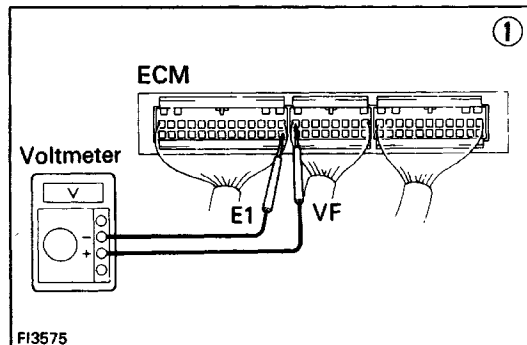
Check wiring between amplifier and ECM or compressor.

BAD

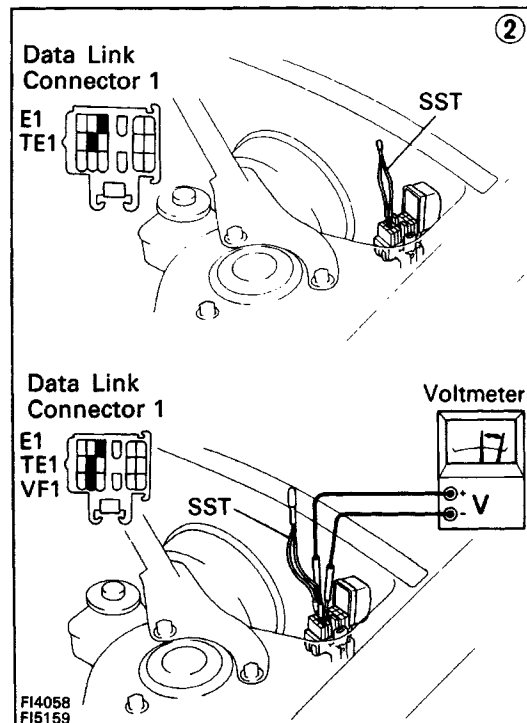
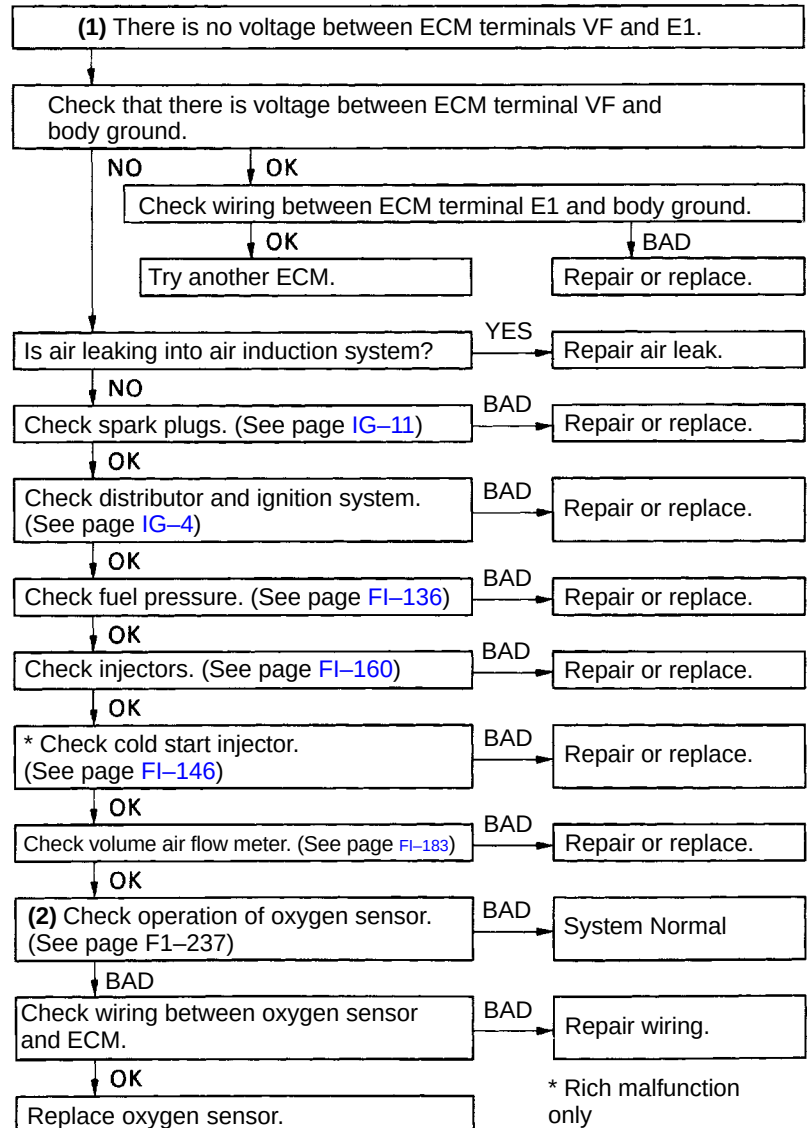
Repair or replace.



FI5408

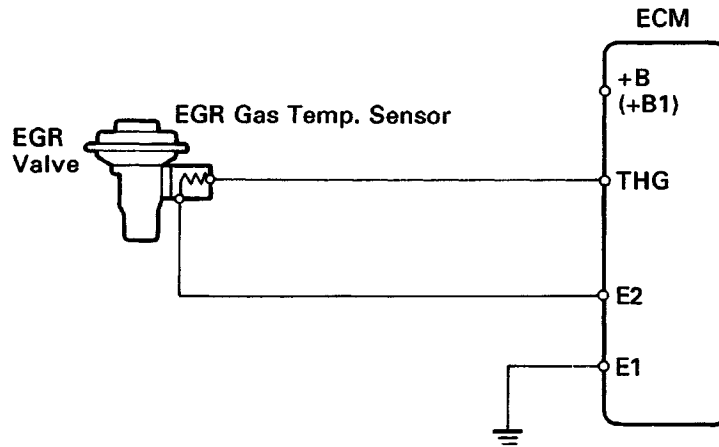


FI3575

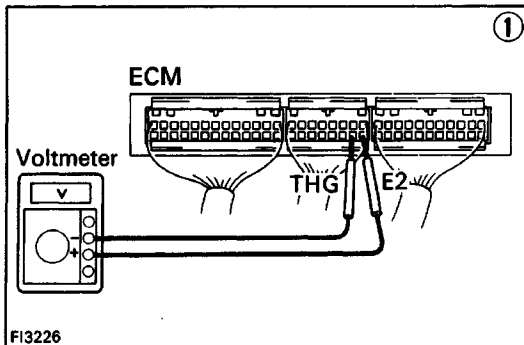
FI4058
FI5159

* Rich malfunction only

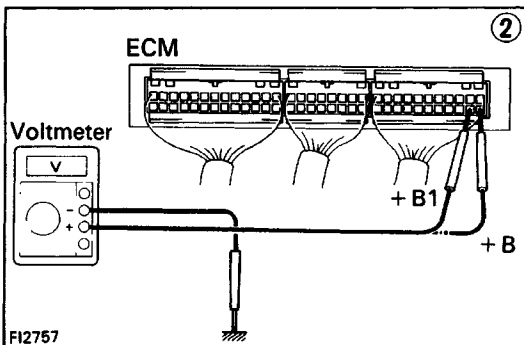
CALIF. only



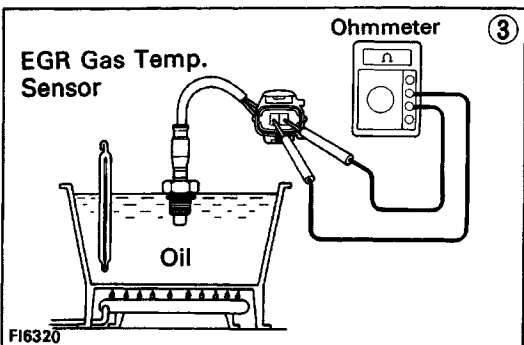
FI2680



FI3226



FI2757



FI6320

(1) There is no voltage between ECM terminals THG and E2.
(IG SW ON)

(2) Check that there is voltage between ECM terminal + B (+B1)
and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-76)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

Repair or replace.

Check EGR system.
(See page EC-21)

BAD

Repair or replace.

OK

(3) Check EGR gas temp.
sensor. (See page FI-236)

BAD

Replace EGR gas temp.
sensor.

OK

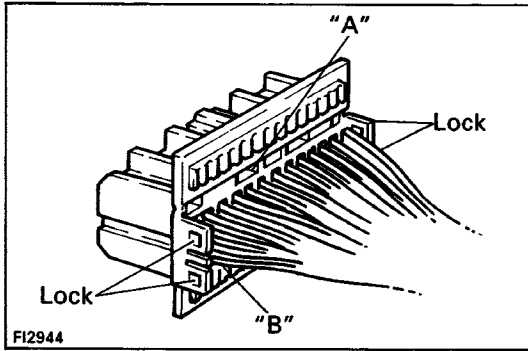
Check wiring between ECM and EGR
gas temp. sensor.

OK

Try another ECM.

BAD

Repair or replace.



MFI SYSTEMS CHECK PROCEDURE

(5S-FE MIT)

PREPARATION

- Disconnect the connectors from the ECM.
- Remove the locks as shown in the illustration so that the tester probe(s) can easily come in.

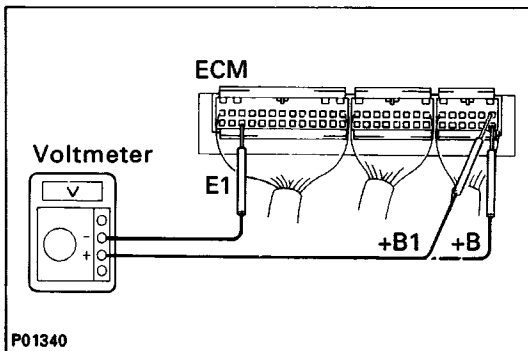
NOTICE: Pay attention to sections "A" and "B" in the illustration which can be easily broken.

- Reconnect the connectors to the ECM.

HINT:

- Perform all voltage measurements with the connectors disconnected.
- Verify that the battery voltage is 11 V or more when the ignition switch is in the "ON" position.

Using a voltmeter with high impedance (10 kΩ/V minimum), measure the voltage at each at each terminal of the wiring connectors.

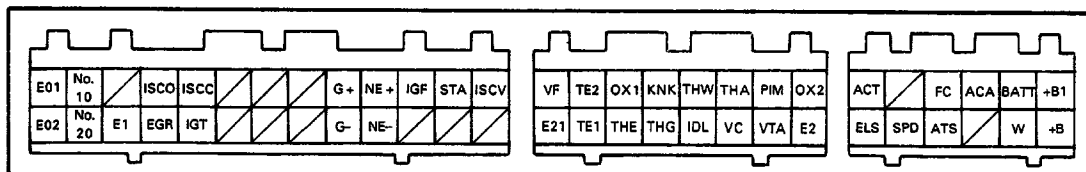


Terminals of ECM (5S-FE M/T)

Symbol	Terminal Name	Symbol	Terminal Name	Symbol	Terminal Name
E01	POWER GROUND	N E+	DISTRIBUTOR	THA	AIR TEMP. SENSOR
E02	POWER GROUND	NE-	DISTRIBUTOR	VC	VACUUM SENSOR THROTTLE POSITION SENSOR
No.10	INJECTOR	IGF	IGNITER	PIM	VACUUM SENSOR
No.20	INJECTOR	—	—	VTA	THROTTLE POSITION SENSOR
—	—	STA	STARTER RELAY	*OX2	SUB-OXYGEN SENSOR
E1	ENGINE GROUND	—	—	E2	SENSOR GROUND
ISCO	ISV VALVE	SCV	A/C IDLE-UP VSV	ACT	A/C AMPLIFIER
EGR	EGR VSV	—	—	ELS	HEAD LIGHT AND DEFOGGER
ISCC	ISV VALVE	VF	DATA LINK CONNECTOR 1	—	—
IGT	IGNITER	E21	SENSOR GROUND	SPD	SPEED SENSOR
—	—	TE2	DATA LINK CONNECTOR 1	FC	CIRCUIT OPENING RELAY
—	—	TE 1	DATA LINK CONNECTOR 1	ATS	A/C AMPLIFIER
—	—	ox i	MAIN OXYGEN SENSOR	ACA	A/C AMPLIFIER
—	—	THE	EVAPORATOR TEMP. SENSOR	—	—
—	—	KNK	KNOCK SENSOR	BATT	BATTERY
—	—	*THG	EGR GAS TEMP. SENSOR	W	MALFUNCTION INDICATOR LAMP
G +	DISTRIBUTOR	THW	ENGINE COOLANT TEMP. SENSOR	+B1	MFI MAIN RELAY
G -	DISTRIBUTOR	IDL	THROTTLE POSITION SENSOR	+B	MFI MAIN RELAY

ECM Terminals

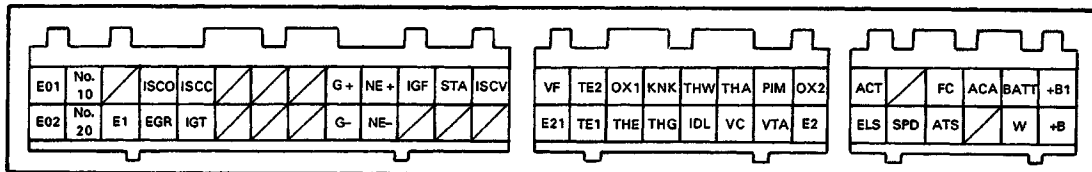
* Calif. only

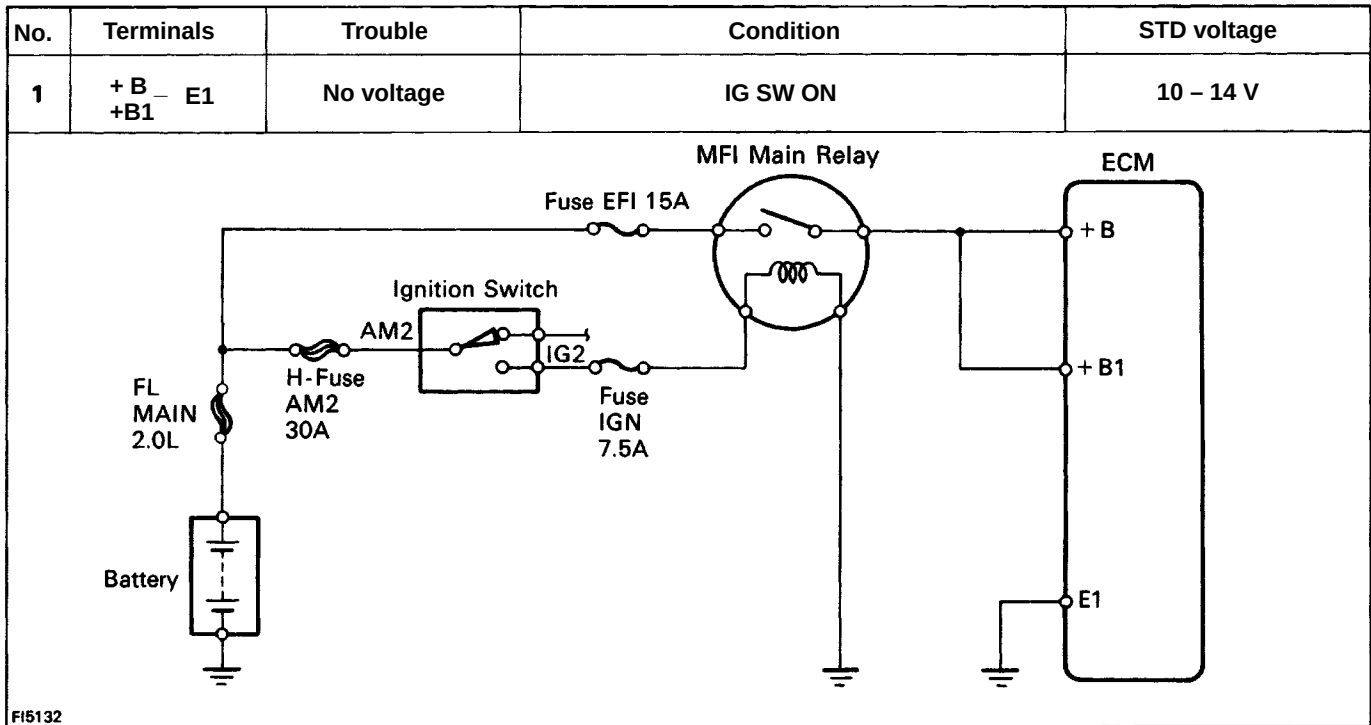


Voltage at ECM Wiring Connectors (5S-FE M/T)

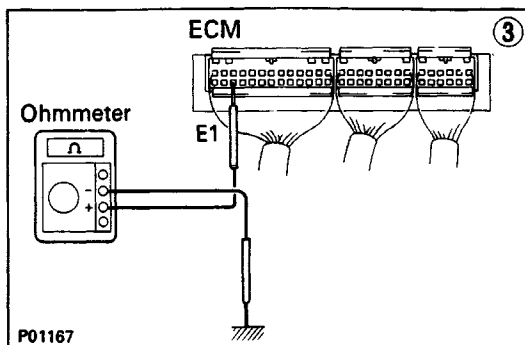
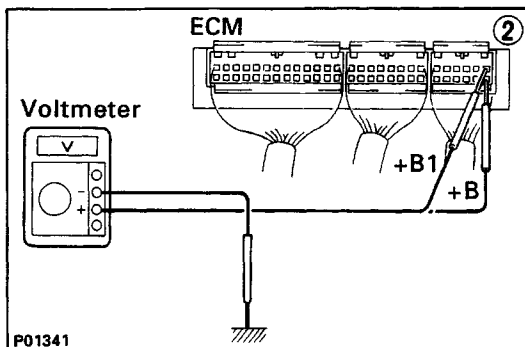
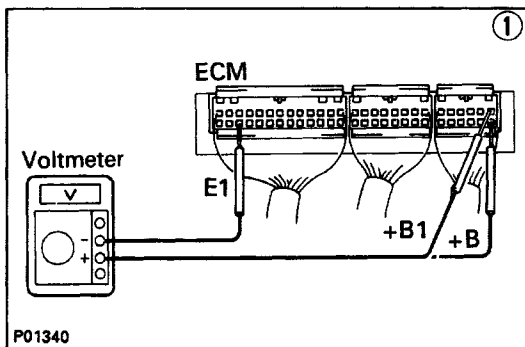
No.	Terminals	Condition		STD voltage (V)	See page
1	+B _ E 1 +B1	IG SW ON		10–14	FI–94
2	BATT – E1			10–14	FI–95
3	IDL – E2	• G SW ON	Throttle valve open	8–14	FI–96
	VC – E2			4.5–5.5	
	VTA – E2		Throttle valve fully closed (Throttle opener must be cancelled first)	0.8–1.2	
			Throttle valve fully open	3.2–4.2	
4	PIM–E2	IG SW ON		3.3–3.9	FI–98
	VC– E2			4.5–5.5	
5	No.10 _ E01 No.20 _ E02				
6	THA – E2	IG SW ON	Intake air temp. 20°C (168°F)	1.9–2.9	FI–100
7	THW – E2		Engine coolant temp. 80°C (176°F)	0.1–1.1	FI–101
8	STA – E 1	Cranking		6–14	FI–102
9	IGT– E1	Cranking or idling		0.8–1.2	FI–103
10	ISCC _ E1 ISCO	IG SW ON	ECM connectors disconnected	8–14	FI–104
11	W–E1	No trouble (malfunction indicator lamp off) and engine running		10–14	FI–105

ECM Terminals





FI5132



(1) There is no voltage between ECM terminals + B or + B1 and E1. (IG SW ON)

(2) Check that there is voltage between ECM terminal +B or + B1 and body ground. (IG SW ON)

NO OK

(3) Check wiring between ECM terminal E1 and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Check fuses, H-fuse, fusible link and ignition switch.

BAD

Repair or replace.

OK

Check MFI main relay. (See page FI-216)

BAD

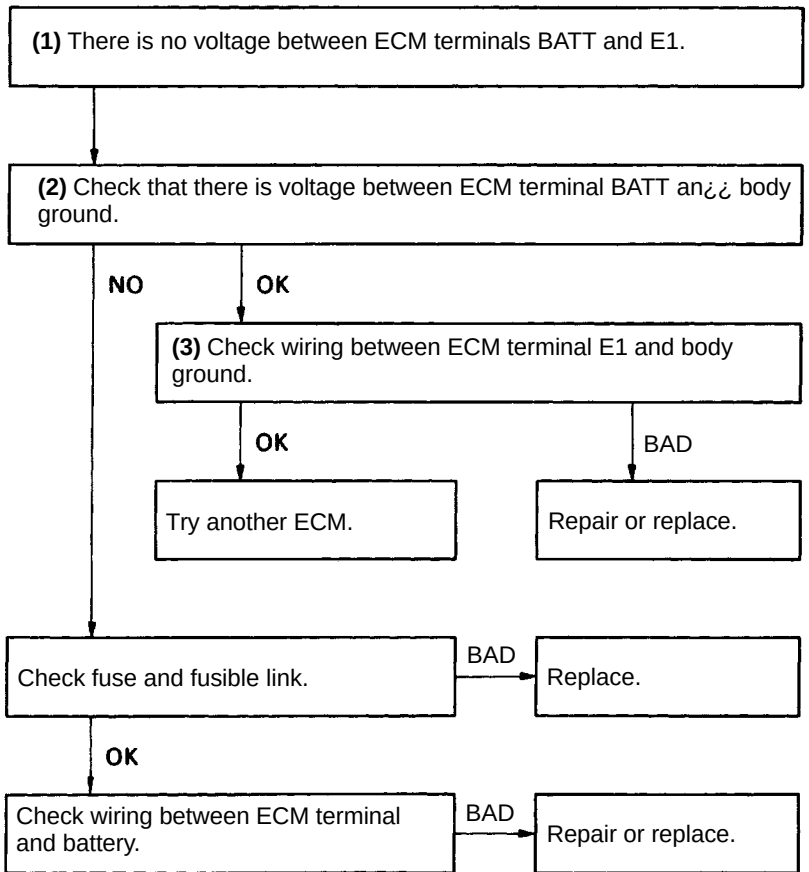
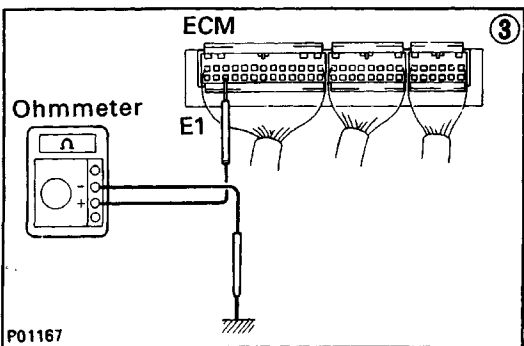
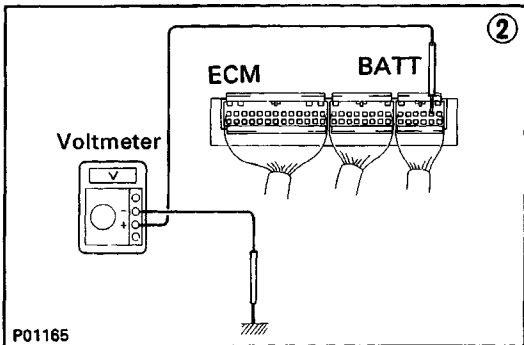
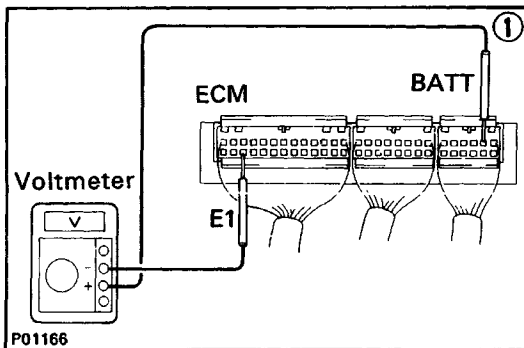
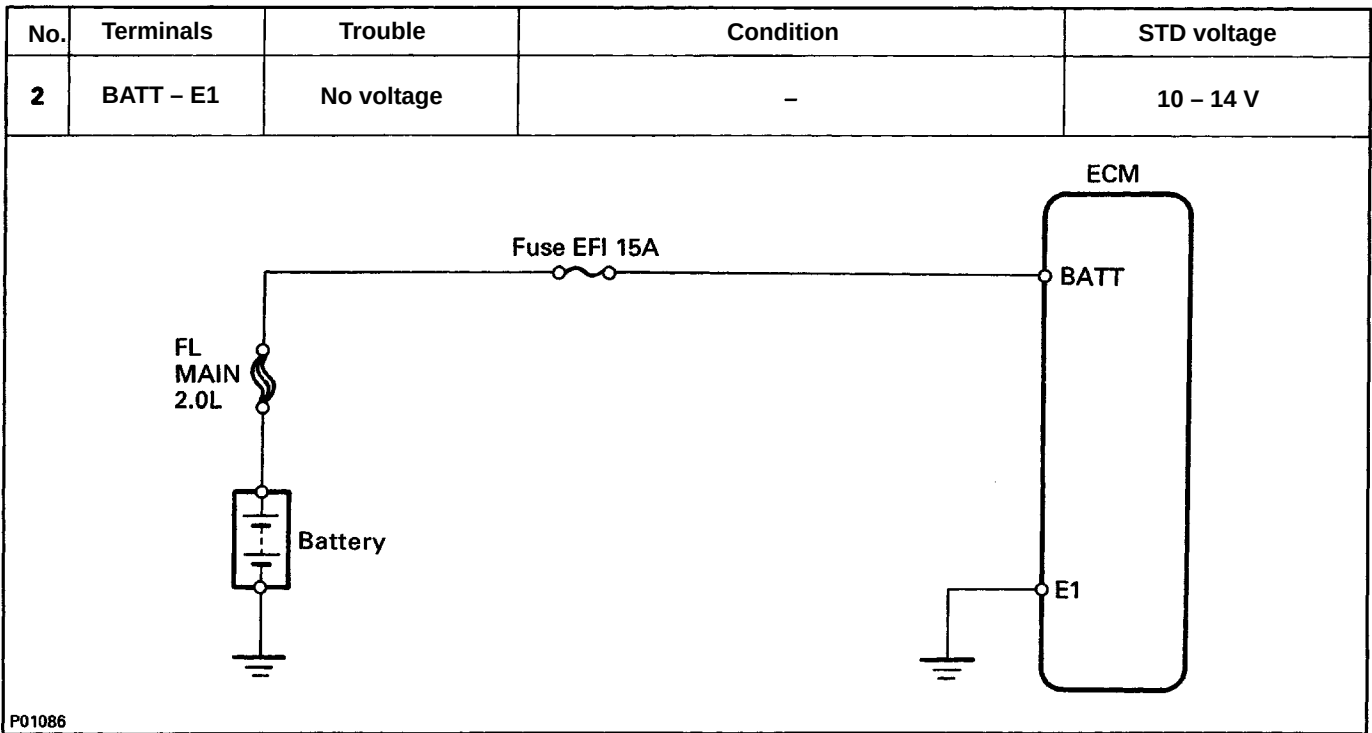
Replace.

OK

Check wiring between MFI main relay and battery.

BAD

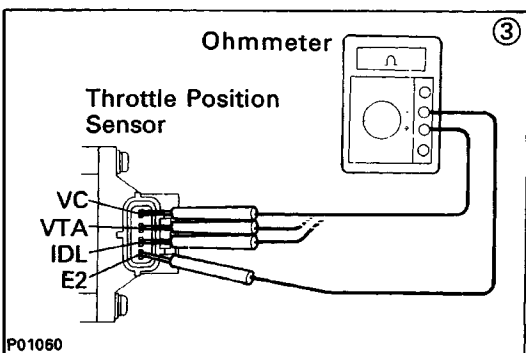
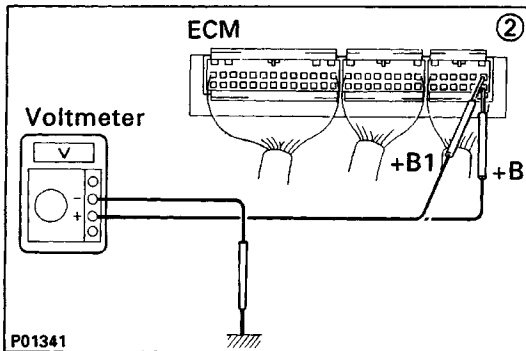
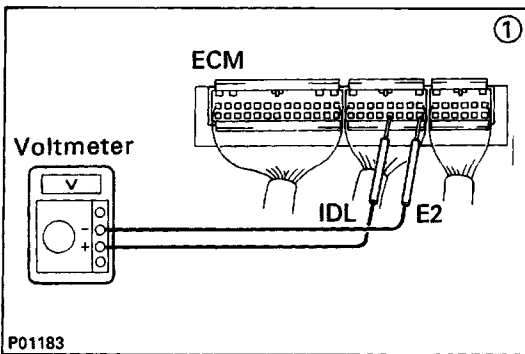
Repair or replace.



No.	Terminals	Trouble	Condition	STD voltage	
3	IDL - E2	No voltage	IG SW ON	Throttle valve open	
	VC - E2			4.5 - 5.5 V	
	VTA - E2			Throttle valve fully closed (Throttle opener must be cancelled first)	0.8 - 1.2 V
				Throttle valve fully open	3.2 - 4.2 V

Throttle Position Sensor

P01419



• 1DL - E2

(1) There is no voltage between ECM terminals IDL and E2.
(IG SW ON) (Throttle valve open)

(2) Check that there is voltage between ECM terminal + B (+B1)
and body ground. (IG SW ON)

NO OK

Check wiring between ECM terminal E1 and body
ground.

OK

Try another ECM.

BAD

Repair or replace.

Refer to No.1.
(See page FI-94)

BAD

Repair or replace.

OK

Check throttle position
sensor. (See page FI-199)

BAD

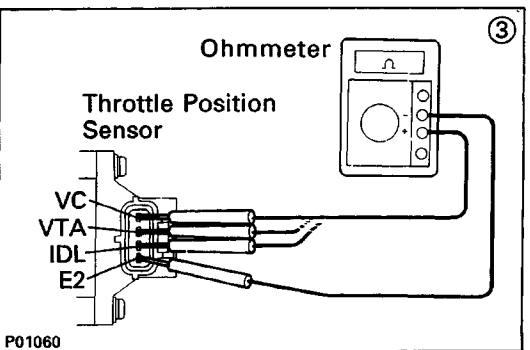
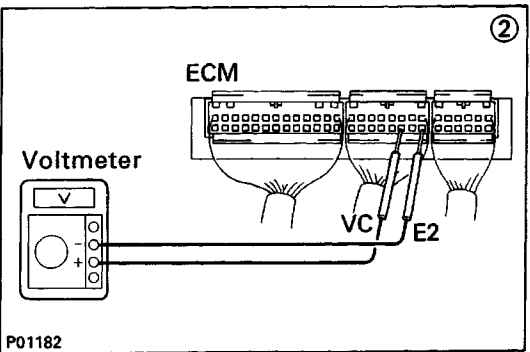
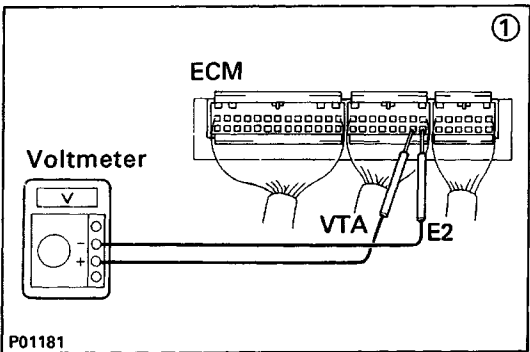
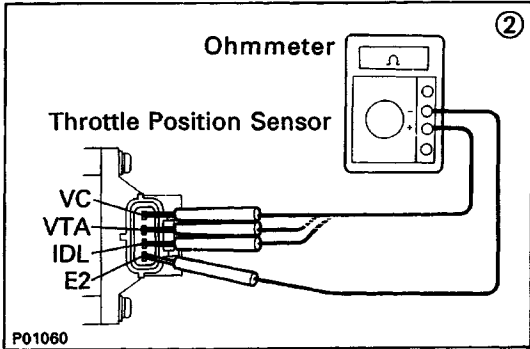
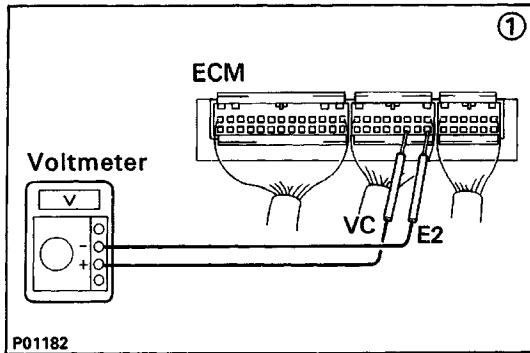
Repair or replace
throttle position sensor.

OK

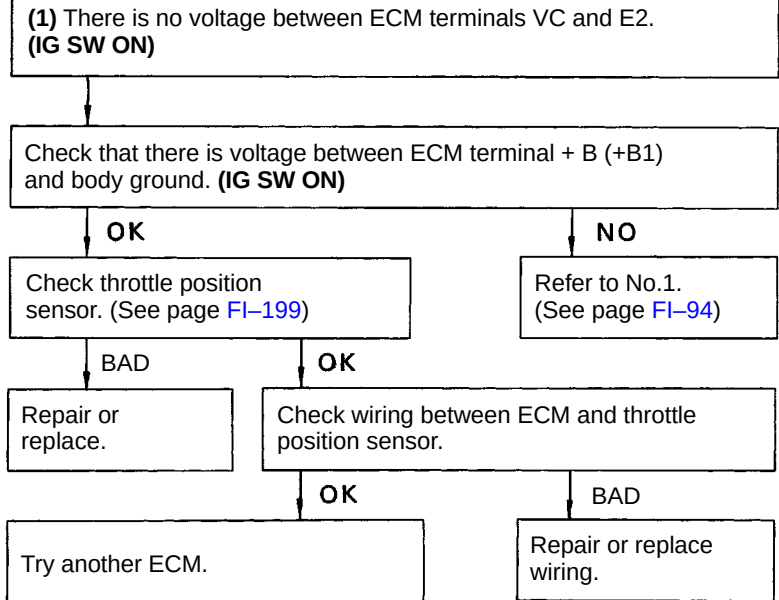
Check wiring between ECM and
throttle position sensor.

OK

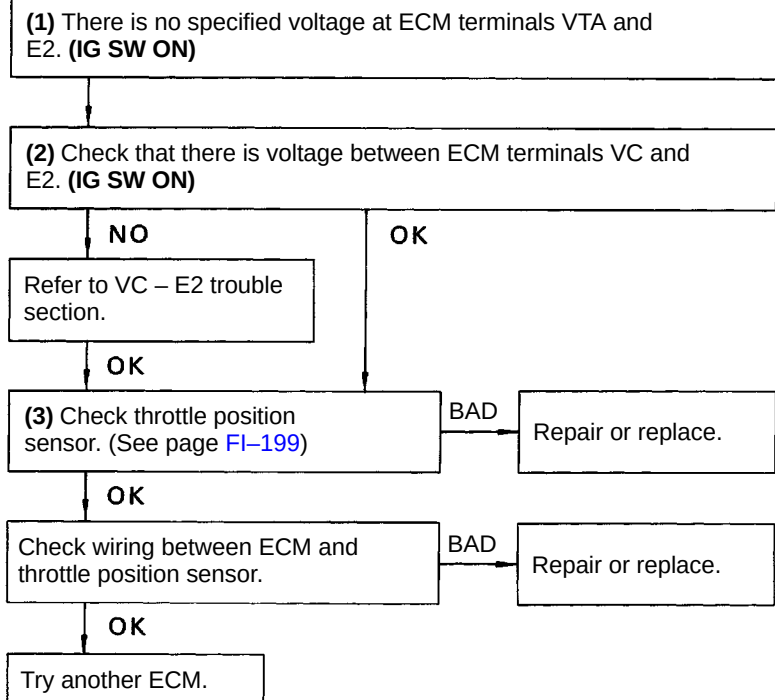
Try another ECM.



• VC-E2



• VTA - E2



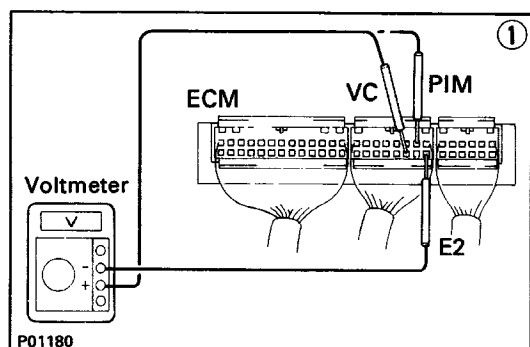
No.	Terminals	Trouble	Condition	STD voltage
4	PIM – E2	No voltage	IG SW ON	3.3 – 3.9 V
	VC – E2			4.5 – 5.5 V

Vacuum Sensor
(Manifold Absolute Pressure Sensor)

ECM

E2, PIM, VCC, E2 (E21), PIM, VC, E1

FI1226



• PIM – E2, VC – E2

(1) There is no voltage between ECM terminals PIM or VC and E2. (IG SW ON)

(2) Check that there is voltage between ECM terminal + B (+B1) and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-94)

(3) Check wiring between ECM terminal E1 and body ground.

OK

BAD

Check vacuum sensor.
(See page FI-234)

Repair or replace.

BAD

Replace vacuum sensor.

OK

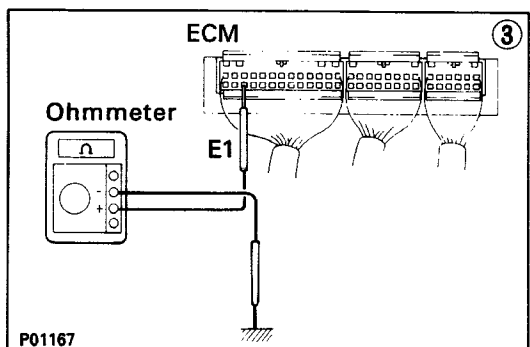
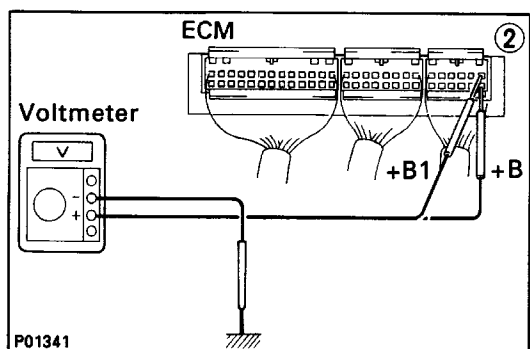
Check wiring between ECM and vacuum sensor.

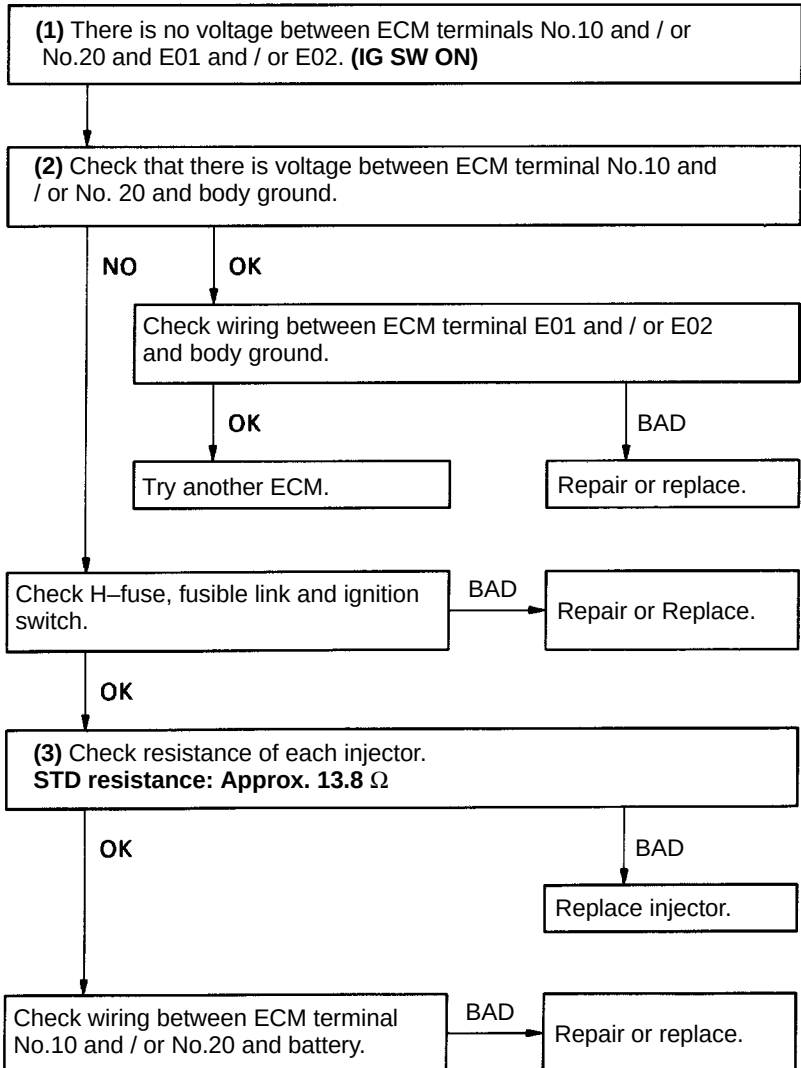
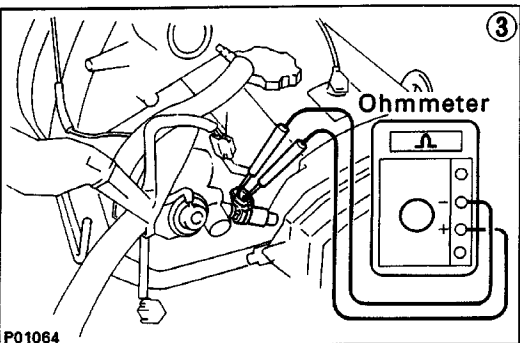
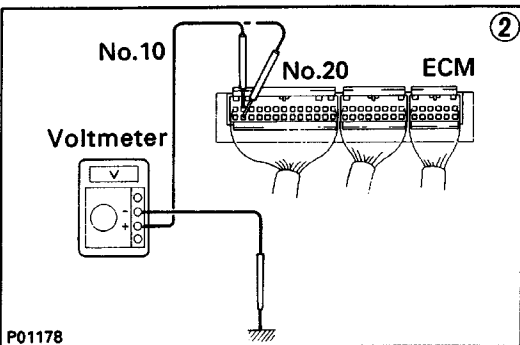
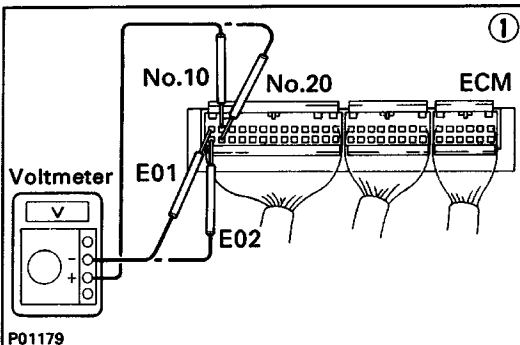
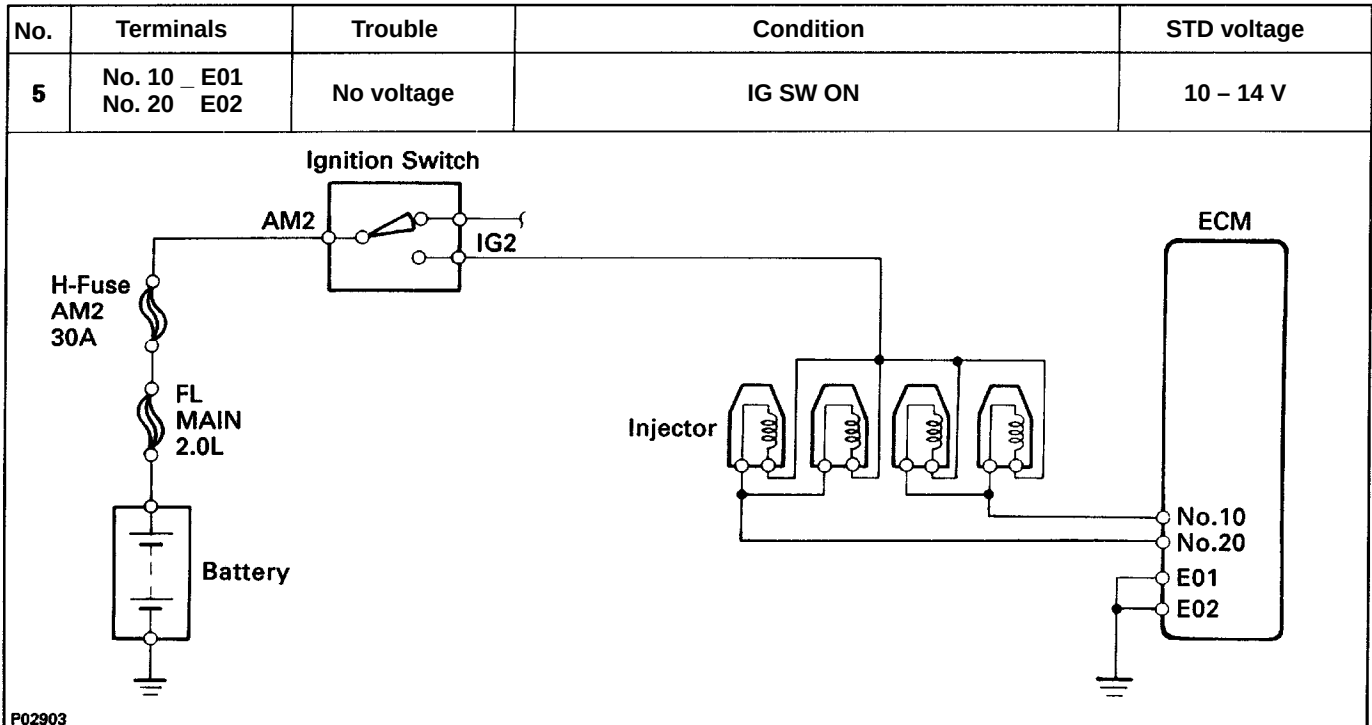
OK

BAD

Try another ECM.

Repair or replace.

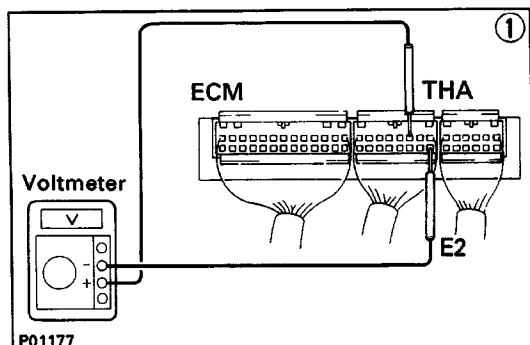




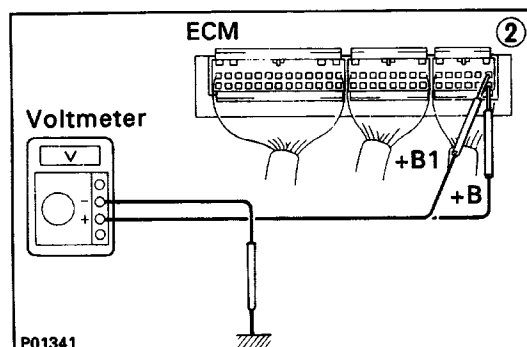
No.	Terminals	Trouble	Condition		STD voltage
6	THA – E2	No voltage	IG SW ON	Intake air temperature 20°C (68°F)	1.9 – 2.9 V

Diagram showing the Intake Air Temp. Sensor connected to the ECM. The sensor has terminals THA and E2. THA is connected to ECM terminal THA. E2 is connected to ECM terminal E2 (E21). The ECM also has terminals +B (+B1) and E1. E1 is connected to ground.

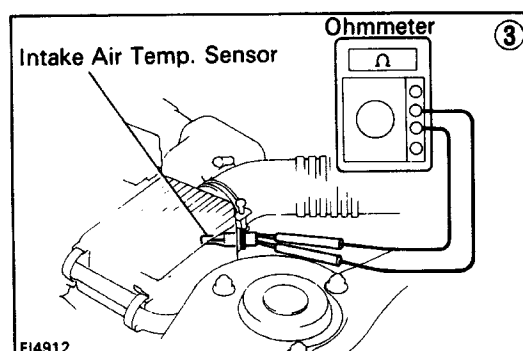
FI3572



P01177



P01341



FI4912

(1) There is no voltage between ECM terminals THA and E2.
(IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1
and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-94)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

Check intake air temp. sensor.
(See page FI-232)

Repair or replace.

BAD

OK

Replace air temp.
sensor.

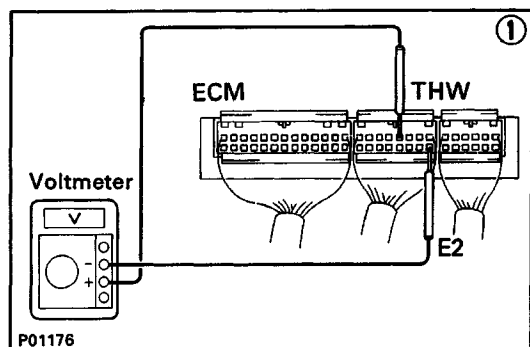
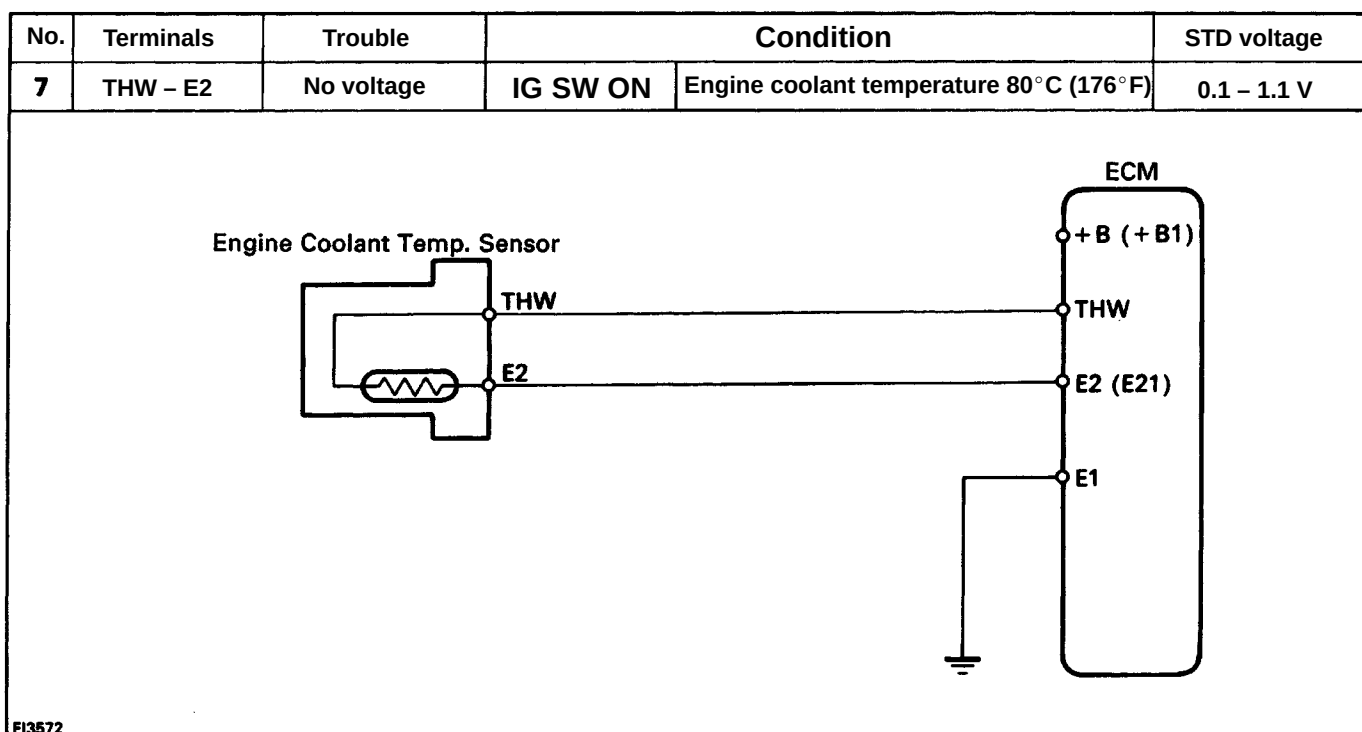
Check wiring between ECM and air
temp. sensor.

OK

BAD

Try another ECM.

Repair or replace.



(1) There is no voltage between ECM terminals THW and E2.
(IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1 and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-94)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

(3) Check engine coolant temp. sensor. (See page FI-231)

Repair or replace.

BAD

Replace engine coolant temp. sensor.

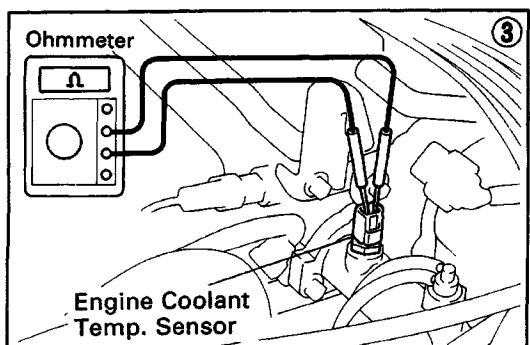
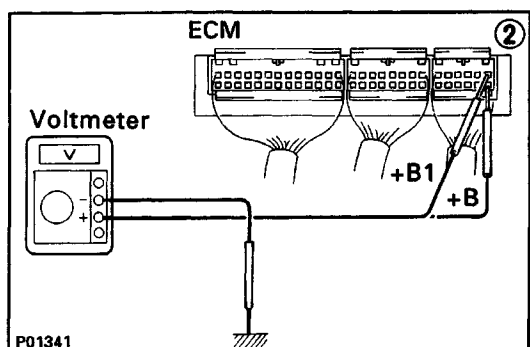
Check wiring between ECM and engine coolant temp. sensor.

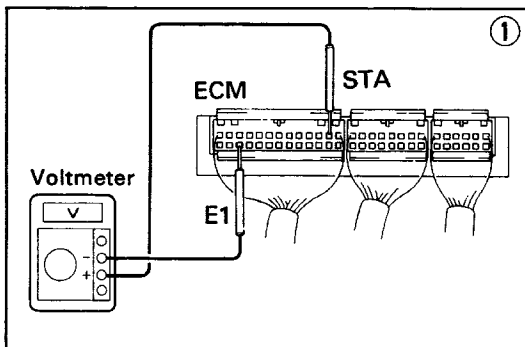
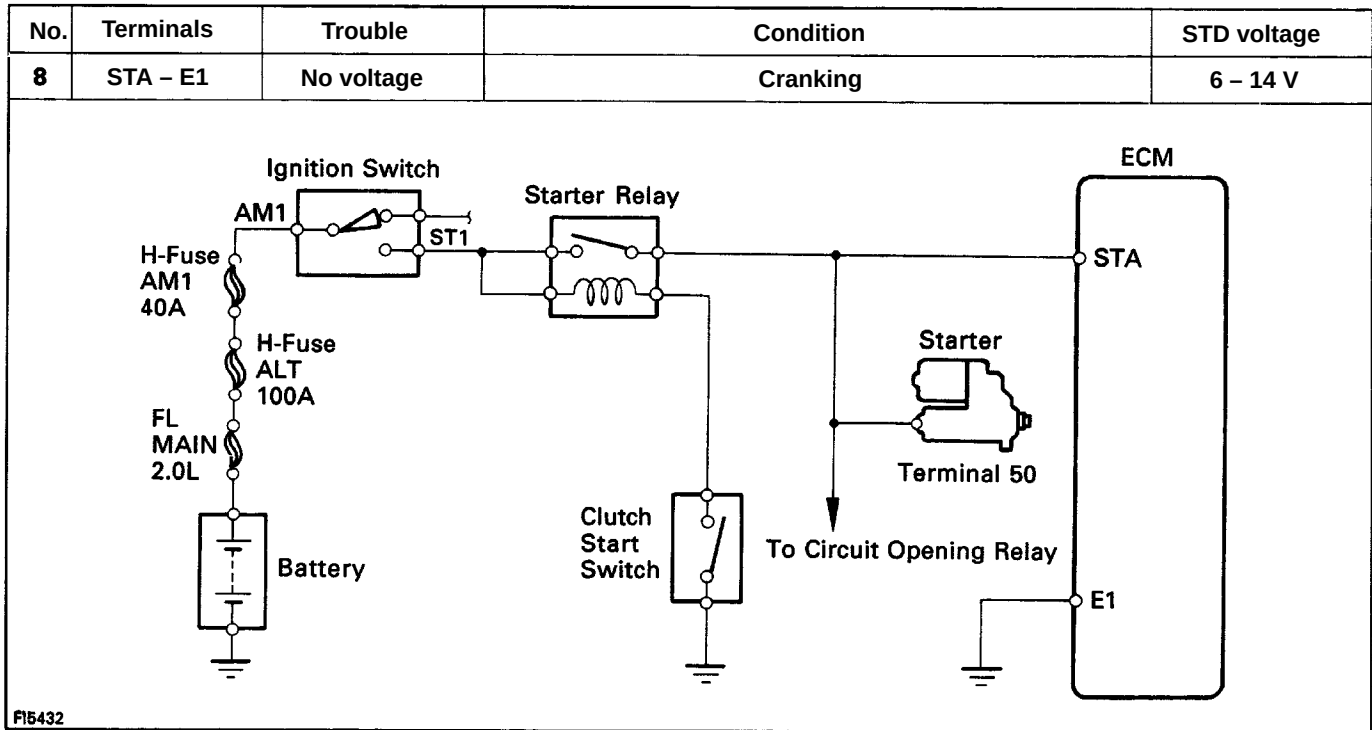
OK

BAD

Try another ECM.

Repair or replace.





(1) There is no voltage between ECM terminals STA and E1.
(IG SW START)

Check starter operation.

OK

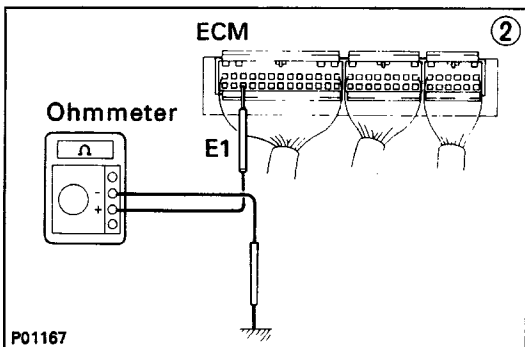
Check wiring between ECM terminal STA and ignition switch terminal ST1.

BAD

OK

BAD

Repair or replace.



P01167

(2) Check wiring between ECM terminal E1 and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Check H-fuse, fusible link, battery, wiring, ignition switch, clutch start switch and starter relay.

BAD

Repair or replace.

OK

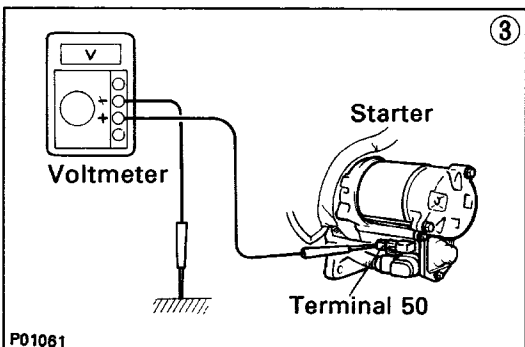
(3) Check that there is voltage at starter terminal 50.
(IG SW START) STD voltage: 6 – 14 V

OK

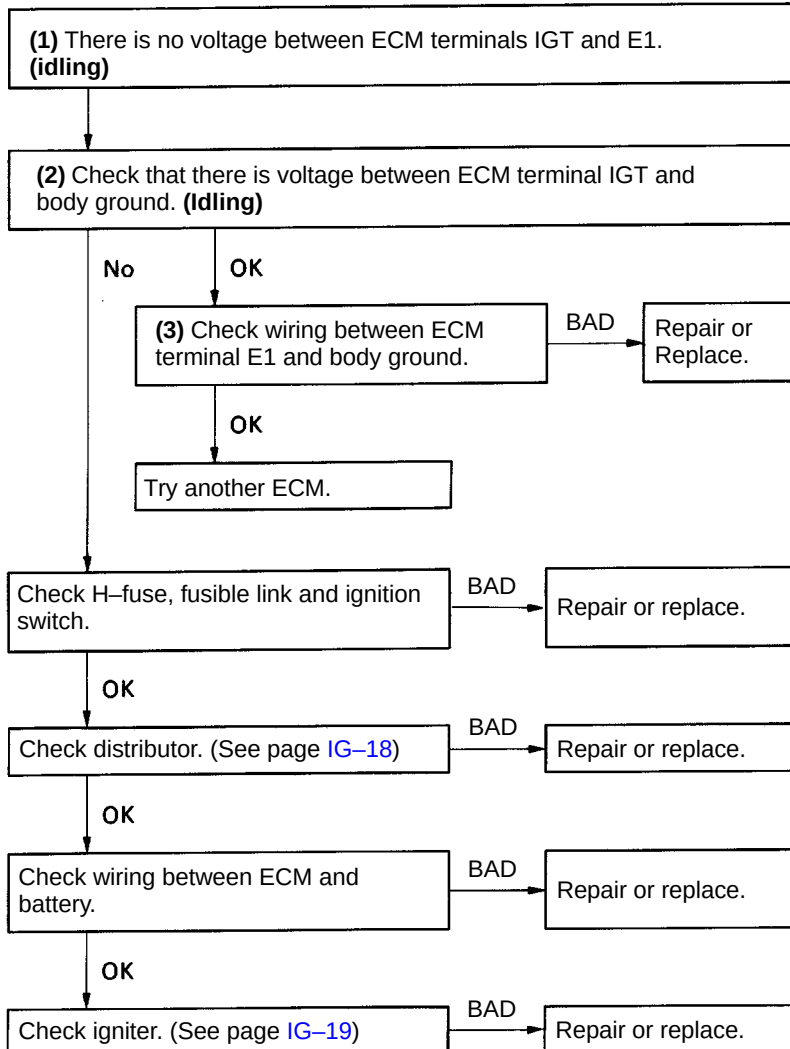
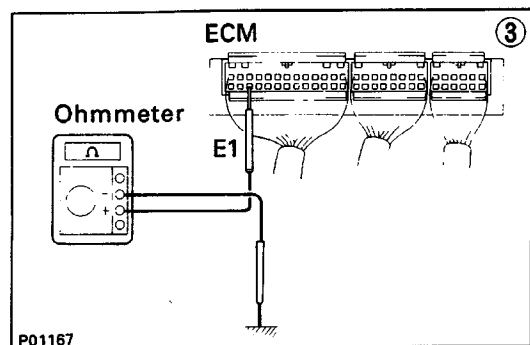
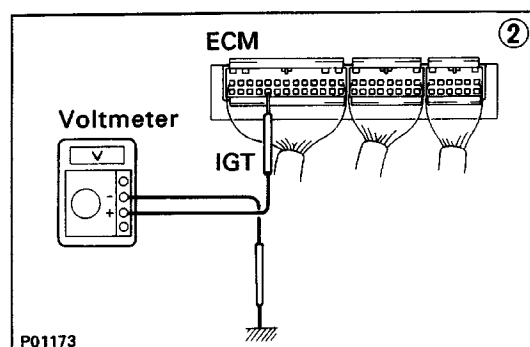
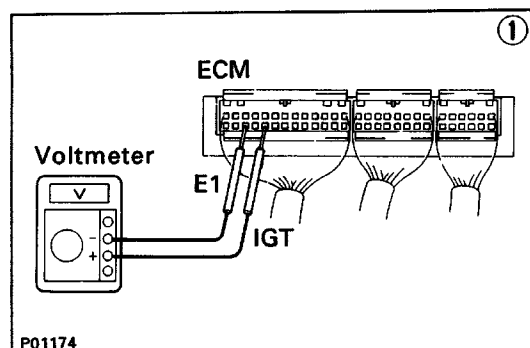
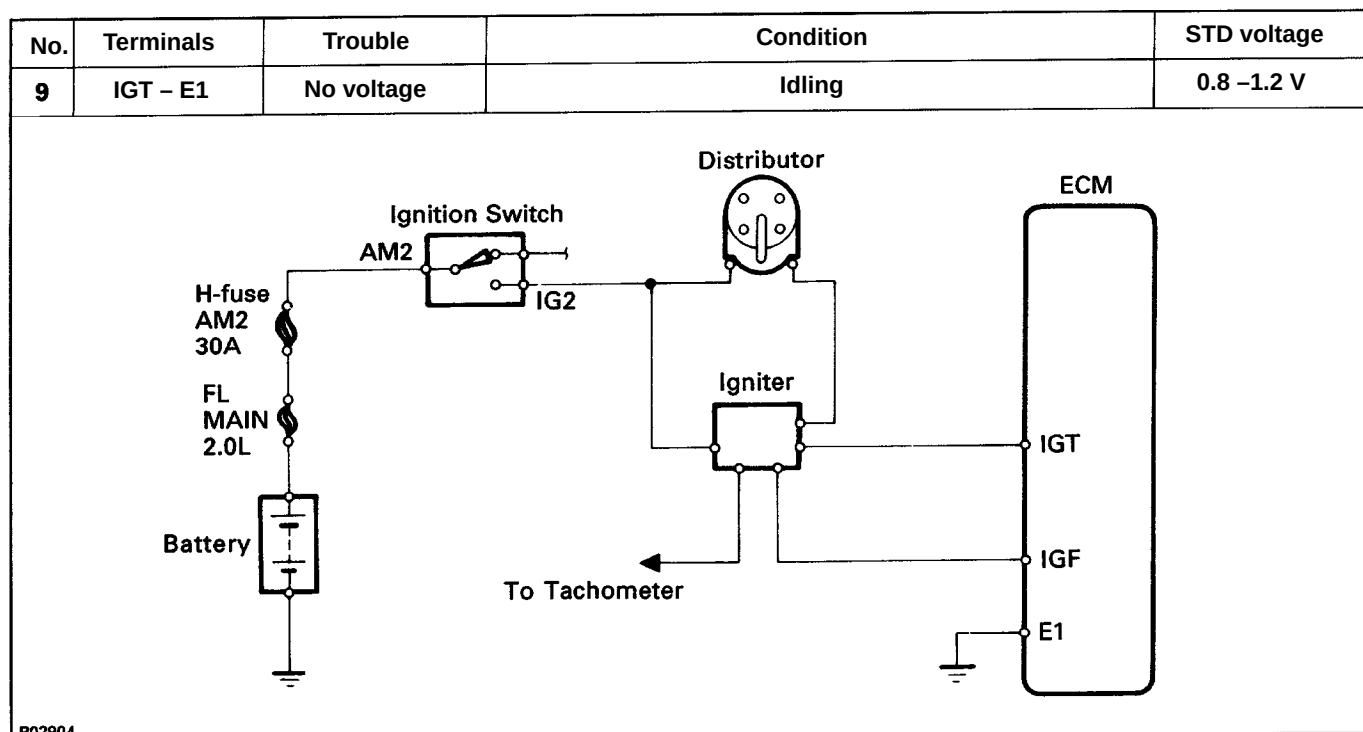
Check starter.

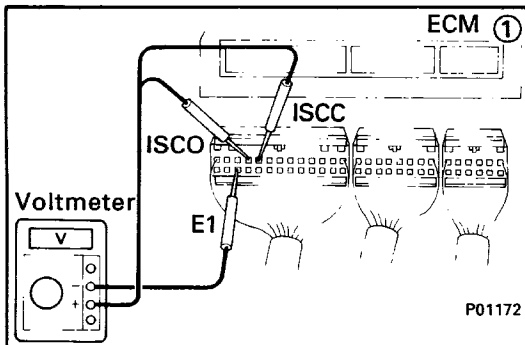
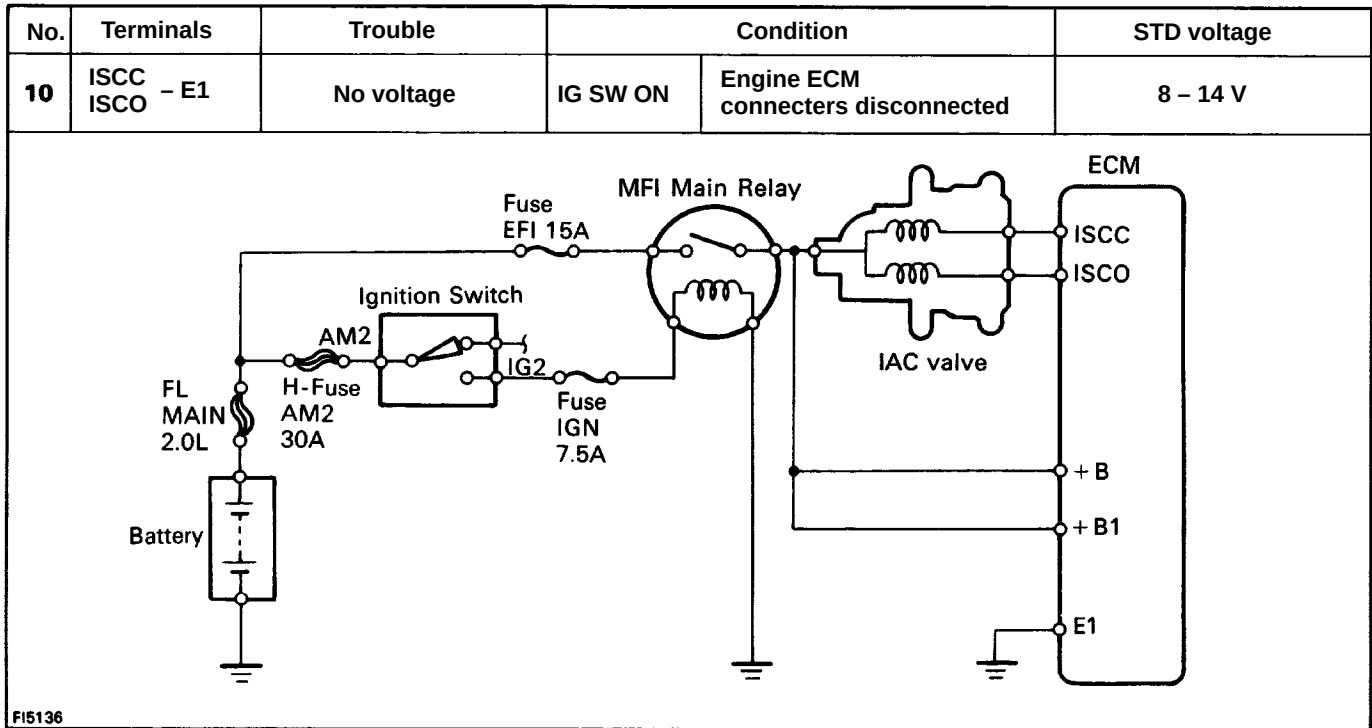
NO

Check wiring between ignition switch terminal ST1 and starter terminal 50.



P01061





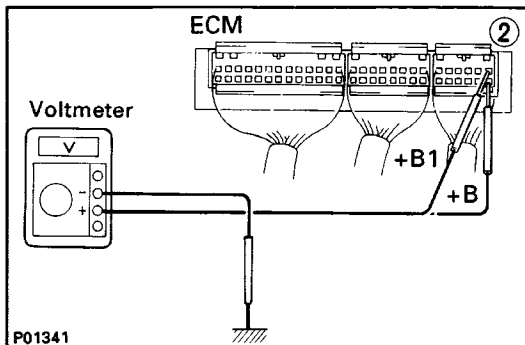
(1) There is no voltage between ECM terminals ISCC or ISCO and E1. (IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1 and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-94)



(3) Check resistance between IAC valve terminals +B and ISCC or ISCO.

STD resistance:
Approx. 19.3 – 22.3Ω

BAD

Replace IAC valve.

OK

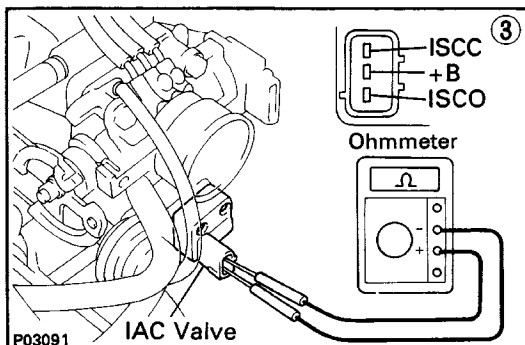
Check wiring between ECM and IAC valve.

BAD

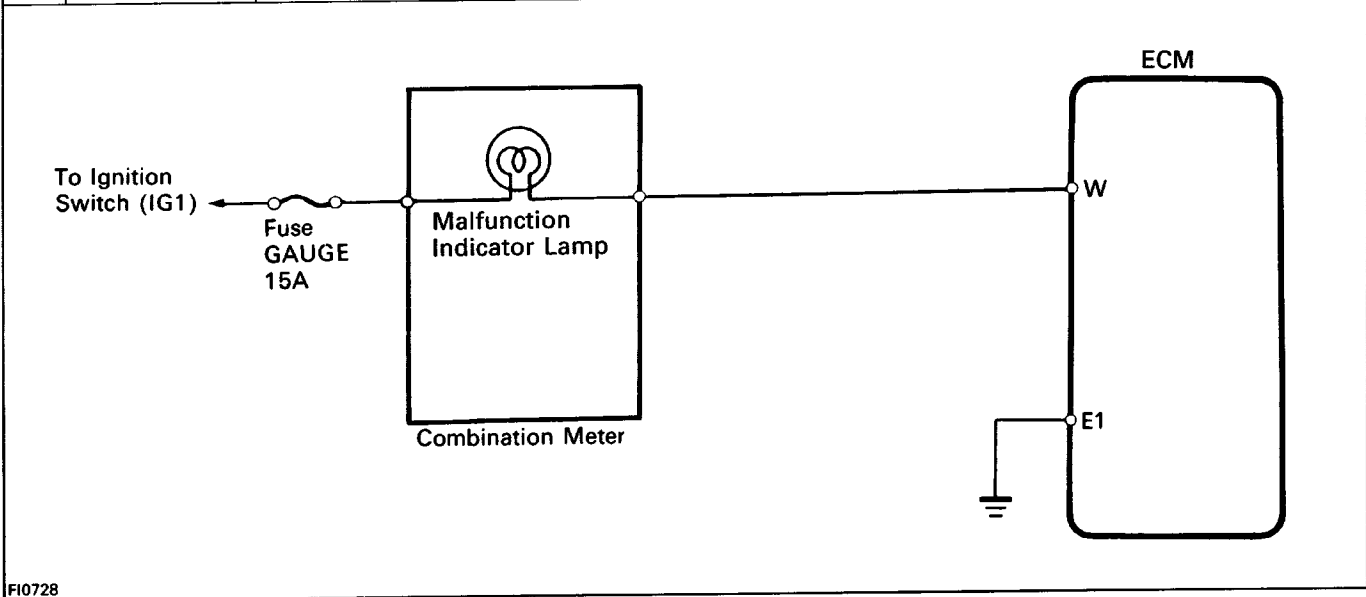
Repair or replace wiring.

OK

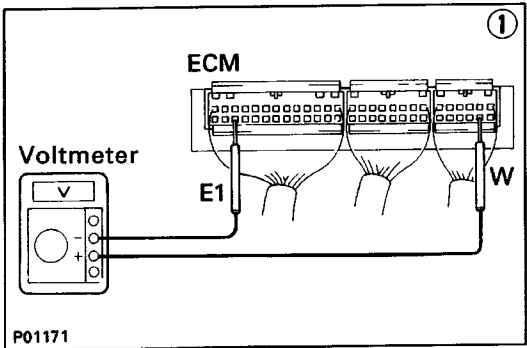
Try another ECM.



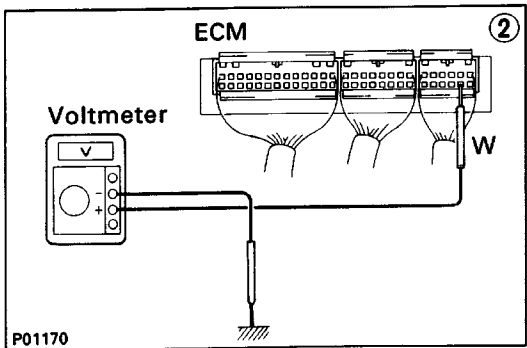
No.	Terminals	Trouble	Condition	STD voltage
11	1AI - E 1	No voltage	No trouble (malfunction indicator lamp off) and engine running.	10 - 14 V



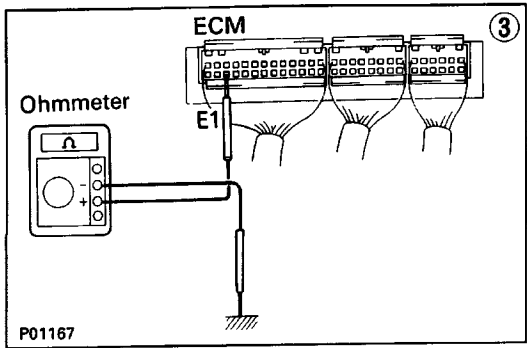
FI0728



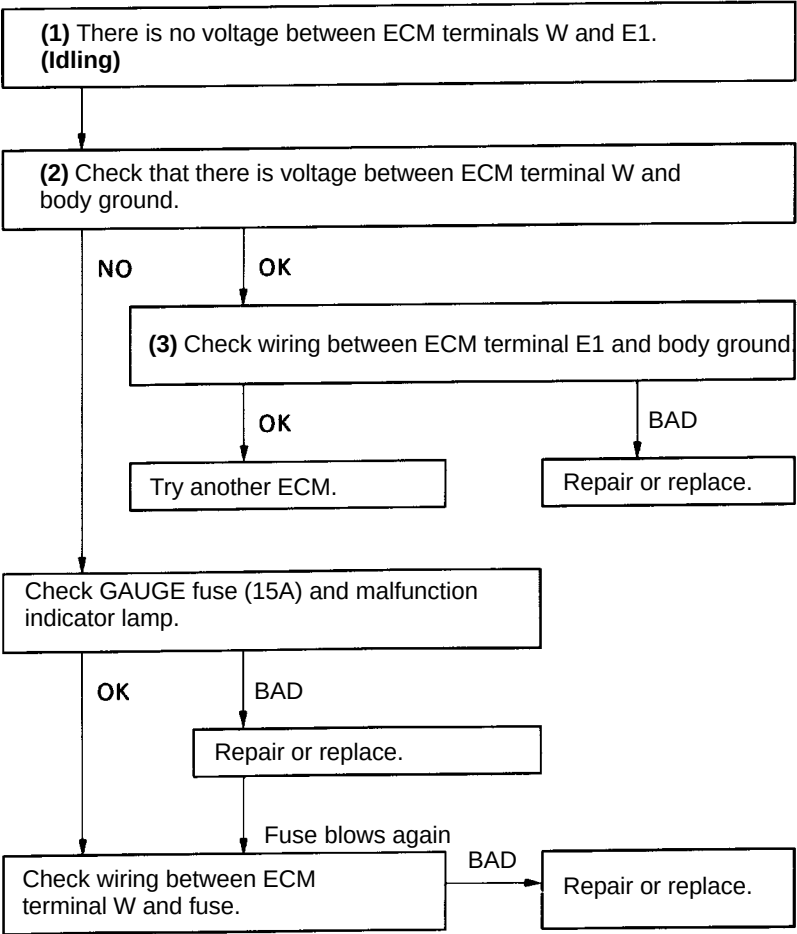
P01171

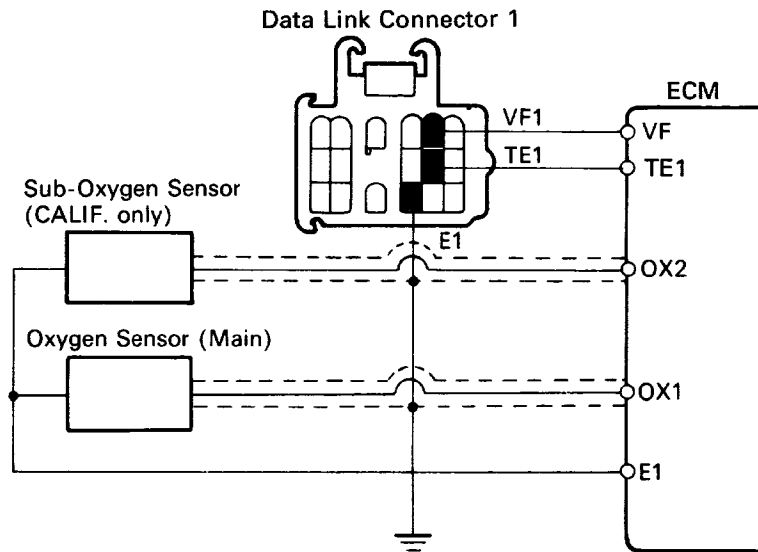


P01170

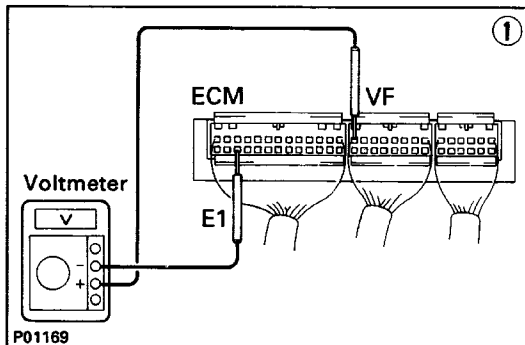


P01167

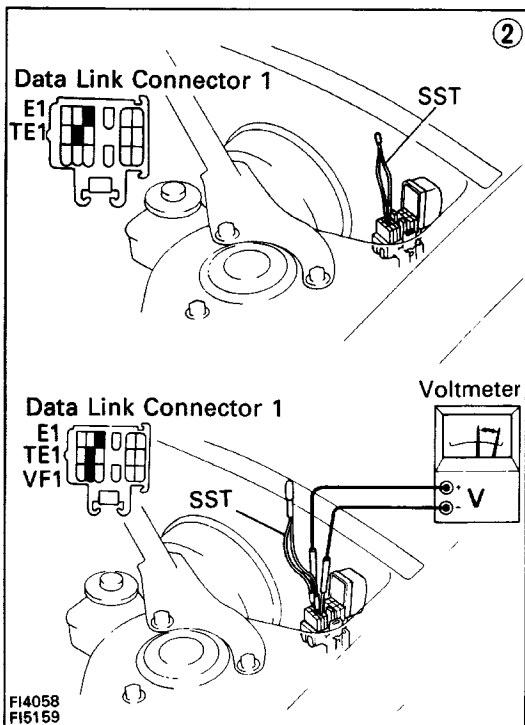




FI3841



P01169

FI4058
FI5159

(1) There is no voltage between ECM terminals VF and E1.

Check that there is voltage between ECM terminal VF and body ground.

NO

OK

Check wiring between ECM terminal E1 and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Is air leaking into air induction system?

YES

Repair or replace.

NO

Check spark plugs. (See page IG-16)

BAD

Repair or replace.

OK

Check distributor and ignition system. (See page IG-4)

BAD

Repair or replace.

OK

Check fuel pressure. (See page FI-128)

BAD

Repair or replace.

OK

Check injectors. (See page FI-169)

BAD

Repair or replace.

OK

Check vacuum sensor (See page FI-234)

BAD

Repair or replace.

OK

(2) Check operation of oxygen sensors. (See pages FI-237 and 239)

OK

System Normal

BAD

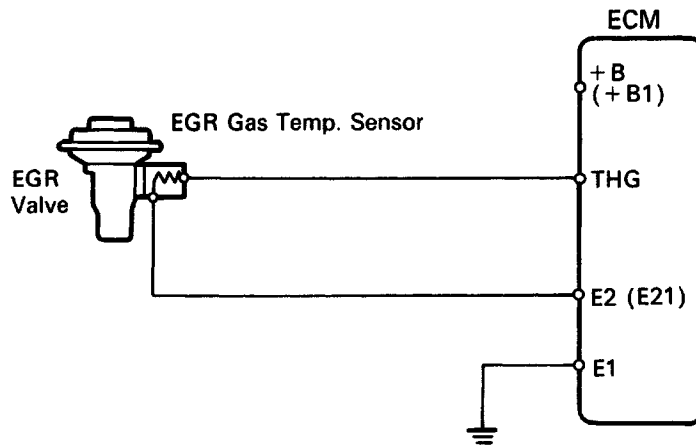
Check wiring between oxygen sensor and ECM.

BAD

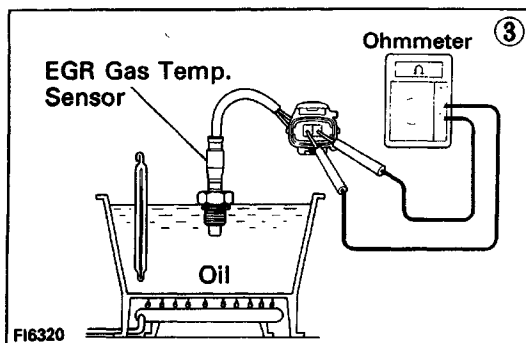
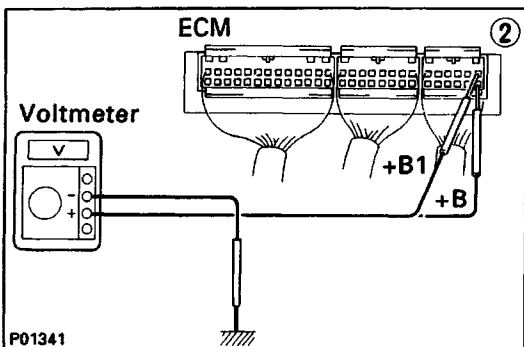
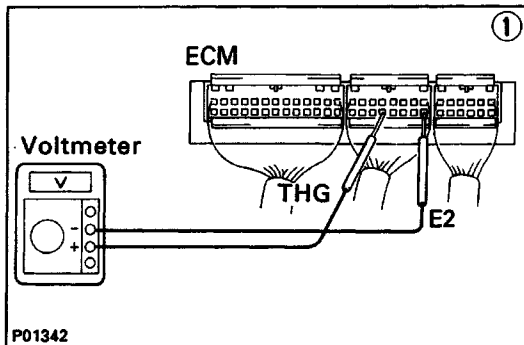
Repair wiring.

OK

Replace oxygen sensors.

CALIF. only

FI2680



(1) There is no voltage between ECM terminals THG and E2.
(IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1
and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-94)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

Repair or replace.

Check EGR system.
(See page EC-36)

BAD

Repair or replace.

OK

(3) Check EGR gas temp.
sensor. (See page FI-236)

BAD

Replace EGR gas
temp. sensor.

OK

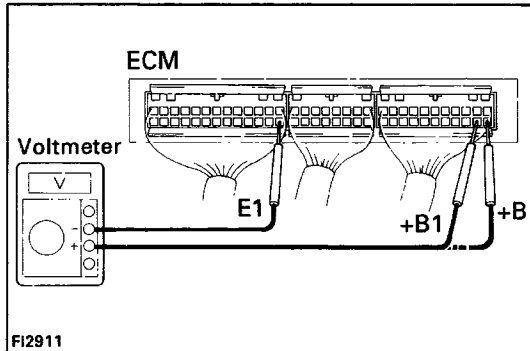
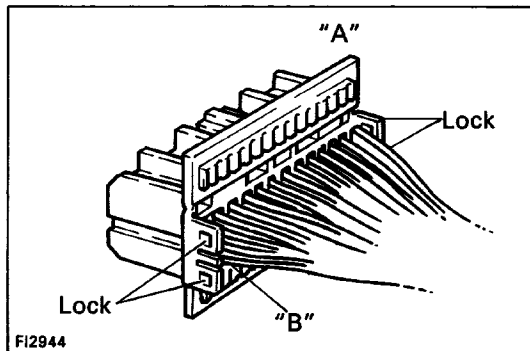
Check wiring between ECM and EGR
gas temp. sensor.

OK

Try another ECM.

BAD

Repair or replace.



MFI SYSTEM CHECK PROCEDURE

(5S-FE A/T)

PREPARATION

- Disconnect the connectors from the ECM.
- Remove the locks as shown in the illustration so that the tester probe(s) can easily come in.

NOTICE: Pay attention to sections "A" and "B" in the illustration which can be easily broken.

- Reconnect the connectors to the ECM.

HINT:

- Perform all voltage measurements with the connectors connected.
- Verify that the battery voltage is 11 V or more when the ignition switch is in the "ON" position. Using a voltmeter with high impedance (10 kΩ/V minimum), measure the voltage at each terminal of the wiring connectors.

Terminals of ECM (5S-FE A/T)

Symbol	Terminal Name	Symbol	Terminal Name	Symbol	Terminal Name
E01	POWER GROUND	S1	SOLENOID		—
E02	POWER GROUND	S2	SOLENOID	ACT	A/C AMPLIFIER
No.10	INJECTOR	SL	SOLENOID	SP1	SPEED SENSOR
	—	E1	ENGINE GROUND	OD1	OD OFF SWITCH
No.20	INJECTOR	VF	DATA LINK CONNECTOR 1	ACA	A/C AMPLIFIER
	—	E21	SENSOR GROUND		—
ISCO	IAC VALVE	TT	DATA LINK CONNECTOR 1	OD2	OD MAIN SWITCH
EGR	EGR VSV	TE1	DATA LINK CONNECTOR 1		—
ISCC	IAC VALVE	OX1	OXYGEN SENSOR		—
P	PATTERN SELECT SWITCH	TE2	DATA LINK CONNECTOR 1		—
	—	*OX2	SUB-OXYGEN SENSOR	W	MALFUNCTION INDICATOR LAMP
	—	KNK	KNOCK SENSOR		—
ISCV	A/C IDLE-UP VSV	THW	ENGINE COOLANT TEMP. SENSOR	BIK	STOP LIGHT SWITCH
IGT	IGNITER	IDL	THROTTLE POSITION SENSOR	ATS	A/C AMPLIFIER
2	PARK/NEUTRAL POSITION SWITCH	THA	AIR TEMP. SENSOR	THE	EVAPORATOR TEMP. SENSOR
L	PARK/NEUTRAL POSITION SWITCH	VTA	THROTTLE POSITION SENSOR	FC	CIRCUIT OPENING RELAY
NE -	DISTRIBUTOR	PI M	VACUUM SENSOR	ELS	HEADLIGHT RELAY DEFOGGER RELAY
G +	DISTRIBUTOR	*THG	EGR GAS TEMP. SENSOR	+B	MFI MAIN RELAY
NE +	DISTRIBUTOR	VC	VACUUM SENSOR, THROTTLE POSITION SENSOR	BA^	BATTERY
G -	DISTRIBUTOR	E2	SENSOR GROUND	+B I	MFI MAIN RELAY
IGF	IGNITER	STA	STARTER SWITCH	*Calif. only	
SP2	SPEED SENSOR	N SW	PARK/NEUTRAL POSITION SWITCH		

ECM Terminals

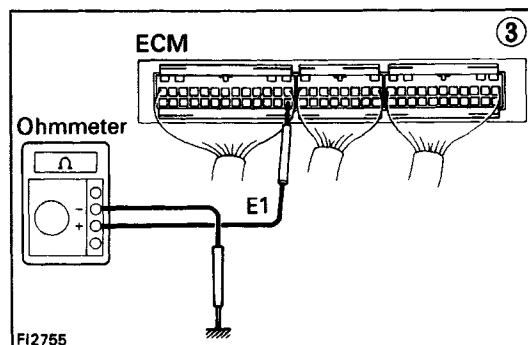
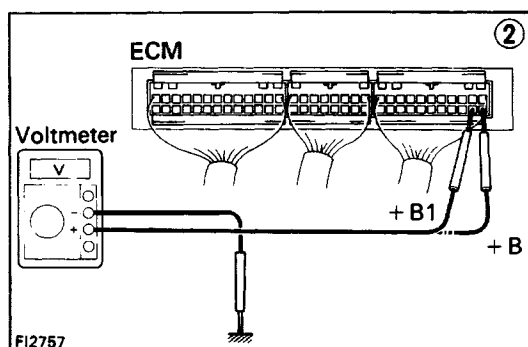
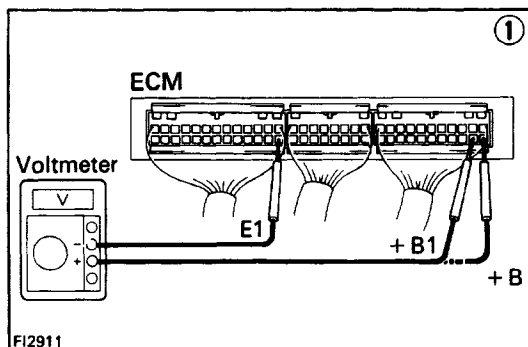
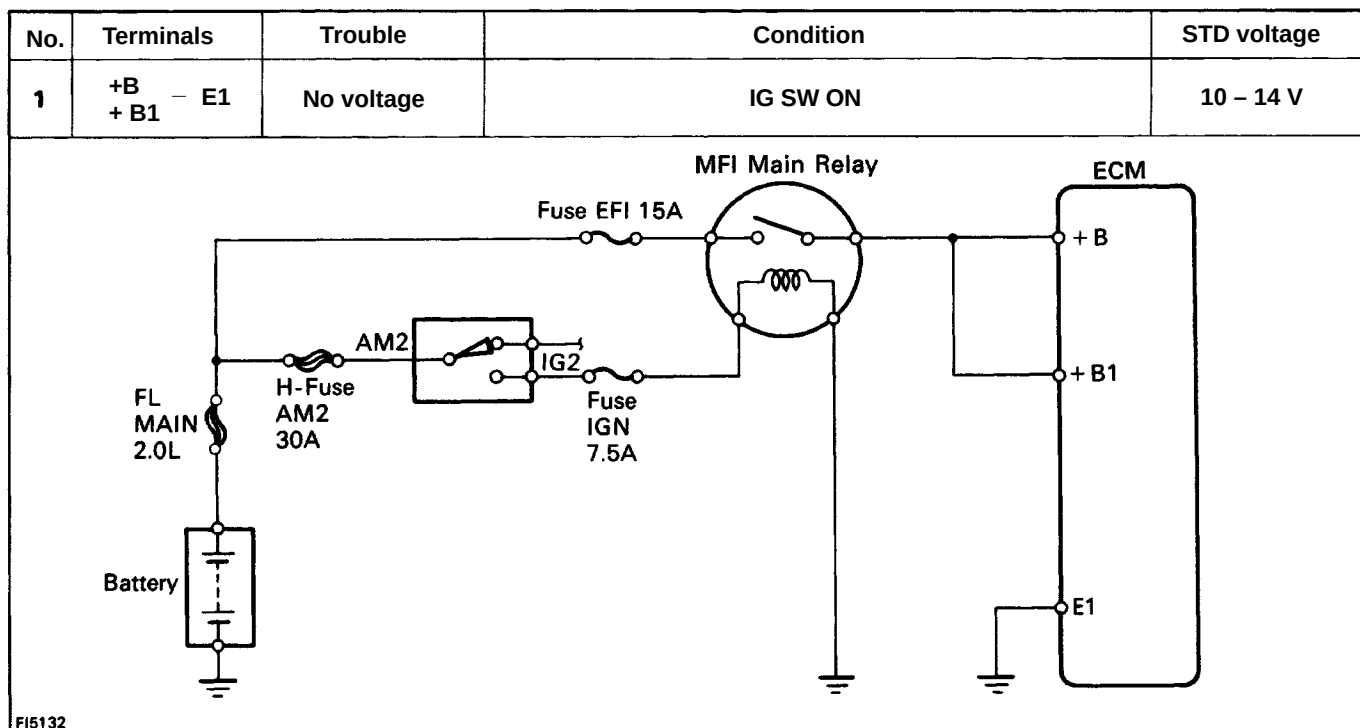
E01	No. 10	No. 20	ISCO	ISCC	ISCV	2	NE -	NE +	IGF	S1	SL	VF	TT	OX1	OX2	THW	THA	PI M	VC	STA	SPD	ACA	OD2	W	B/K	THE	ELS	BATT
E02			EGR	P	IGT	L	G +	G -	SP2	S2	E1	E21	TE1	TE2	KNK	IDL	VTA	THG	E2	NSW	ACT	OD1			ATS	FC	+B1	+B

Voltage at ECM Wiring Connectors (5S-EE A/T)

No.	Terminals	Condition		STD voltage (V)	See page
1	+B–E1 + B1	IG SW ON		10–14	FI–110
2	BATT – E 1	—		10–14	F–111
3	IDL – E2	IG SW ON	Throttle valve open	8–14	F–112
	VC – E2		—	4.5–5.5	
	VTA – E2		Throttle valve fully closed (Throttle opener must be cancelled first)	0.8–1.2	
			Throttle valve fully open	3.2–4.2	
4	PIM – E2	IG SW ON		3.3–3.9	FI–114
	VC – E2			4.5–5.5	
5	No. 10 E01 – No. 20 E02				
6	THA – E2	IG SW ON	Intake air temp. 20°C (68°F)	1.9–2.9	FI–116
7	THW – E2		Engine coolant temp. 80°C (176°F)	0.1 –1.1	FI–117
8	STA – E 1	Cranking		6–14	FO–118
9	IGT – E1	Cranking or idling		0.8–1.2	FI–119
10	ISCC – EI ISCO	IG SW ON	ECM connectors disconnected	8–14	FI–120
11	W – E1	No trouble (malfunction indicator lamp off) and engine running		10–14	FI–121

ECM Terminals

E01	No. 10	No. 20	ISCO	ISCC	ISCV	2	NE –	NE +	IGF	S1	SL	VF	TT	OX1	OX2	THW	THA	PIM	VC	STA	SPD	ACA	OD2	W	B/K	THE	ELS	BATT
E02			EGR	P	IGT	L	G +	G –	SP2	S2	E1	E21	TE1	TE2	KNK	IDL	VTA	THG	E2	NSW	ACT	OD1			ATS	FC	+B1	+B



(1) There is no voltage between ECM terminals +B or +B1 and E1. (IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1 and body ground. (IG SW ON)

NO

OK

(3) Check wiring between ECM terminal E1 and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Check fuses, H-fuse, fusible link and ignition switch.

BAD

Repair or replace.

OK

Check MFI main relay. (See page FI-216)

BAD

Replace.

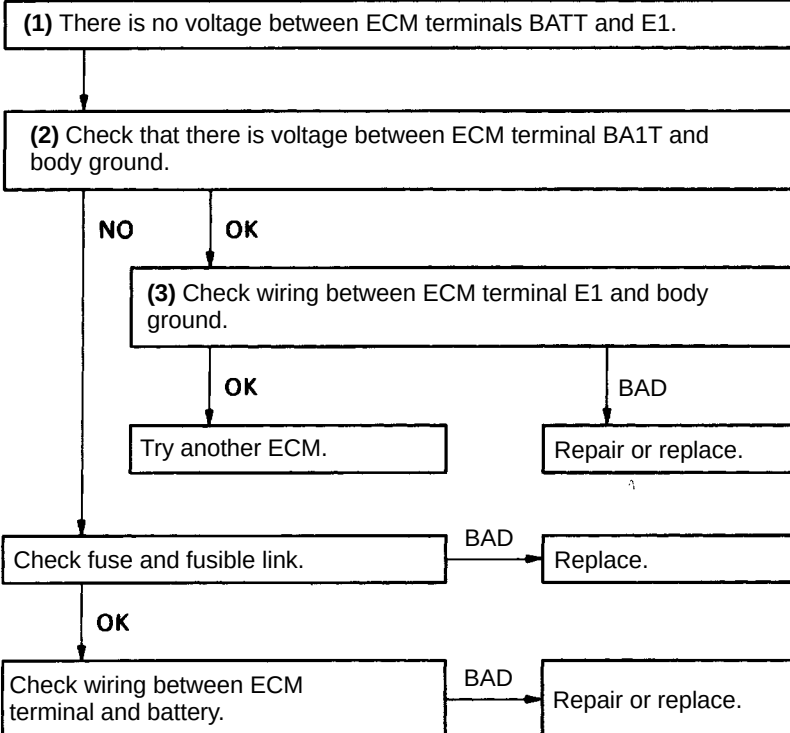
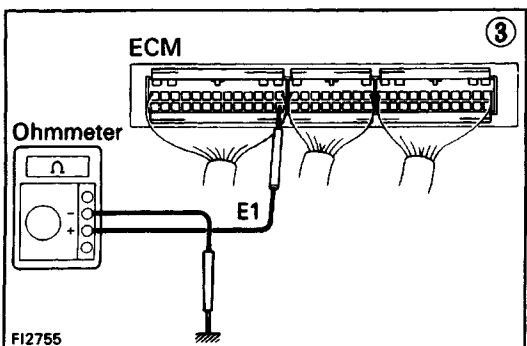
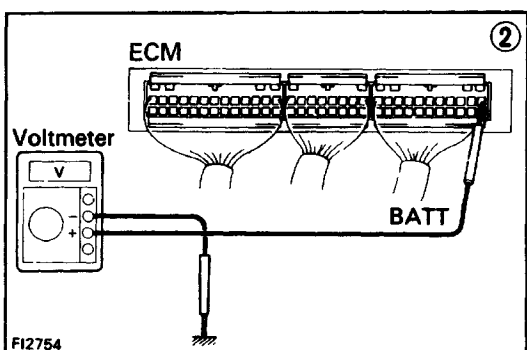
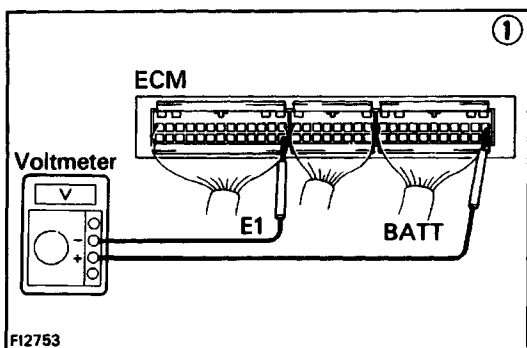
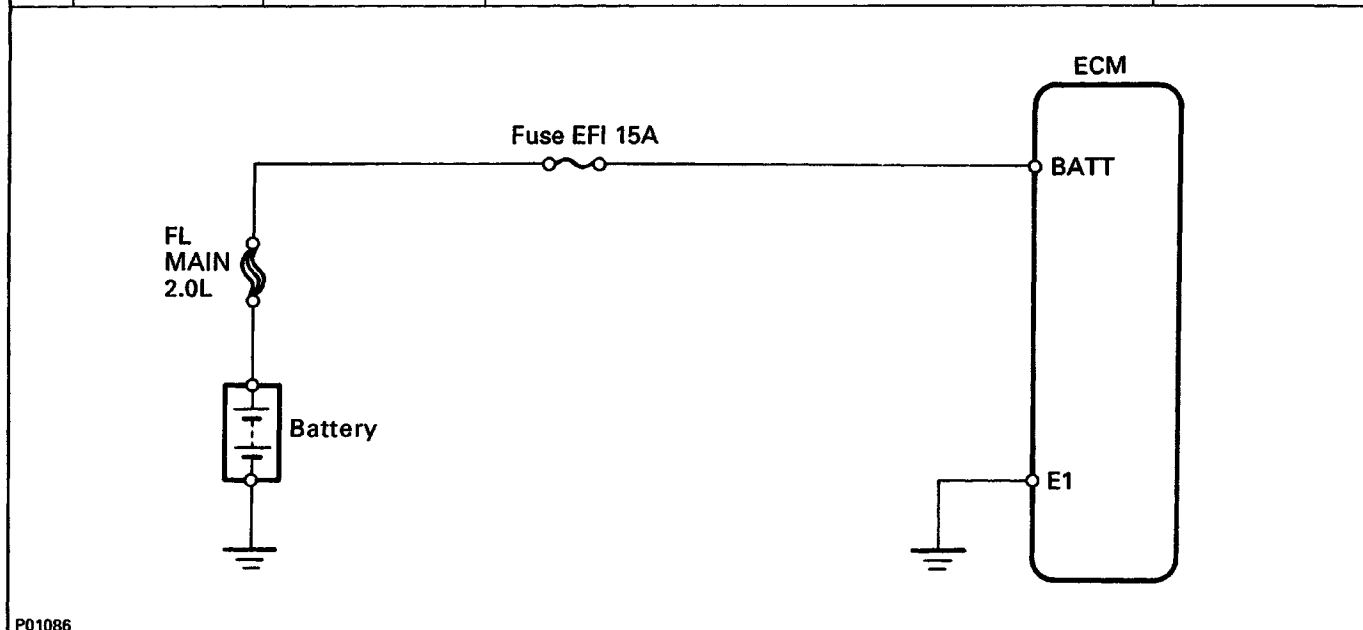
OK

Check wiring between MFI main relay and battery.

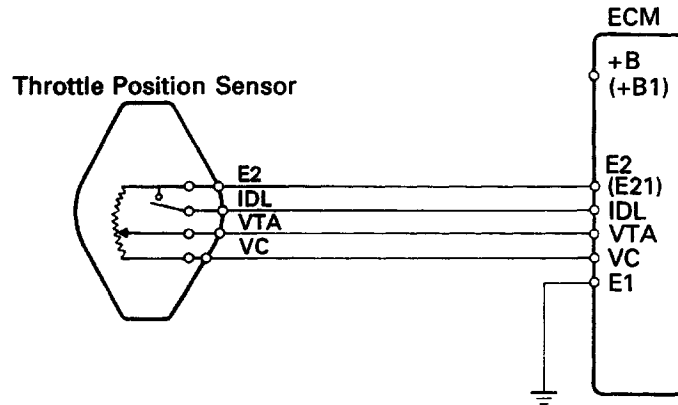
BAD

Repair or replace.

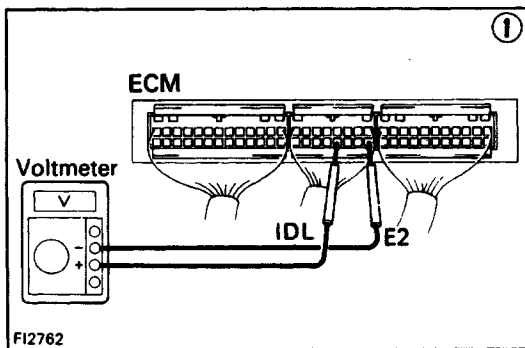
No.	Terminals	Trouble	Condition	STD voltage
2	BATT – E1	No voltage	–	10 – 14V



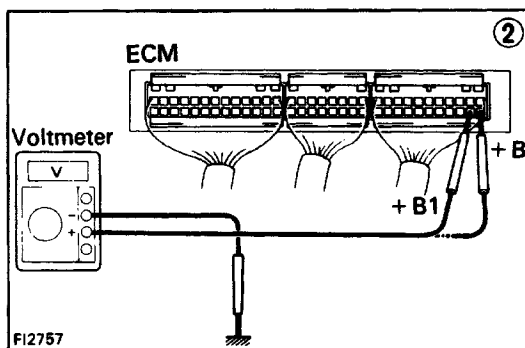
No.	Terminals	Trouble	Condition		STD voltage
3	IDL – E2	No voltage	IG SW ON	Throttle valve open	8 – 14V
	VC – E2			–	4.5 – 5.5 V
	VTA – E2			Throttle valve fully closed (Throttle opener must be cancelled first)	0.8 –1.2 V
				Throttle valve fully open	3.2 – 4.2 V



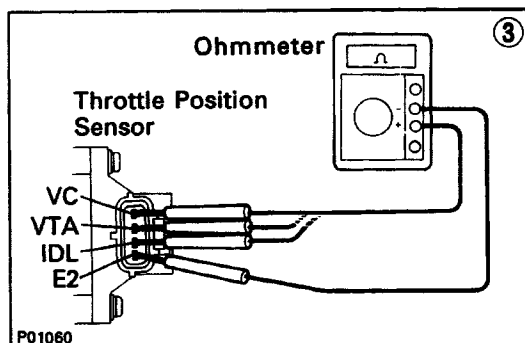
P01419



FI2762

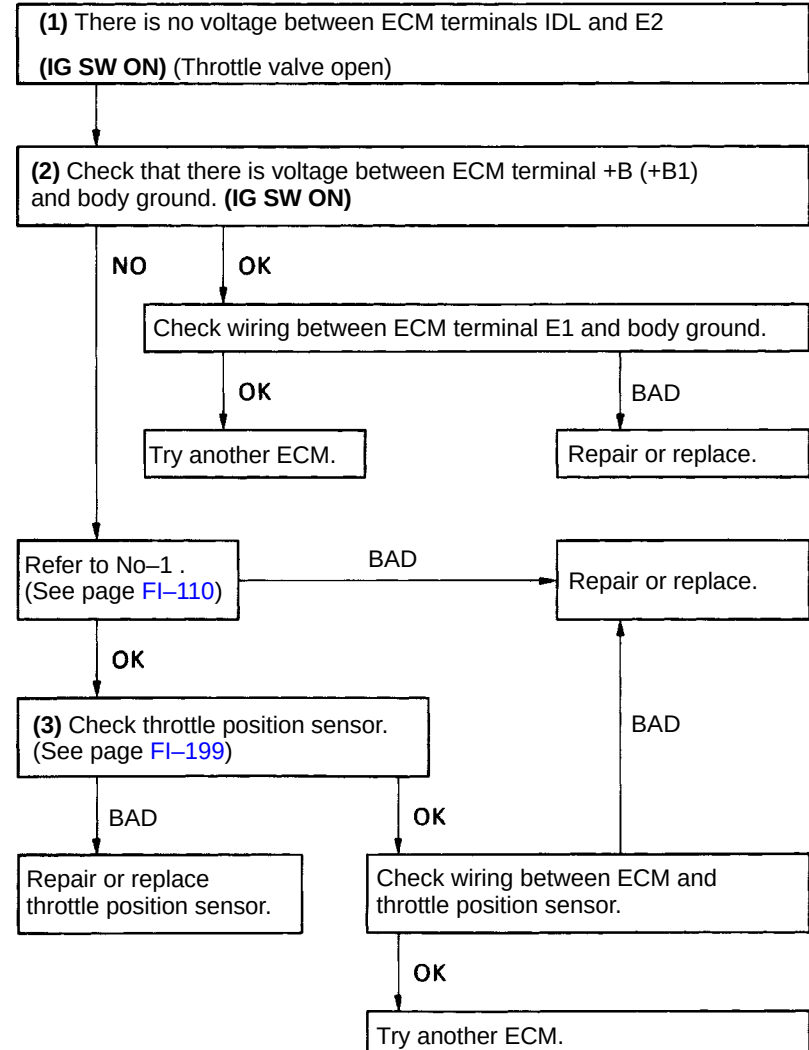


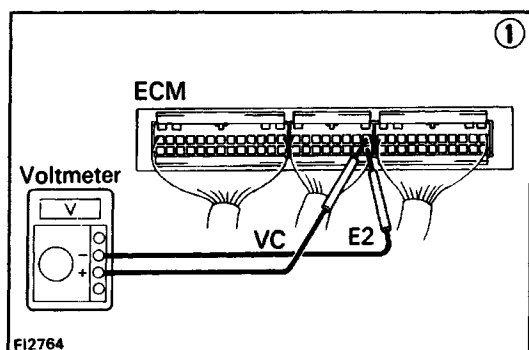
FI2757



P01060

• IDL – E2





• VC-E2

(1) There is no voltage between ECM terminals VC and E2.
(IG SW ON)

Check that there is voltage between ECM terminal +B (+B1) and body ground. (IG SW ON)

OK

NO

(2) Check throttle position sensor.
(See page FI-199)

Refer to No.1.
(See page FI-110)

BAD

OK

Repair or replace.

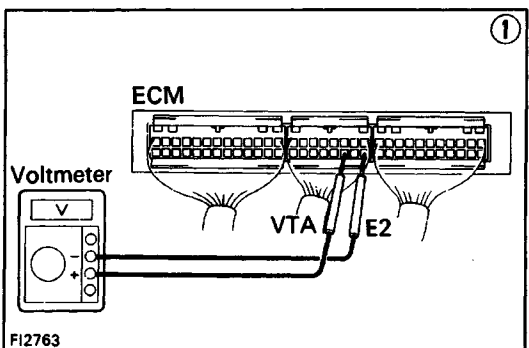
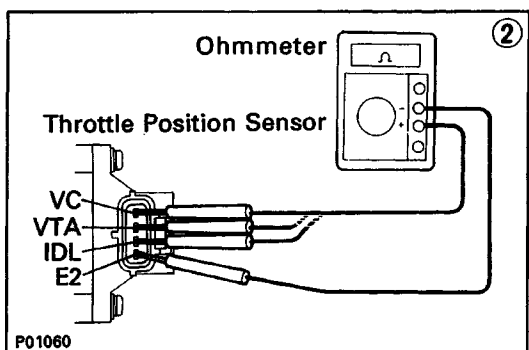
Check wiring between ECM and throttle position sensor.

OK

BAD

Try another ECM.

Repair or replace wiring.



• VTA - E2

(1) There is no specified voltage at ECM terminals VTA and E2.
(IG SW ON)

(2) Check that there is voltage between ECM terminals VC and E2
(IG SW ON).

NO

OK

Refer to VC - E2 trouble section.

OK

(3) Check throttle position sensor.
(See page FI-199)

BAD

Repair or replace.

OK

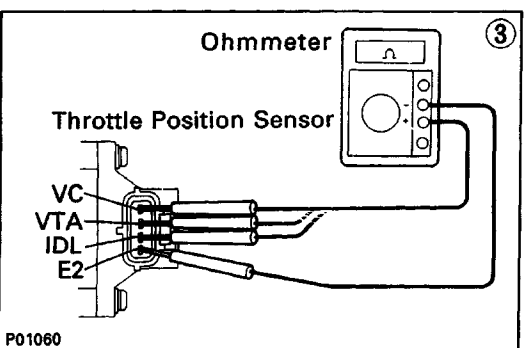
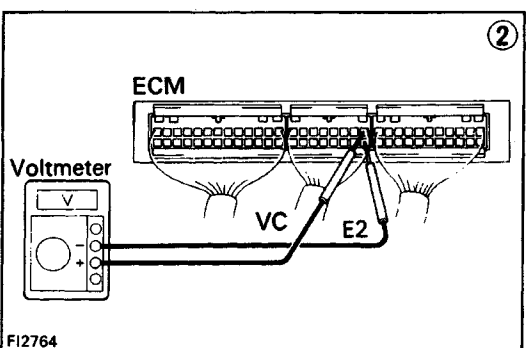
Check wiring between ECM and throttle position sensor.

BAD

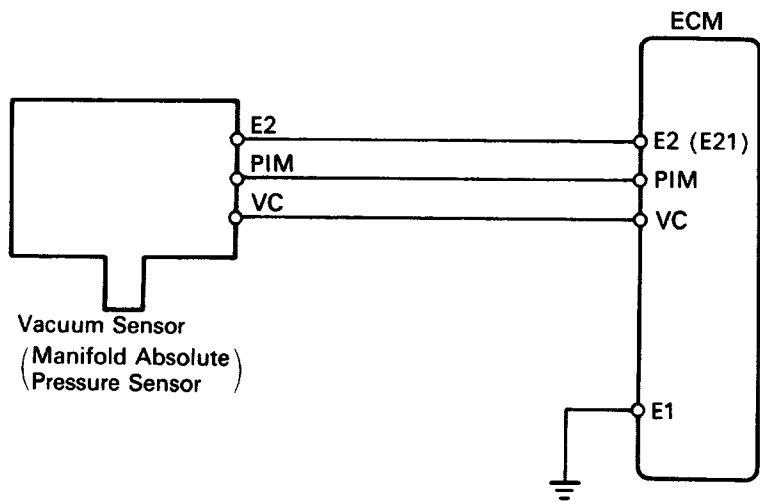
Repair or replace.

OK

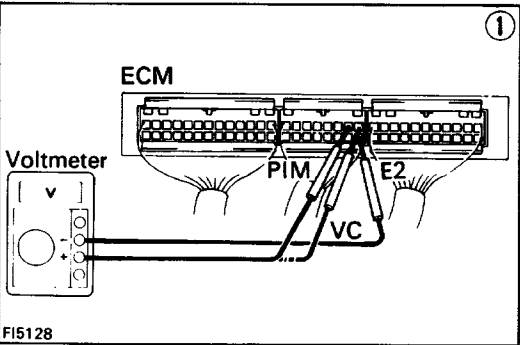
Try another ECM.



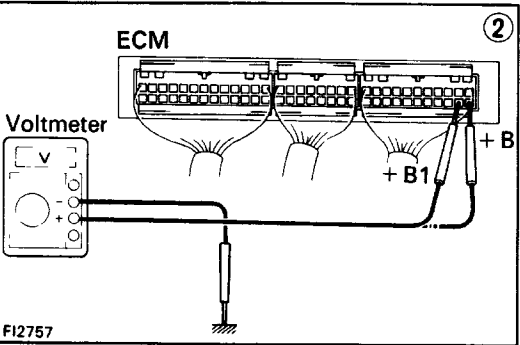
No.	Terminals	Trouble	Condition	STD voltage
4	PIM – E2	No voltage	IG SW ON	3.3 – 3.9 V
	VC – E2			4.5 – 5.5 V



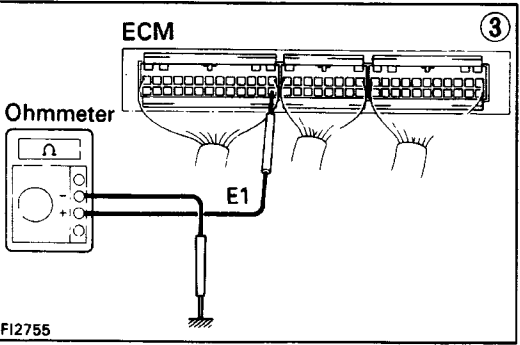
FI1226



FI5128

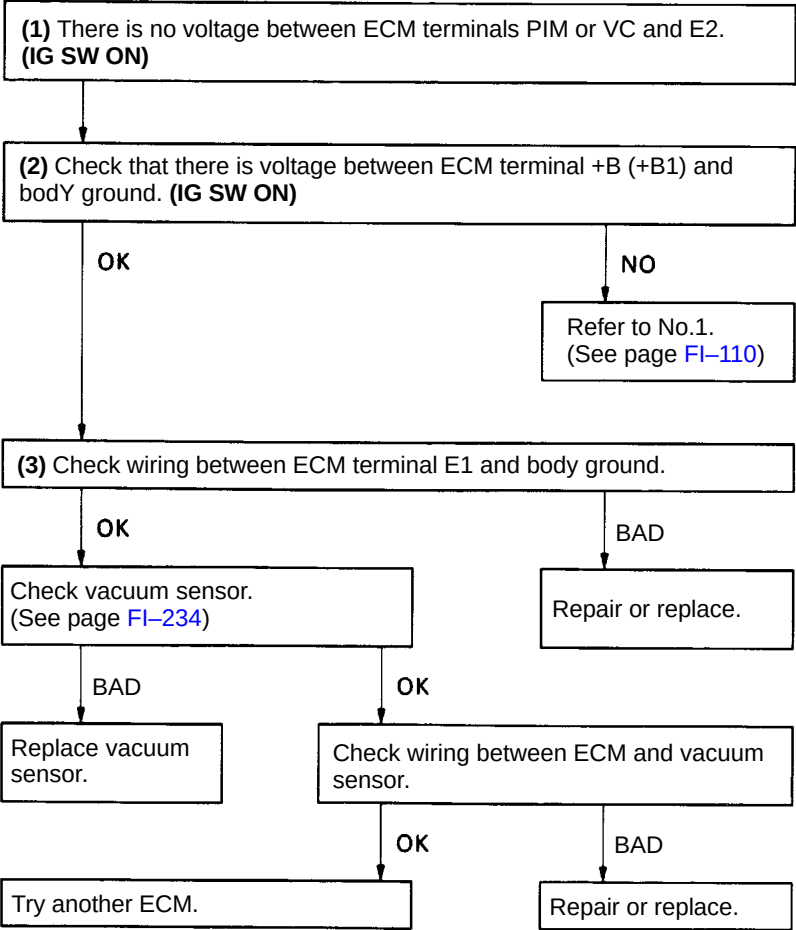


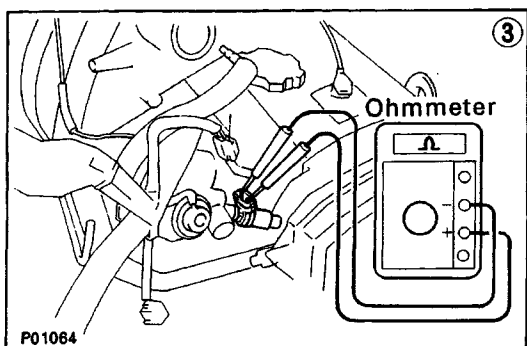
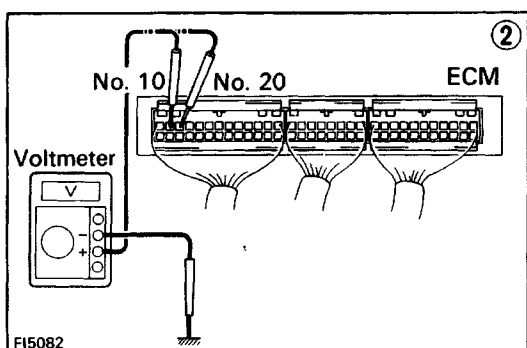
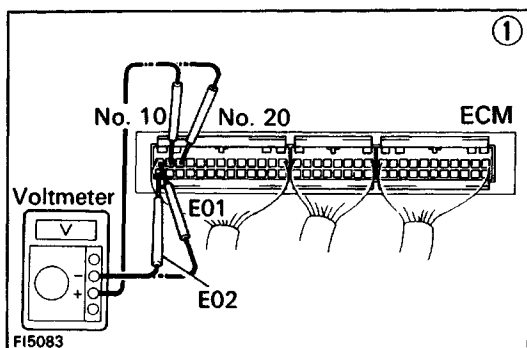
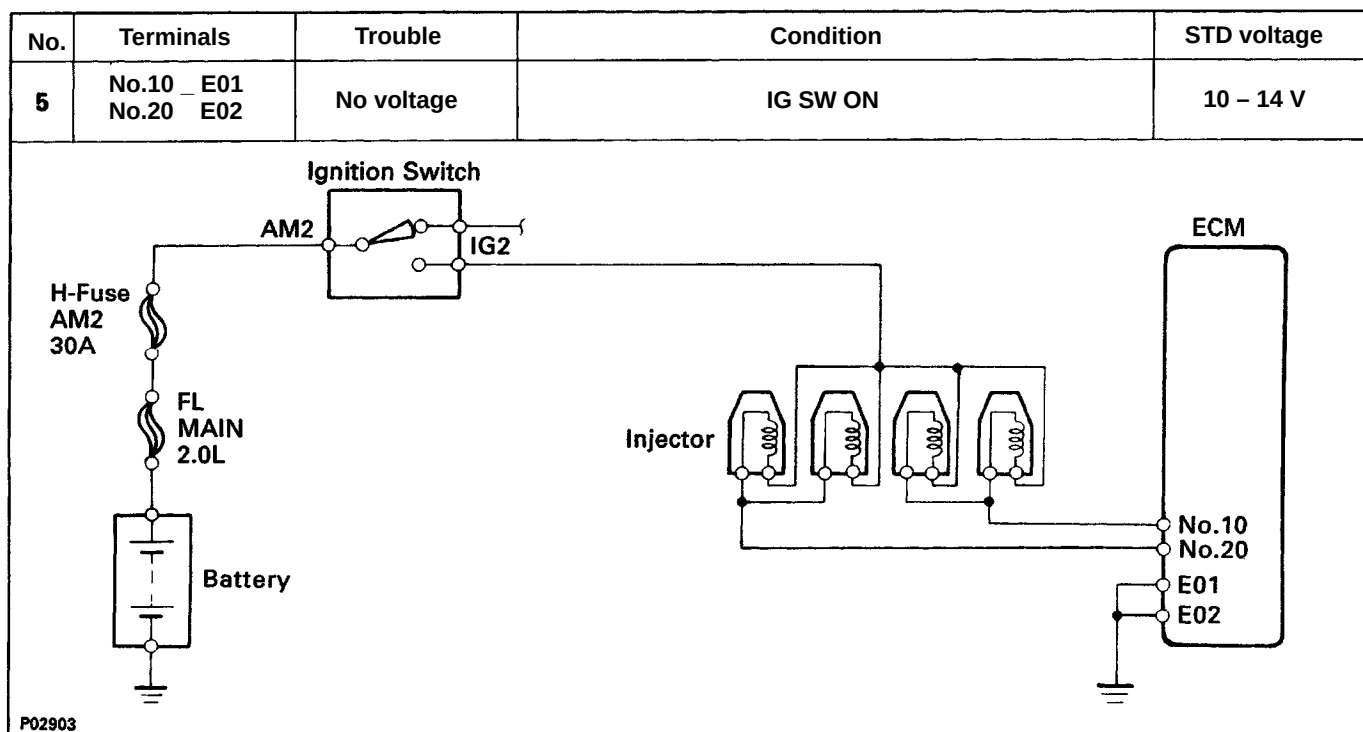
FI2757



FI2755

• PIM – E2, VC – E2





(1) There is no voltage between ECM terminals No.10 and/or No.20 and E01 and/or E02. (IG SW ON)

(2) Check that there is voltage between ECM terminal No.10 and/or No.20 and body ground.

NO OK

Check wiring between ECM terminal E01 and/or E02 and body ground.

OK

Try another ECM.

BAD

Repair or replace.

Check H-fuse, fusible link and ignition switch.

BAD

Repair or replace.

OK

(3) Check resistance of each injector.
STD resistance: Approx. 13.8 Ω

OK

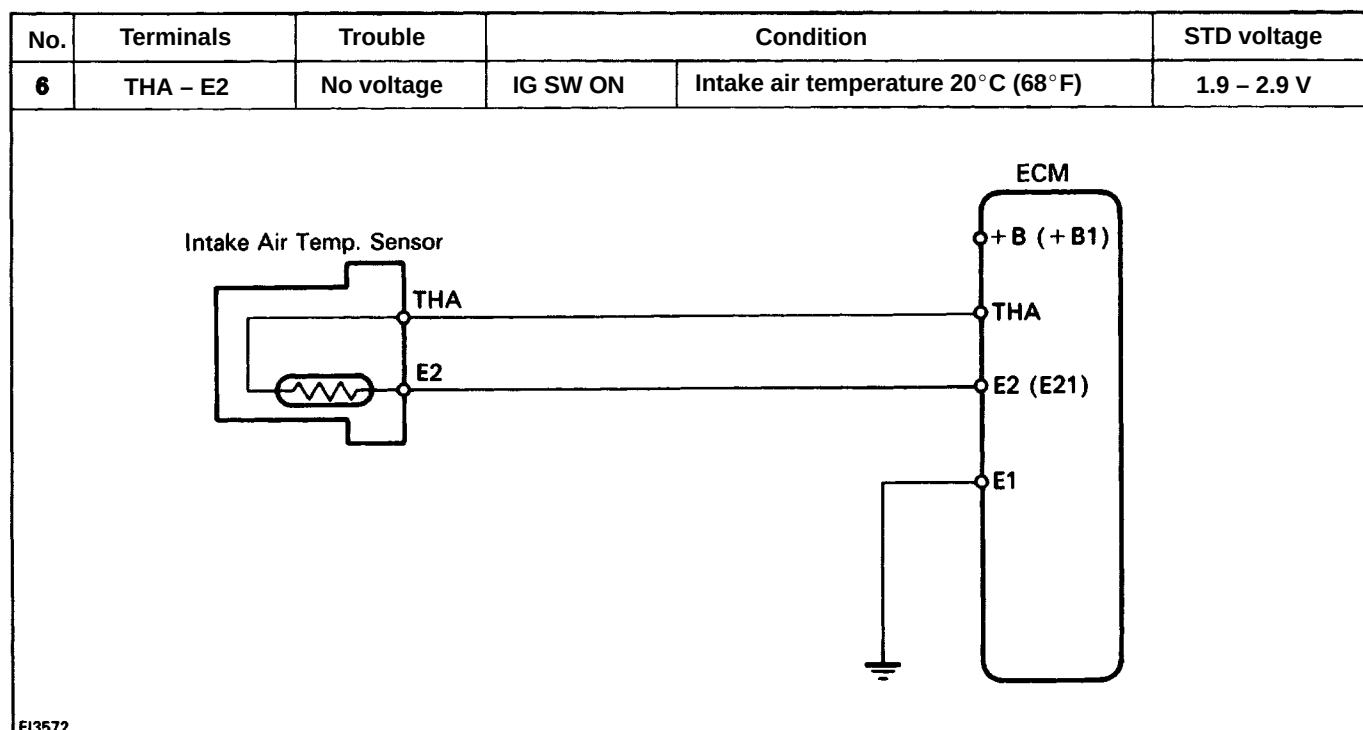
OK

Replace injector.

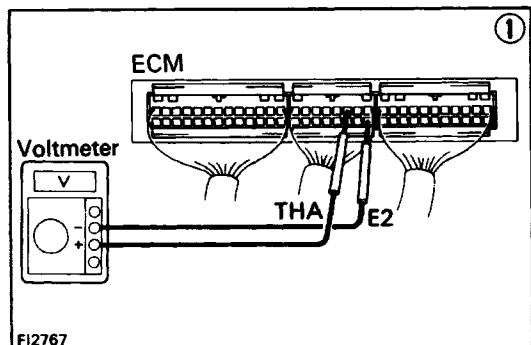
Check wiring between ECM terminal No.10 and/or No.20 and battery.

BAD

Repair or replace.



FI3572



FI2767

(1) There is no voltage between ECM terminals THA and E2 (IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1 and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-110)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

(3) Check intake air temp. sensor. (See page FI-231)

Repair or replace.

BAD

Replace intake air temp. sensor.

OK

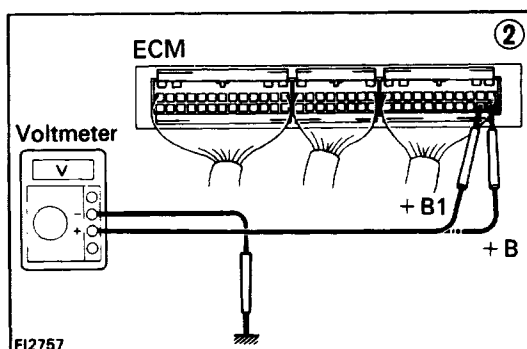
Check wiring between ECM and air temp. sensor.

Try another ECM.

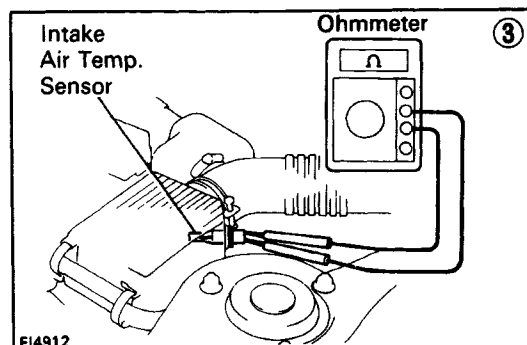
OK

BAD

Repair or replace.



FI2757



FI4912

No.	Terminals	Trouble	Condition		STD voltage
7	THW – E2	No voltage	IG SW ON	Engine coolant temperature 80° C (176° F)	0.1 – 1.1 V

Engine Coolant Temp. Sensor

ECM

THW

E2

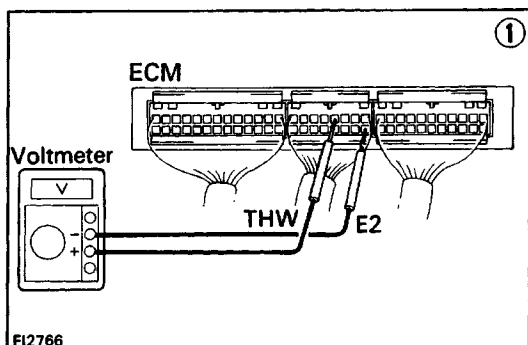
E1

+B (+B1)

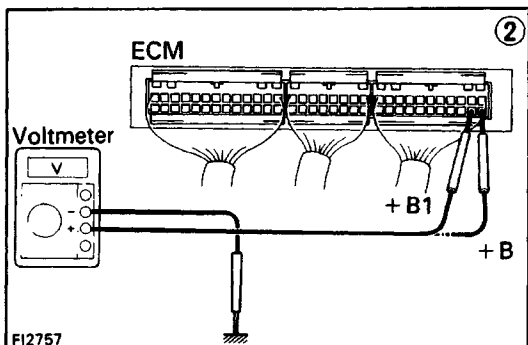
THW

E2 (E21)

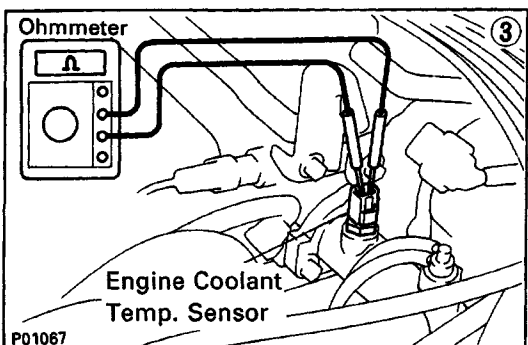
FI357



FI2766



FI2757



P01067

(1) There is no voltage between ECM terminals THW and E2.
(IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1 and body ground. (IG SW ON)

OK

NO

Refer to No.1
(See page FI-110)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

(3) Check engine coolant temp. sensor. (See page FI-231)

Repair or replace.

BAD

OK

Replace engine coolant temp. sensor

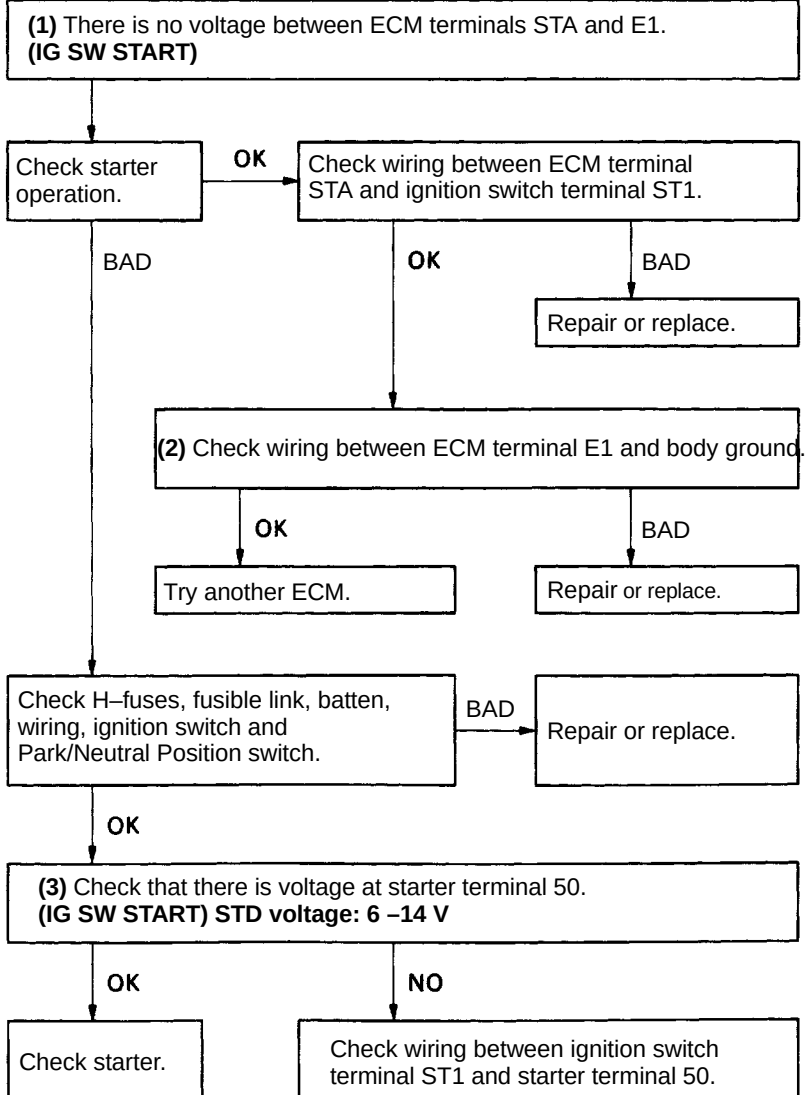
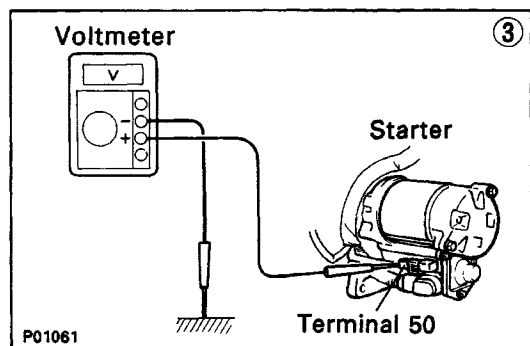
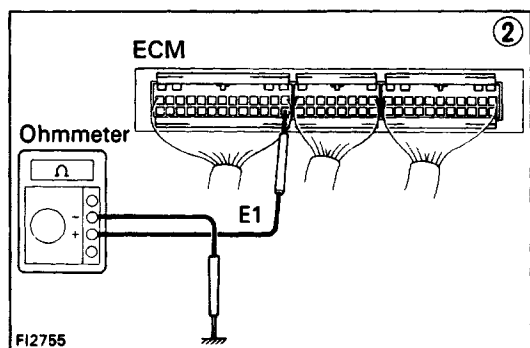
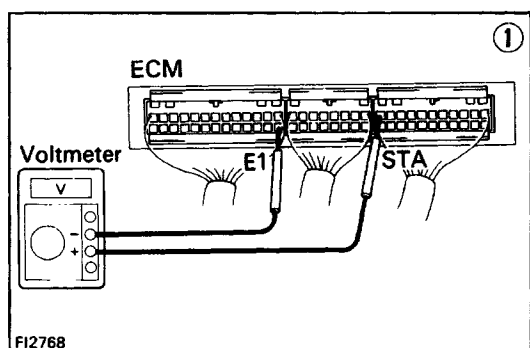
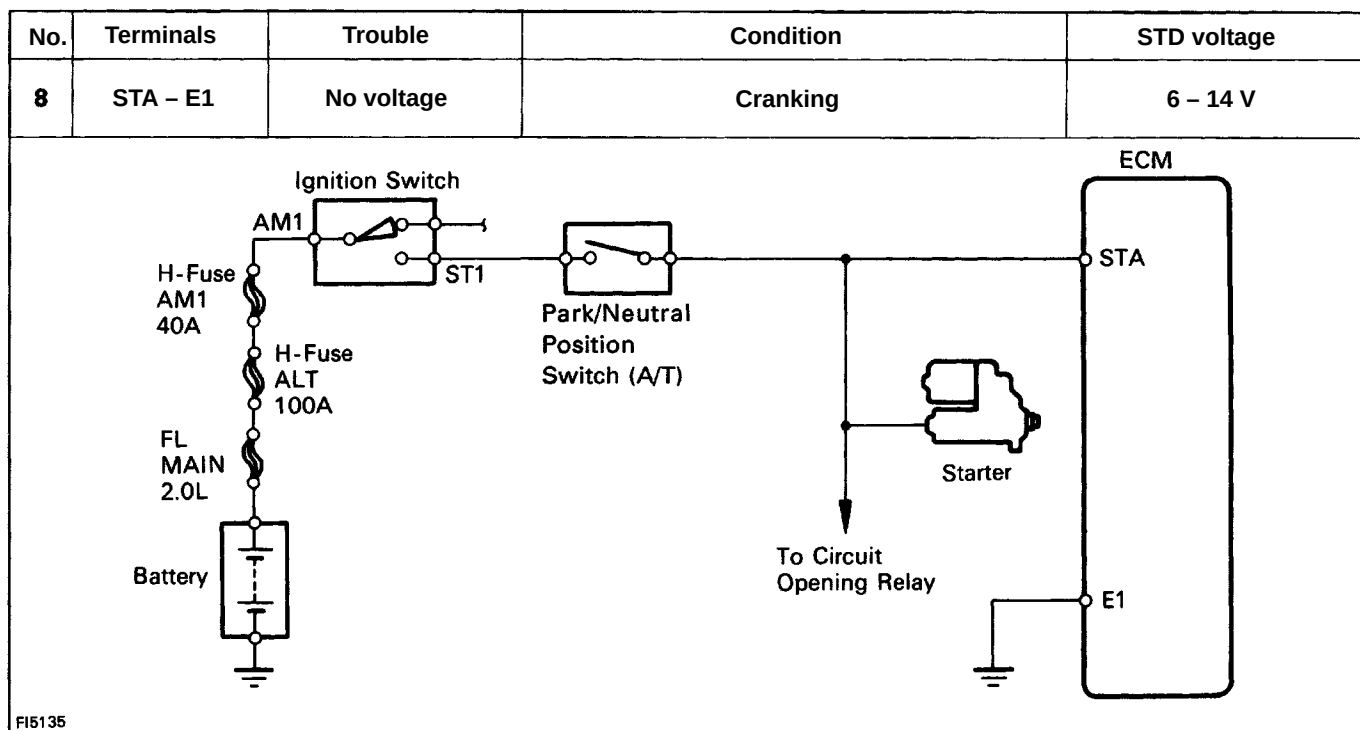
Check wiring between ECM and water temp. sensor.

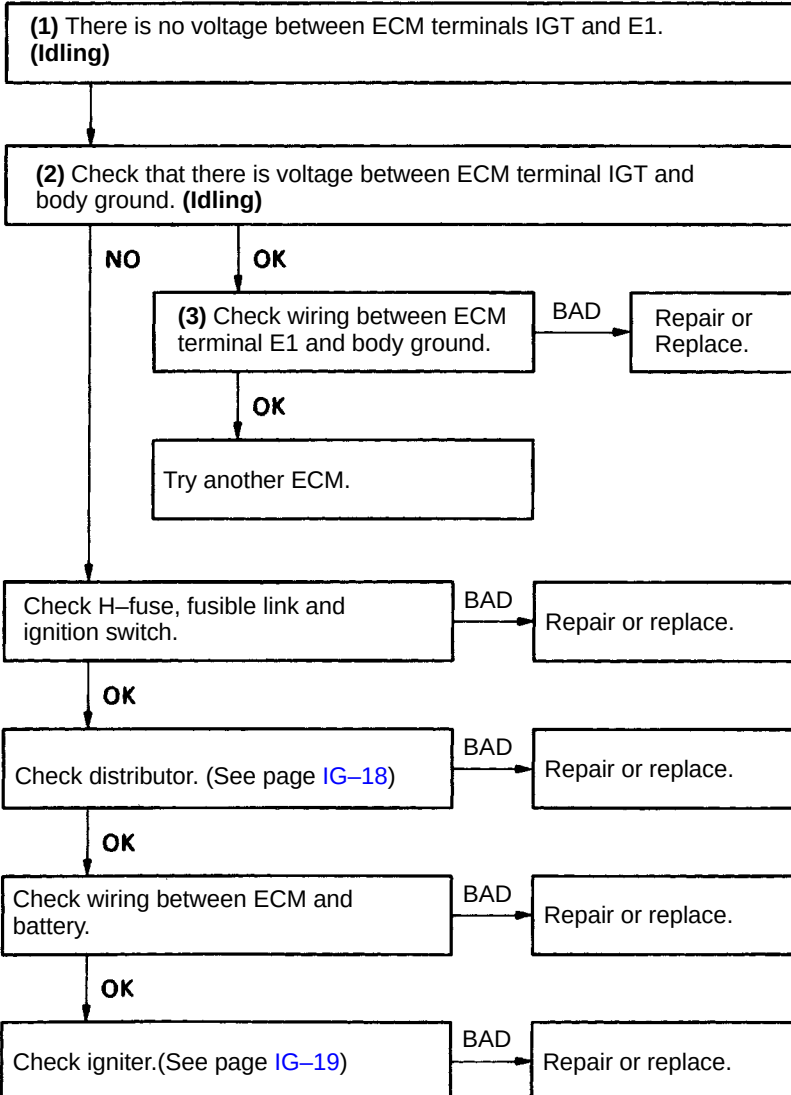
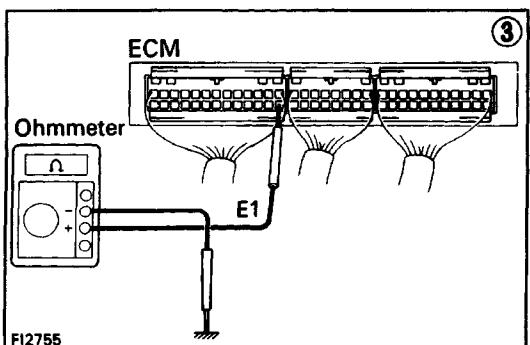
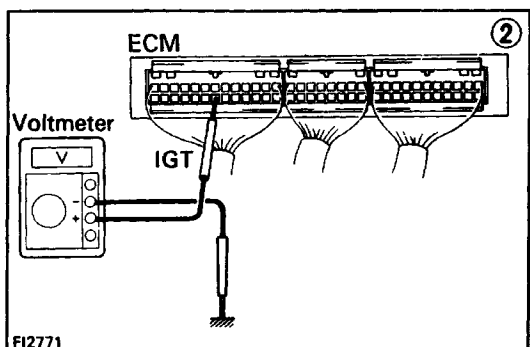
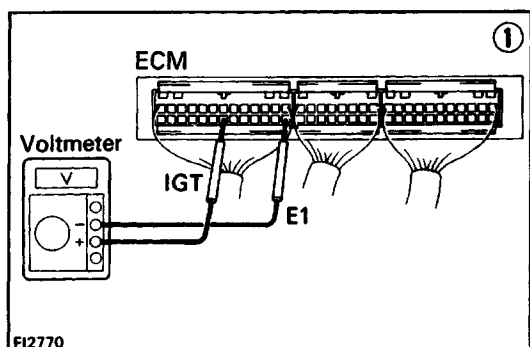
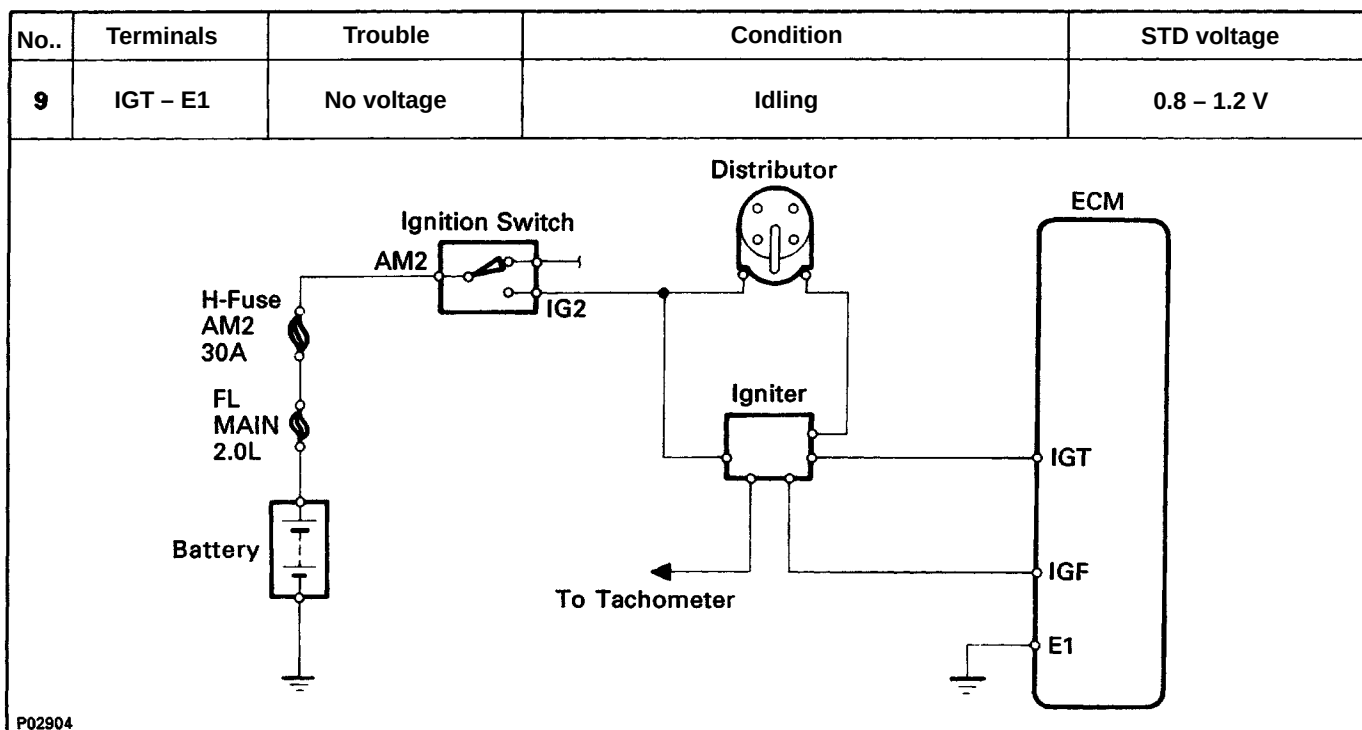
OK

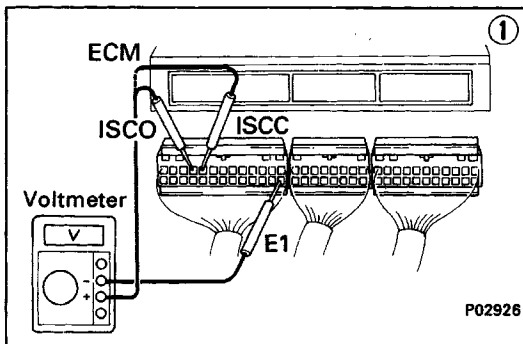
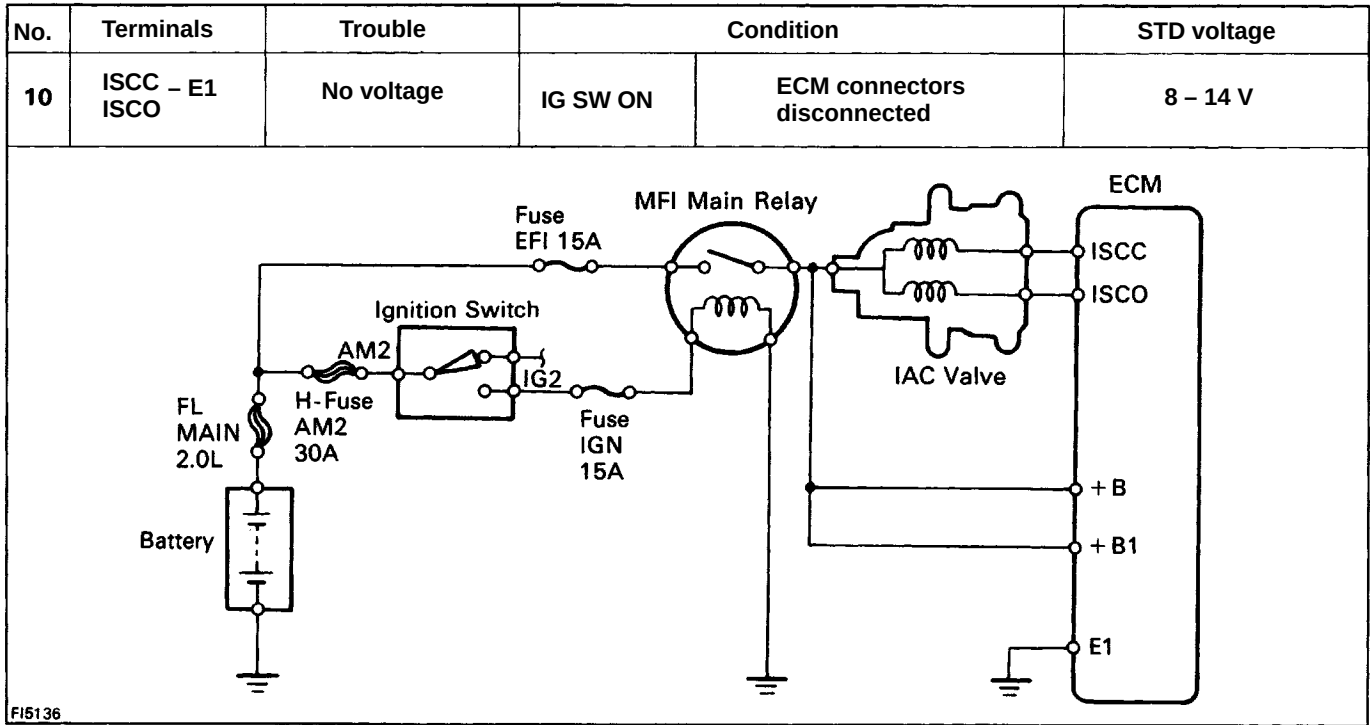
BAD

Try another ECM.

Repair or replace.







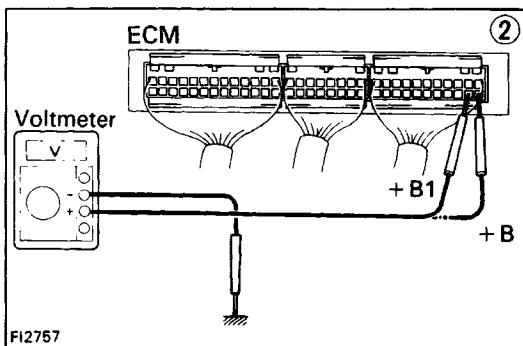
(1) There is no voltage between ECM terminals ISCC or ISCO and E1. (IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1 and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-110)



(3) Check resistance between IAC valve terminals +B and ISCC or ISCO. STD resistance: 19.3 - 22.3Ω

BAD

Replace IAC valve.

OK

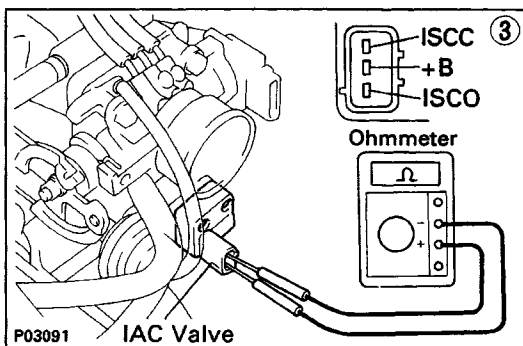
Check wiring between ECM and IAC valve.

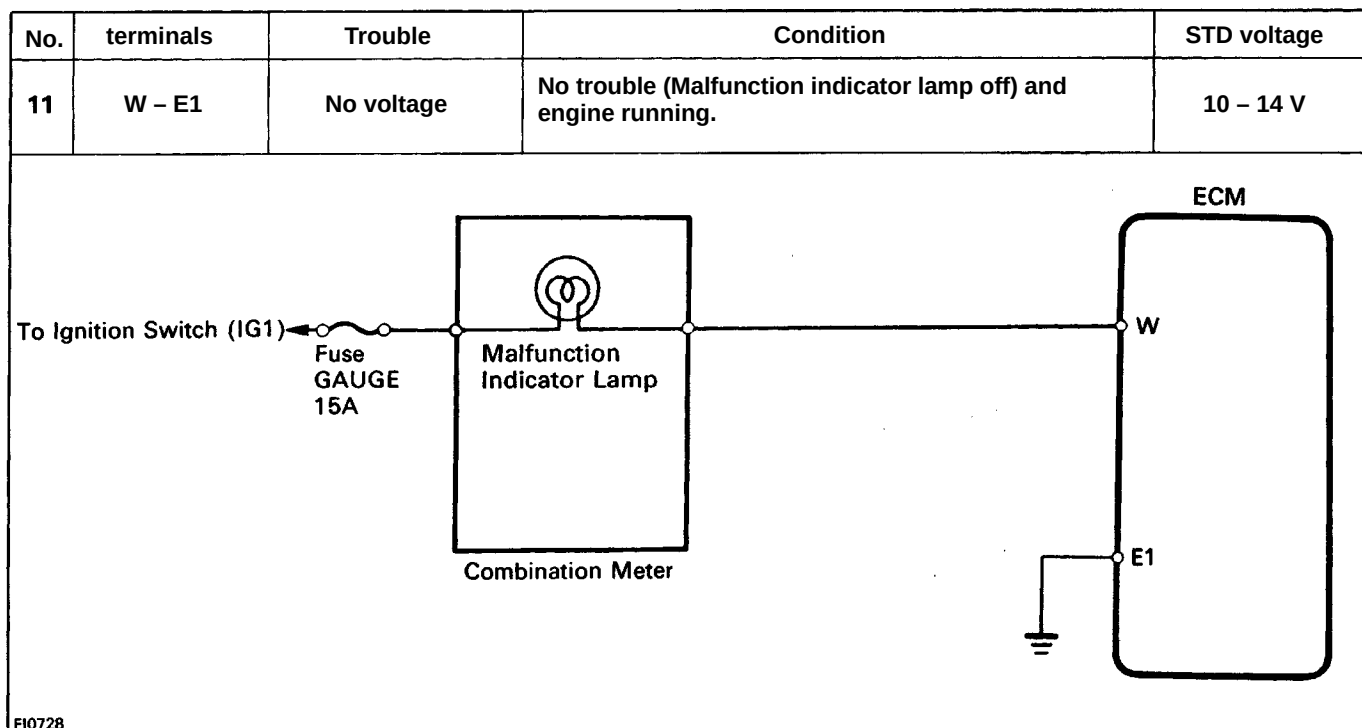
BAD

Repair or replace wiring.

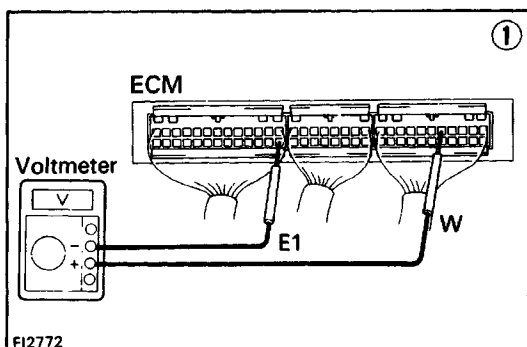
OK

Try another ECM.

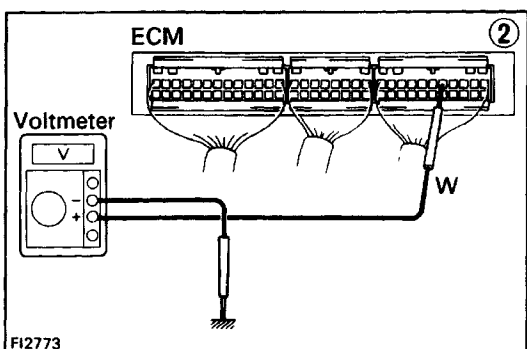




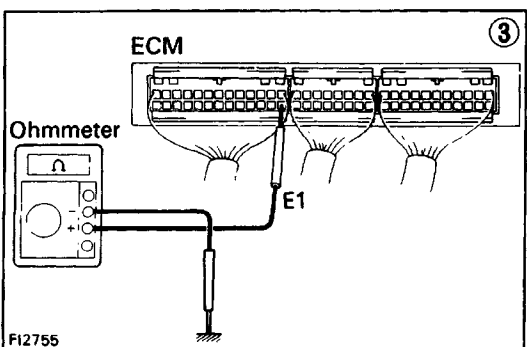
FI0728



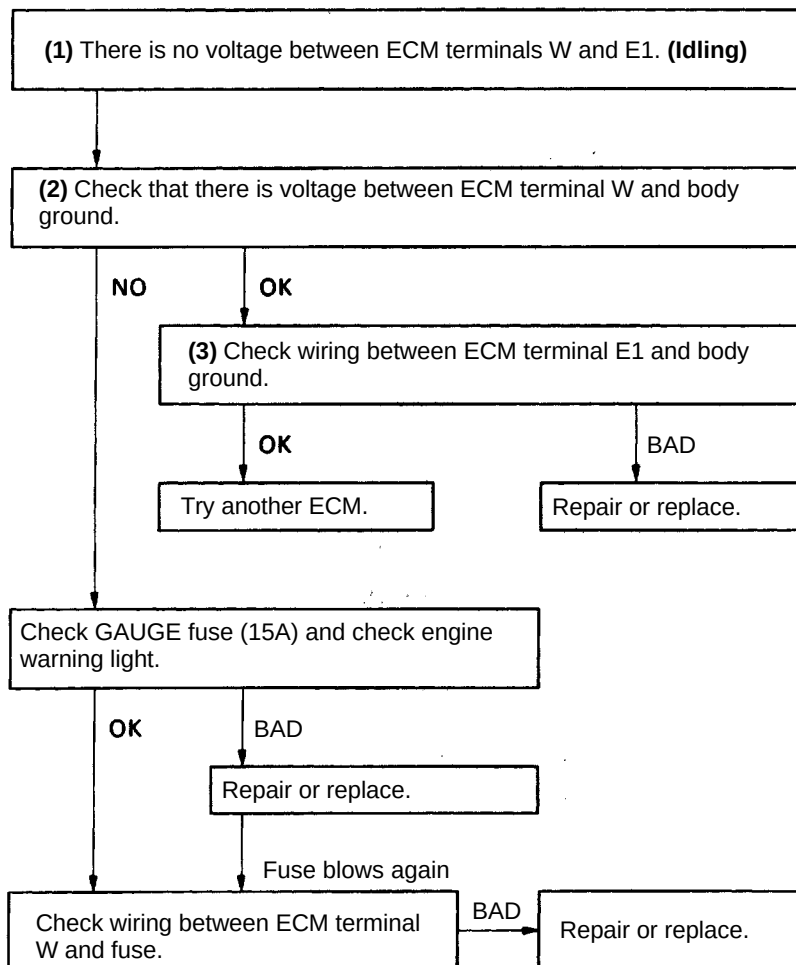
FI2772

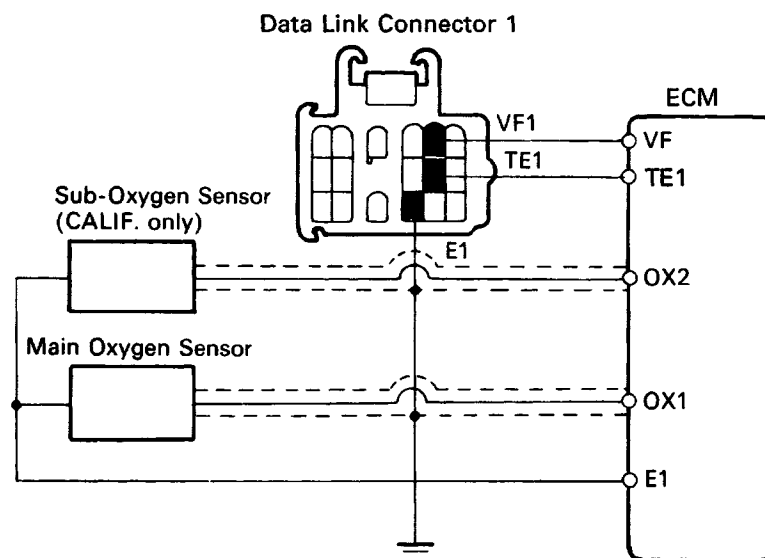


FI2773

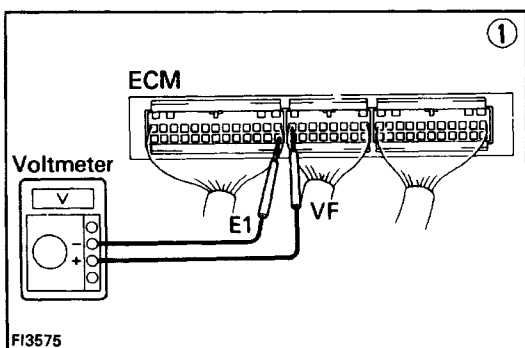


FI2755

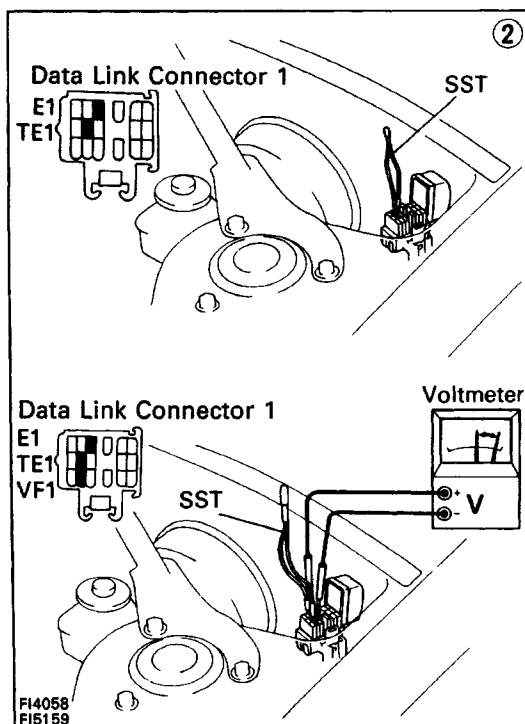




FI3841



FI3575

FI4058
FI5159

(1) There is no voltage between ECM terminals VF and E1.

Check that there is voltage between ECM terminal VF and body ground.

NO

OK

Check wiring between ECM terminal E1 and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Is air leaking into air induction system?

YES

Repair air leak.

NO

Check spark plugs. (See page IG-16)

BAD

Repair or replace.

OK

Check distributor and ignition system. (See page IG-4)

BAD

Repair or replace.

OK

Check fuel pressure. (See page FI-128)

BAD

Repair or replace.

OK

Check injectors. (See page FI-169)

BAD

Repair or replace.

OK

Check vacuum sensor. (See page FI-234)

BAD

Repair or replace.

OK

(2) Check operation of oxygen sensors. (See pages FI-237 and 239)

BAD

System Normal

BAD

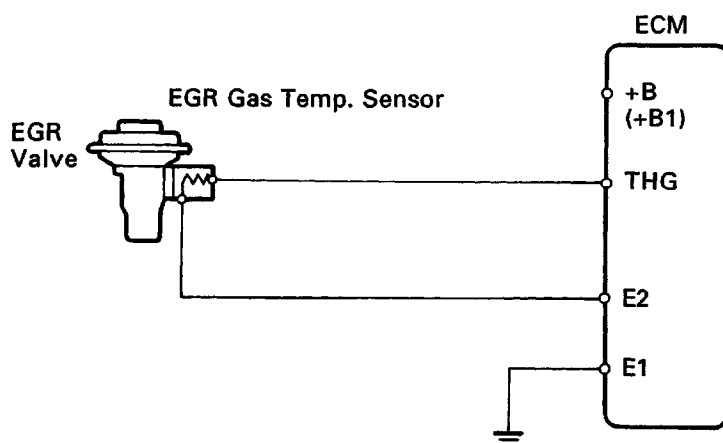
Check wiring between oxygen sensor and ECM.

BAD

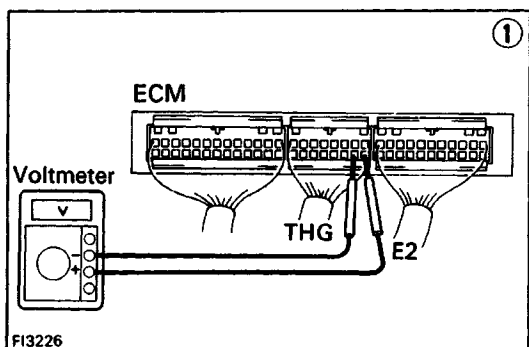
Repair wiring.

OK

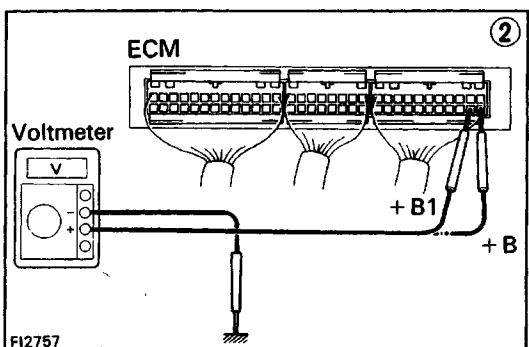
Replace oxygen sensors.

CALIF. only

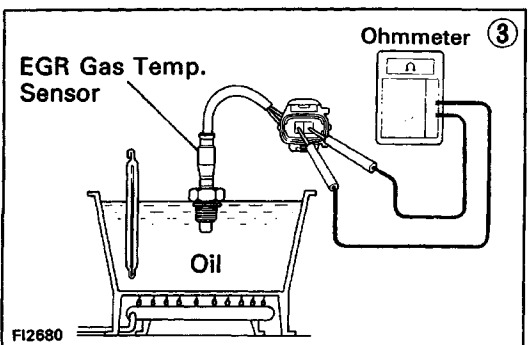
FI6320



FI3226



FI2757



FI2680

(1) There is no voltage between ECM terminals THG and E2. (IG SW ON)

(2) Check that there is voltage between ECM terminal +B or +B1 and body ground. (IG SW ON)

OK

NO

Refer to No.1.
(See page FI-110)

Check wiring between ECM terminal E1 and body ground.

OK

BAD

Repair or replace.

Check EGR system.
(See page EC-21)

BAD

Repair or replace.

OK

(3) Check EGR gas temp. sensor. (See page FI-236)

BAD

Replace EGR gas temp. sensor.

OK

Check wiring between ECM and EGR gas temp. sensor.

OK

Try another ECM.

BAD

Repair or replace.